A MODULAR AND COST-EFFECTIVE FRAMEWORK FOR WIRELESS FOUNDATION MODELS THROUGH WORLD KNOWLEDGE INTEGRATION

A DISSERTATION
SUBMITTED TO THE DEPARTMENT OF
ELECTRICAL AND COMPUTER ENGINEERING
AND THE COMMITTEE ON GRADUATE STUDIES OF
ROWAN UNIVERSITY
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY

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October 2026

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I certify that I have read this dissertation and that in my opinion it is fully adequate, in scope and in quality, as a dissertation for the degree of Doctor of Philosophy.

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Abstract

THE development of foundation models for wireless communications is limited by the cost of training large RF models from scratch, the scarcity of labeled signals, and the fact that models built for one task or dataset rarely carry over to another. This dissertation asks how far wireless foundation-model capabilities can be reached without training a large RF model from scratch, by borrowing world knowledge from vision self-supervision, pretrained language models, and learning-to-learn.

It proposes a modular framework organized around three research questions. For representation (RQ1), self-supervised, vision-based encoders (DenoMAE and DenoMAE2.0, co-authored work used here as the representation) give label-efficient features, reaching 82.4% on matched 10-class data, and the CageDroneRF benchmark shows that vision models work on real over-the-air RF captures (0.986 mAP@0.5 for detection); the representation collapses at extreme noise and transfers weakly to an unseen dataset. For reasoning (RQ2), Plug-and-Play and DiSC-AMC convert signal statistics into compact symbolic prompts with retrieved examples: a fine-tuned 7B model reaches 83.0% in distribution and 82.5% out of distribution, and a training-free 7B model reaches 71.9% at 0.5K tokens, above the proprietary o3-mini baseline at 2.9K tokens. For adaptation (RQ3), Meta-Task's learnable auxiliary task improves few-shot vision benchmarks but gives no net gain on cross-dataset wireless classification.

The components compose: DiSC-AMC retrieves over a learned encoder and lifts RadioML

accuracy up to eight times over the encoder alone, and the adaptation test runs all three modules together. Across them one finding recurs: the generality of the underlying representation bounds what reasoning and adaptation can add. The domain-specific encoder that wins on matched in-distribution data transfers worse to a new dataset than a generic one, and neither reasoning nor a learned regularizer repairs it. The framework's cost-effectiveness is supported by token- and parameter-efficiency results, not by a complete latency, energy, or monetary comparison, and the results are bounded by single evaluation runs and the absence of published split and run manifests.

Acknowledgments

I WOULD like to express my sincere gratitude to my advisor, Professor Huaxia Wang, for their unwavering support and guidance throughout my PhD journey. Their insightful feedback and encouragement were instrumental in shaping this thesis.

I am also grateful to the members of my dissertation committee, Professor Jie Li, Professor Qianqian Zhang, Professor Shen-Shyang Ho and Professor Nikhil Muralidhar, for their valuable time and constructive comments.

My research would not have been possible without the brilliant collaborators and colleagues in our research group. I especially thank Atik Faysal for the stimulating discussions and shared discoveries.

I extend my heartfelt thanks to my family and friends for their endless love, patience, and support. To my parents, Azam and Farhad, and my partner, Rozhina, thank you for believing in me every step of the way.

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Nomenclature

AI Artificial Intelligence

ANN Artificial Neural Network

API Application Programming Interface

DenoMAE Denoising Masked Autoencoder

DL Deep Learning

GAN Generative Adversarial Network

GPU Graphics Processing Unit

LLM Large Language Model

MAE Masked Autoencoder

ML Machine Learning

MSE Mean Squared Error

NLP Natural Language Processing

ReLU Rectified Linear Unit

RL Reinforcement Learning

SGD Stochastic Gradient Descent

t-SNE t-Distributed Stochastic Neighbor Embedding

UML Unified Modeling Language

WPFM Wireless Physical-Layer Foundation Model

Chapter 1

Introduction

THE rapid evolution of wireless communications, driven by technologies such as the upcoming 6G standard[4], necessitates a transformation in how Artificial Intelligence (AI) addresses challenges unattainable by traditional optimization techniques. The remarkable success of foundation models in computer vision (CLIP[5], MAE[6]) and natural language processing (GPT[7], BERT[8]) has demonstrated the transformative power of learning universal representations that generalize across tasks with minimal fine-tuning. These models leverage massive unlabeled datasets through self-supervised learning to discover rich, transferable features that encode fundamental properties of their respective domains. Yet, the current AI landscape in wireless networks remains dominated by models developed and trained for isolated, task-specific applications[4], leading to significant limitations in knowledge sharing, reuse, and adaptability. This dissertation addresses a fundamental question: *Can we develop practical pathways toward Wireless Physical-Layer Foundation Models (WPFMs) by strategically approximating and integrating world knowledge through modular, cost-effective architectures?*

The limitations of current wireless AI systems are particularly evident in Automatic Modulation Classification (AMC), a critical component of spectrum management, signal

identification, and cognitive radio systems[9]. Conventional deep learning approaches face three major challenges: (1) **Imbalance across conditions:** Models often fail to distinguish between low and high Signal-to-Noise Ratio (SNR) conditions[9], resulting in uneven classification performance where low SNR signals obscured by noise require specialized handling compared to high SNR signals; (2) **Data scarcity and training overhead:** Acquiring high-quality labeled radio frequency (RF) signal data demands specialized expertise and equipment, while traditional deep learning necessitates large labeled datasets[10], creating barriers to scalability; (3) **Lack of interpretability and generalization:** Task-specific models struggle to adapt quickly to new tasks, generalize across diverse channel characteristics, and lack capacity for human-understandable interaction[9].

Building wireless foundation models from scratch to address these challenges is prohibitively expensive [5]. Unlike images or text, wireless signals exhibit unique characteristics: they are inherently noisy, their structure depends on both transmitter design and channel physics, and their semantic meaning is encoded in subtle statistical patterns. These properties demand an economical approach that leverages existing world knowledge rather than massive isolated training. This dissertation presents such a solution: a modular and cost-effective framework that achieves foundation model capabilities through the strategic integration of world knowledge. Our approach validates three essential components that work synergistically: (1) a **rich feature encoder** (DenoMAE) that uses self-supervised learning to approximate world knowledge of signal representations; (2) an **intelligent reasoning engine** (LLM integration) that performs open-set classification by converting signal features into natural language prompts; and (3) an **adaptation module** (Meta-Task) that enables rapid knowledge transfer to new tasks with minimal data. By demonstrating proof-of-concept for these core components, we provide both a practical framework and a conceptual blueprint for achieving foundation model capabilities in wireless communications without prohibitive training costs.

1.1 Objectives and Motivation

The main objective of this dissertation is to establish foundational principles and practical methodologies for developing Wireless Physical-Layer Foundation Models (WPFMs). These models serve as universal representations of wireless signals, allowing for generalization across various modulation schemes, channel conditions, and tasks. Instead of pursuing a cost-prohibitive strategy of building massive models from scratch, this work demonstrates that the capabilities of foundation models can be achieved more economically through a strategic and modular design. This design integrates self-supervised learning, reasoning based on large language models (LLMs), and meta-learning adaptation.

The motivation behind this approach is inspired by the success of foundation models in various domains. Models such as GPT-4 [11] and CLIP [5] have demonstrated that pre-training on large, diverse datasets leads to the development of powerful and transferable representations. These successes rely on three essential components: vast amounts of unlabeled data, self-supervised pretext tasks that capture domain-invariant structures, and flexible architectures that allow for adaptation to specific tasks.

In the field of wireless communications, petabytes of unlabeled signal data are generated daily. However, the unique characteristics of these signals, particularly their noisy and physics-dependent nature, pose distinct challenges. This dissertation aims to address the key challenge of developing WPFMs that can comprehend and represent these complex signals without incurring the prohibitive costs associated with isolated training, while still achieving the generalization capabilities seen in their vision and language counterparts.

To address this, the remainder of this section is structured to first articulate the long-term vision for WPFMs. It then deconstructs this vision into a set of three research questions that guide the technical work. Finally, it outlines the specific technical approach and contributions developed in this dissertation to answer these questions.

1.1.1 Vision: Wireless Foundation Models

We envision wireless foundation models analogous to CLIP in vision or GPT in language: pre-trained on diverse, unlabeled signals to learn representations encoding fundamental wireless signal properties, then adapted to specific tasks (classification, denoising, channel estimation) with minimal fine-tuning. Ideally, such models would:

- **Generalize Across Modulations:** Recognize modulation schemes never encountered during training through compositional understanding of signal properties, enabled by rich learned representations
- **Handle SNR Variation Robustly:** Keep performance usable across noise conditions, a property this dissertation tests from -10 dB to $+10$ dB and finds to fail at extreme noise (-30 dB)
- **Operate Cost-Effectively:** Enable training-free or few-shot deployment through LLM integration and self-supervised pre-training, avoiding exhaustive retraining for each scenario
- **Adapt Rapidly with Minimal Data:** Use meta-learning to specialize for new deployment scenarios from few labeled examples, a property this dissertation supports on few-shot vision benchmarks but not on cross-dataset wireless AMC
- **Provide Interpretable Reasoning:** Convert signal features to natural language prompts for LLM-based classification, enabling human-understandable explanations and prompt-based optimization
- **Leverage World Knowledge:** Integrate pre-trained LLMs' reasoning capabilities to approximate and transfer existing knowledge rather than learning everything from scratch

This list is the target, not the result. The chapters that follow test how far each property can be reached with modular, reused components, and the answers are bounded: some properties are supported (open-set reasoning with compact prompts), some only in part (rapid adaptation), and some fail at stated limits (extreme noise, an unseen dataset).

1.1.2 Research Questions

This dissertation addresses three interconnected research questions toward realizing cost-effective wireless foundation models:

RQ1: Can vision-based, self-supervised learning effectively transfer world knowledge to the wireless domain?

- How can wireless signals be represented as images (e.g., spectrograms, constellation diagrams) to leverage pre-trained vision models like ViT and YOLO?
- What self-supervised objectives (e.g., masked autoencoding) are most effective for learning rich, transferable representations from these signal-images?
- Can multimodal learning, combining different image representations (e.g., IQ samples and constellation diagrams), improve feature quality and robustness?
- How effectively do features learned by vision transformers transfer to wireless-specific tasks like modulation classification and signal detection?
- To what extent can this cross-modal approach reduce the dependency on large labeled datasets in the wireless domain?

RQ2: Can LLMs enable open-set classification of wireless signals through world knowledge transfer?

- Can statistical signal features (cumulants, spectral properties) be converted to natural language prompts for LLM reasoning?
- Do pre-trained LLMs' in-context learning capabilities transfer to wireless signal classification without fine-tuning?
- What bridges the semantic gap between quantitative signal properties and linguistic descriptions?
- How competitive is training-free LLM-based classification compared to extensively trained task-specific models?
- Can LLMs provide interpretable reasoning for wireless classification decisions, enabling human-understandable explanations?

RQ3: Can meta-learning enable rapid adaptation to new tasks with minimal labeled data?

- What episodic training strategies maximize generalization for wireless classification tasks?
- Can task-aware regularization within meta-learning prevent overfitting while maintaining adaptability?
- How can we design episode generation strategies that account for signal-specific characteristics like SNR variation and modulation similarity?
- What design principles enable efficient knowledge transfer across different channel conditions and modulation schemes?
- How many labeled examples are minimally required for effective adaptation to new wireless tasks?

1.2 Scope

This work focuses on automatic modulation classification (AMC) as the primary testbed for developing foundational principles for wireless physical-layer foundation models. The scope is defined by the following domain parameters and limitations.

1.2.1 Primary Domain: Wireless Communication

Our core application domain is automatic modulation classification under challenging conditions:

- **Signal Representations:** We explore time-domain IQ samples, constellation diagrams, and noise patterns as complementary modalities
- **Operating Conditions:** SNR ranges mainly from -10 dB to $+10$ dB in training and evaluation, with out-of-distribution queries at -15 to -11 dB, RadioML at -20 to $+20$ dB, and an extreme-noise probe at -30 dB
- **Modulation Schemes:** A 10-class synthetic set (4ASK, 4PAM, 8ASK, 16PAM, CPFSK, DQPSK, GFSK, GMSK, OOK, OQPSK) and the 24-class RadioML 2018.01a benchmark
- **Channel Models:** AWGN for the synthetic data; RadioML adds fading and hardware impairments; CageDroneRF adds real over-the-air channels
- **Datasets:** a synthetic 10-class benchmark for controlled experiments, RadioML 2018.01a for cross-dataset tests, and the CageDroneRF real-RF benchmark

1.2.2 Scope Limitations

While this work establishes foundations for wireless AI, certain aspects remain future work:

- **End-to-End Communication:** Focus is on physical layer modulation classification, not complete communication system design
- **Hardware Implementation:** Validation uses simulated data, a standard benchmark, and recorded over-the-air captures (CageDroneRF); no end-to-end hardware-in-the-loop deployment is evaluated
- **Adversarial Robustness:** While noise robustness is central, adversarial attacks and countermeasures are not extensively explored
- **Multi-User/MIMO Scenarios:** Primarily single-user, single-antenna scenarios; multi-antenna and interference management are secondary
- **Foundation Model Scale:** Due to computational constraints, models are smaller than vision/language foundation models; principles established here enable future scaling

1.3 Contributions

This dissertation makes four primary contributions: self-supervised, vision-based representation (DenoMAE, DenoMAE2.0, and the CageDroneRF benchmark), LLM-based reasoning (Plug-and-Play and DiSC-AMC), and meta-learning adaptation (Meta-Task). Authorship is stated plainly in Section 6.6: Plug-and-Play, DiSC-AMC, and Meta-Task are first-author work, CageDroneRF is shared first-author work, and DenoMAE and DenoMAE2.0 are co-authored works led by A. Faysal that this dissertation uses as the representation.

1.3.1 Toward Wireless Foundation Models: A Modular, Cost-Effective Framework

Central Thesis Contribution: The dissertation asks how far wireless foundation-model capabilities can be reached without training a large RF model from scratch, and proposes a modular framework that borrows world knowledge from vision self-supervision, pretrained language models, and learning-to-learn. It answers with three bounded results:

- **RQ1, Represent (yes, within bounds):** Image-based self-supervision gives label-efficient RF features: 82.4% on matched 10-class data, a 22.1-point gain from pre-training at -10 dB, and 0.986 mAP@0.5 for drone detection on real RF captures. It collapses at extreme noise (-30 dB) and transfers weakly to an unseen dataset.

- **RQ2, Reason (yes, with compact symbols and retrieved examples):** Plug-and-Play first established training-free feasibility; DiSC-AMC [12] then resolved its token-efficiency bottleneck, letting a fine-tuned 7B model reach 83.0% in distribution and 82.5% out of distribution, and a training-free 7B model reach 71.9% at 0.5K tokens against 69.92% for the proprietary o3-mini model at 2.9K tokens. Its accuracy is only as good as the embedding it retrieves over.

- **RQ3, Adapt (partly):** Meta-Task's learnable regularizer improves few-shot vision benchmarks (for example 65.4% to 71.3% for Prototypical Networks on Tiered-ImageNet) but gives no net gain over plain fine-tuning on cross-dataset wireless AMC (22.2% vs. 23.0%).

- **Composition:** DiSC-AMC composes an RQ1 encoder with an RQ2 reasoner, lifting RadioML accuracy up to eight times over the encoder alone, and the RadioML adaptation test runs all three modules together. The RQ3 result points back to RQ1: on a new dataset, representation quality limits accuracy more than regularization does.

- **Cost-effectiveness principle:** The framework concentrates new investment where a task needs it and reuses an available base elsewhere. This is supported by token- and parameter-efficiency results, not by a complete latency, energy, or monetary comparison (Section 6.5).

Because the modules have explicit interfaces, the framework also supports diagnosis: a failure can be traced to the representation, the reasoning, or the adaptation stage. That is how the dissertation's central finding emerges, namely that the generality of the underlying representation bounds what reasoning and adaptation can add (Section 6.3.1).

1.4 Structure

This dissertation is organized to guide readers from foundational concepts through technical contributions toward the unifying vision of wireless foundation models. The thesis follows a standard academic structure with six chapters, each building progressively on previous content.

- **Chapters 2 and 3: Self-Supervised Representation Learning (DenoMAE, DenoMAE2.0 & CDRF):** Addresses RQ1. Presents DenoMAE and DenoMAE2.0, co-authored self-supervised representations (led by A. Faysal) that treat signals as images and, in DenoMAE, noise as a modality, with a matched comparison on the same data and an account of where the representation breaks: extreme noise and a new dataset. Presents CageDroneRF, a real over-the-air RF benchmark and toolkit (shared first-author work) on which vision models are tested for detection, hierarchical classification, and open-set recognition.

- **Chapter 4: LLM-Based Reasoning for Open-Set AMC (Plug-and-Play and DiSC-AMC):** Presents Contribution 2, addressing RQ2. Introduces Plug-and-Play, which

first demonstrates training-free wireless classification through LLM reasoning over statistical feature-to-language prompts, and diagnoses its token-efficiency bottleneck. Then introduces DiSC-AMC [12], which resolves that bottleneck through feature discretization, self-supervised context retrieval, and candidate-space pruning, achieving competitive or superior accuracy relative to dedicated supervised encoders at a fraction of the token and parameter budget, both fine-tuned and training-free.

- **Chapter 5: Meta-Learning for Adaptation (Meta-Task):** Presents Contribution 3, addressing RQ3. Introduces Meta-Task, which reformulates regularization as a learnable task in few-shot learning through a Task-Decoder, shows gains on few-shot vision benchmarks, and then tests it directly on wireless AMC through the DiSC-AMC pipeline, where it gives no net gain over plain fine-tuning, a bounded answer that points back to representation quality.

- **Chapter 6: Conclusions and Future Work:** Synthesizes key findings and answers the three research questions, including a bounded, partly-affirmative answer to RQ3 grounded in a null result from testing Meta-Task directly on wireless AMC. States explicit authorship and role for each collaborative work used as evidence (Section 6.6), distinguishing first-author contributions (Plug-and-Play, DiSC-AMC, Meta-Task, shared-first-author CDRF) from co-authored representations used as evidence (DenoMAE, DenoMAE2.0). Discusses remaining limitations and outlines concrete future research directions, including a first (negative) fusion result, vision-language reasoning, and scoped open gaps identified during evaluation.

Chapter 2

DenoMAE: A Multimodal Autoencoder for Denoising Modulation Signals

2.1 Introduction

In recent years, the deep learning paradigm has shifted to address the high costs and scarcity of labeled data and the intensive data demands of supervised models. Self-supervised learning has emerged as an efficient solution, allowing models to leverage abundant raw data in pretraining. This approach drastically reduces the need for labeled data during fine-tuning while maintaining strong downstream performance [13]. Self-supervised methods have shown remarkable success across fields like natural language processing [14] and computer vision [15, 5], paving the way for new applications in fields with sparsely labeled data, including communications.

In communication systems, particularly wireless networks, maintaining signal integrity is crucial, yet it's often compromised by limited data availability and varying noise levels during transmission. Estimating modulation signals becomes especially challenging

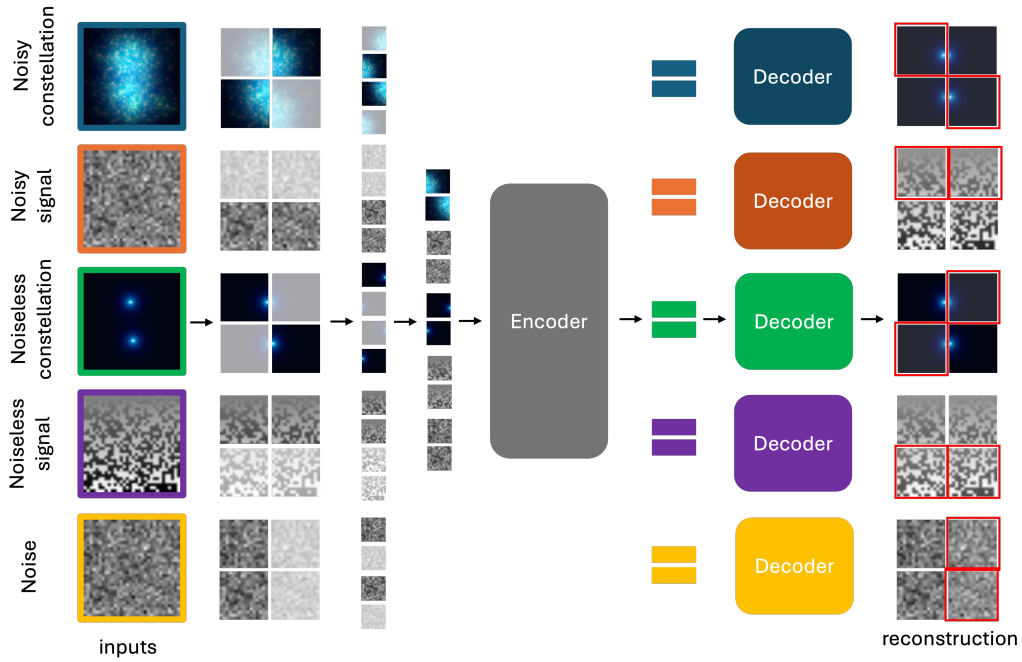


Figure 2.1: DenoMAE Pretraining Strategy: We apply a random 75% masking (not to scale in the illustration) across all input modalities. The remaining 25% of visible patches are processed by a shared encoder, while each modality utilizes a dedicated decoder to reconstruct its masked patches. Only the encoder is reused for fine-tuning the downstream tasks.

as noise levels fluctuate with changing SNRs [16]. This limitation becomes particularly problematic when models encounter previously unseen modulation types [17]. While existing models can often classify known modulation schemes, they struggle to generalize to new or noisy conditions, resulting in misclassification and necessitating extensive retraining [18, 19]. This problem highlights an urgent need for robust, adaptive models that can generalize across diverse conditions with minimal reliance on labeled data.

One approach to improving generalizability and robustness is through pretraining methods that not only capture robust representations but also denoise signals effectively, enhancing classification accuracy in downstream tasks and providing scalable solutions in dynamic communication environments. Masking-based methods have traditionally been

effective for denoising and improving signal clarity, but they are often limited to single-modality approaches. Though effective, these methods are inherently constrained by the information within their specific modality, which restricts their potential to leverage cross-modal interactions for more robust learning [20, 21]. Recent advances in multimodal learning address this limitation by integrating data from multiple modalities, enhancing model robustness and enabling more flexible zero-shot learning capabilities. In works like [22, 23], attention mechanisms and masked modeling pretraining across modalities have successfully reconstructed masked signals, proving beneficial in improving classification under complex conditions [24, 25].

Building on recent advancements, we adapt these techniques and present DenoMAE, a multimodal autoencoder tailored for the unique challenges in communication systems. DenoMAE's innovative approach involves treating noise as an additional modality, allowing the model to generate all other modalities from a single input modality. This unique framework helps the network gain a comprehensive understanding of noise across modalities, significantly enhancing its denoising capabilities. A random masking strategy, which hides 75% of the input modalities, is employed to challenge the model in reconstructing the missing patches, reducing computational complexity, and introducing beneficial uncertainty conducive to effective learning.

Within the broader framework of this dissertation, DenoMAE serves as the foundational representation learning component. While Chapter 4 addresses training-free classification through LLMs and Chapter 5 enables rapid adaptation, DenoMAE provides the robust feature encoder that learns universal signal representations from minimal labeled data. This modular approach demonstrates that wireless foundation models need not follow the massive-scale paradigm of vision and language models, but can achieve strong performance through efficient self-supervised pretraining.

Attribution. DenoMAE and its successor DenoMAE2.0 (Section 2.6) are co-authored

works led by Atik Faysal [26, 1]: the core idea and the main contribution are his, and the author's part was experiment implementation and paper writing. This chapter therefore presents them as the self-supervised representation that the rest of the dissertation builds on and is compared against, not as first-author claims. The dissertation's own first-author contributions begin with Chapter 4, and CageDroneRF (Chapter 3) is shared first-author work.

The DenoMAE paper makes three contributions:

- We propose a novel multimodal paradigm that incorporates noise as an additional modality, enabling more effective pretraining on modulation signals and enhancing model resilience to noise.
- We demonstrate that our pre-trained model achieves significant improvements in classification accuracy under low SNR conditions, with accuracy gains of up to 22.1% over non-pre-trained models.
- It reaches competitive accuracy while using one-tenth of the unlabeled data for pre-training and one-third of the labeled data for fine-tuning relative to the cited baselines (a descriptive comparison across differing protocols, Table 2.2; the matched comparison is in Section 2.6.1), which underscores its data efficiency.

Framework Position and Research Questions: Within the dissertation's modular framework for wireless foundation models, DenoMAE serves as the representation learning component, addressing **RQ1**: *Can vision-based, self-supervised learning effectively transfer world knowledge to the wireless domain?* By treating wireless signals as images (constellation diagrams) and applying masked autoencoding [27], DenoMAE and DenoMAE2.0 carry vision-based self-supervision into the wireless domain, and the noise-as-modality design lets the encoder learn denoised representations from only 1,000 labeled

samples. The answer developed in this chapter is *yes, within bounds*: pretraining clearly helps on matched data and at moderate noise, but the representation collapses at extreme noise and transfers weakly to a new dataset (Section 2.7). Together with CDRF's evidence on real RF captures (Chapter 3), this chapter supplies the representation on which the reasoning (Chapter 4) and adaptation (Chapter 5) chapters build, and whose limits those chapters in turn expose.

2.2 Related Work

DenoMAE builds on masked autoencoders (MAE) [27] and recent progress in multimodal pretraining [22]. Prior AMC work has largely focused on single-modality pipelines like CNN classifiers [28], transformer backbones [10], or handcrafted features but often trained and evaluated at fixed SNRs. In contrast, we treat noise as a first-class modality. This design encourages the encoder to disentangle signal structure from channel effects during pretraining, yielding features that transfer more reliably across SNRs and channel conditions. Empirically, it reduces labeled-data requirements while improving low-SNR accuracy, addressing a long-standing robustness gap in AMC.

2.3 Methodology

In this work, we propose DenoMAE, an innovative pretraining method for modulation classification that leverages recent advancements in multisensory pretraining. By incorporating noise as an additional modality during pretraining, DenoMAE significantly reduces data requirements while enhancing performance in downstream classification tasks.

2.3.1 DenoMAE Framework

DenoMAE consists of four primary components: input embedding, encoder, shared latent space, and decoder. Figure 2.1 illustrates the overall architecture.

2.3.1.1 Input Embedding

For each modality $m \in \{1, \ldots, n\}$, we represent the input as a C-channel image $X_m \in \mathbb{R}^{C \times H \times W}$. We divide each image into non-overlapping patches of size $p \times p$, resulting in $N = (H/p)^2$ patches per modality. Each patch is flattened and linearly projected to a D-dimensional embedding space:

$$E_m = [e_1^m, ..., e_N^m] + E_{pos}, \quad e_i^m \in \mathbb{R}^D$$

where E_{pos} represents learnable positional embeddings.

2.3.1.2 Encoder

The encoder f_{enc} is a series of L transformer blocks. We randomly mask a high proportion (e.g., 75%) of the input patches for each modality. The encoder processes the visible patches:

$$H_m = f_{enc}([e_{i_1}^m, ..., e_{i_k}^m]), \quad i_j \in \text{visible patches}$$

2.3.1.3 Shared Latent Space

To enable cross-modal learning, we project the encoded representations from all modalities into a shared latent space. We use a linear projection followed by layer normalization:

$$Z_m = \text{LN}(W_m H_m + b_m)$$

The shared latent representation Z is the concatenation of all modality-specific projections:

$$Z = [Z_1; ...; Z_5]$$

2.3.1.4 Decoder

The decoder f_{dec} reconstructs the masked patches for all modalities simultaneously. It takes the shared latent representation Z as input and processes it through a series of transformer blocks. The output is then split into modality-specific representations and projected back to the original patch space:

$$\hat{X}_m = f_{\text{proj}_m}(f_{dec}(Z))$$

where f_{proj_m} is a modality-specific projection layer.

2.3.1.5 Loss Function

We employ a multi-modal reconstruction loss that combines individual losses for each modality:

$$\mathcal{L} = \sum_{m=1}^{5} w_m \mathcal{L}_m$$

where w_m is the weight for modality m, and $\mathcal{L}_m$ is the reconstruction loss for that modality. We use the mean squared error (MSE) as the reconstruction loss:

$$\mathcal{L}_m = \frac{1}{|\mathcal{M}_m|} \sum_{i \in \mathcal{M}_m} \|\hat{X}_{m,i} - X_{m,i}\|_2^2$$

2.4 Experimental Setup

We begin this section by detailing the parameters used for sample generation, followed by the classifier's configuration. Finally, we present the parameters applied to our downstream

tasks.

2.4.1 Constellation Diagram Generation

Constellation diagrams offer a more informative approach to modulation classification compared to raw time-series signals, capturing richer detail essential for accurate interpretation [29, 30]. The process of generating constellation diagrams for modulation signal classification involves multiple stages. We begin by mapping modulated signals onto a 7×7 complex plane, which provides sufficient space to capture signal samples while maintaining computational efficiency for SNR ranges of -10 dB to 10 dB. The basic constellation diagram is then enhanced through a multi-step process: first, as a gray image that handles varying pixel densities where multiple samples may occupy single pixels; second, as an enhanced grayscale image that employs an exponential decay model ($B_{i,j}$) to account for both the precise position of samples within pixels and their influence on neighboring pixels. This model considers the sample point's power (P), the distance between sample points and pixel centroids ($d_{i,j}$), and an exponential decay rate (α) [31]. To make the representation compatible with DenoMAE, which expects RGB input, we generate a three-channel image by creating three distinct enhanced grayscale images from the same data samples, each utilizing different exponential decay rates.

2.4.2 Parameters in Sample Generation

The dataset comprises signals from ten modulation formats, each originally of length $L_0 = 1024$. To conform to the transformer's input dimensions, signals undergo a two-step preprocessing: (1) reshaping to $\mathbf{S}_1 \in \mathbb{R}^{32\times32}$, and (2) interpolation to $\mathbf{S}_2 \in \mathbb{R}^{224\times224}$. For formats with $L_0 > 1024$ (e.g., $L_{\text{GMSK}} = 8196$), signals are initially downsampled to L_0. To match the three-channel structure of constellation images $\mathbf{I} \in \mathbb{R}^{3\times224\times224}$, $\mathbf{S}_2$ is replicated

across three channels, yielding $\mathbf{S}_3 \in \mathbb{R}^{3\times224\times224}$. All signals are sampled at $f_s = 200$ kHz, ensuring consistency across modulation formats.

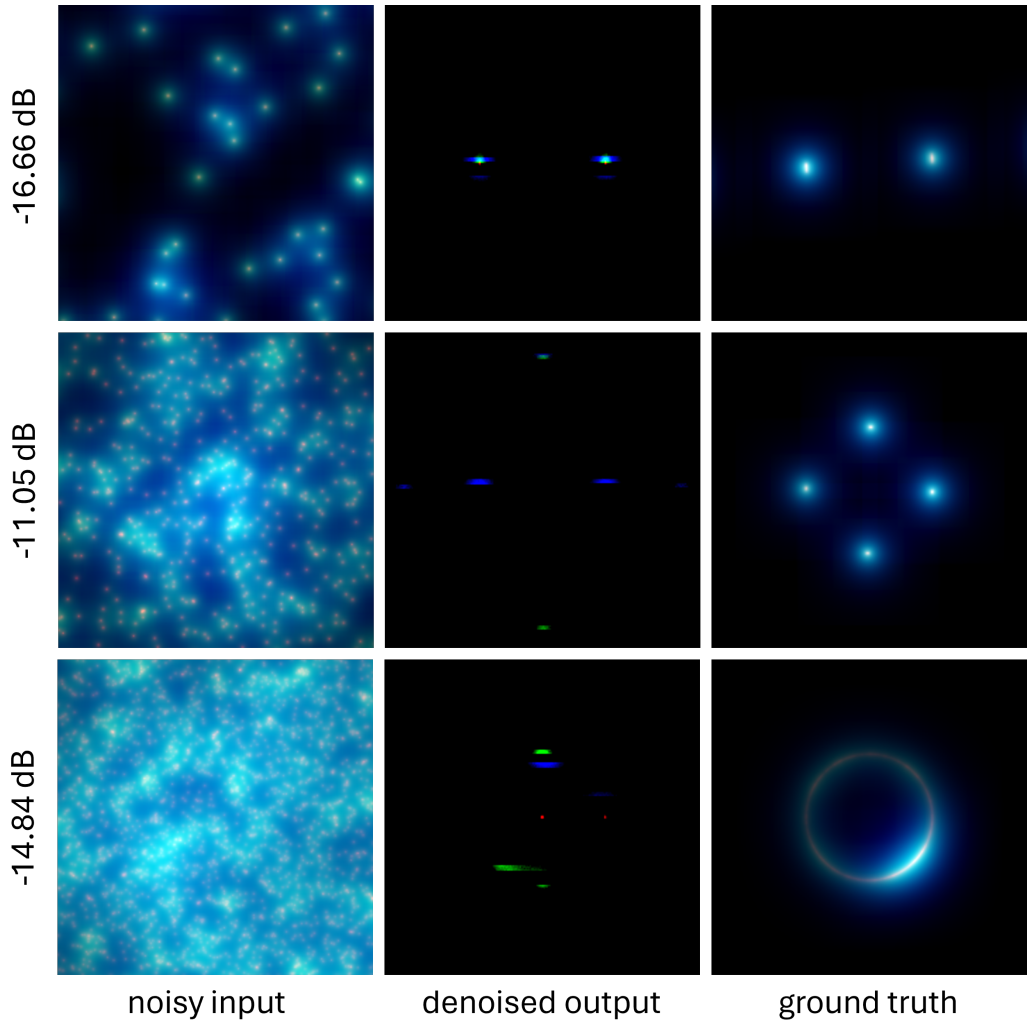


Figure 2.2: Extrapolation-ability of DenoMAE on out-of-bound much lower SNRs.

2.4.3 Parameters for DenoMAE

The MultiMAE architecture processes n =5 modalities of input images $\mathbf{X} \in \mathbb{R}^{3\times224\times224}$, segmented into 16×16 patches. The model employs a transformer-based encoder-decoder structure with $d_{\text{model}} = 768$ embedding dimensions. The encoder consists of $L_e = 12$ layers, while the decoder uses $L_d = 4$ layers, both utilizing $h = 12$ attention heads. During training, a random masking strategy with $p_{\text{mask}} = 0.75$ is applied. The model is optimized using AdamW with learning rate $\eta = 10^{-4}$ over $E = 100$ epochs, using batches of size $B = 64$. This flexible architecture allows for easy adaptation to various multi-modal denoising tasks.

2.4.4 Finetuning Parameters

The fine-tuning task adapts the pre-trained DenoMAE model for classification across $C = 10$ classes. The encoder $E(\cdot)$ from DenoMAE, with $d_{\text{model}} = 768$, is used as a backbone, followed by a classifier head $H(\cdot)$ consisting of a two-layer MLP with hidden dimension 512 and dropout $p = 0.5$. For an input $\mathbf{X} \in \mathbb{R}^{3\times224\times224}$, the model computes $\hat{y} = H(G(E(\mathbf{X})))$, where $G(\cdot)$ is global average pooling. The model is trained for $E = 150$ epochs using Adam optimizer with learning rate $\eta = 10^{-4}$ and batch size $B = 32$. Training data undergoes normalization but no augmentation.

2.5 Results and Discussion

This work centers on denoising and reconstructing constellation diagrams, a modality used exclusively during downstream classification tasks. In the following sections, we provide detailed insights into those findings.

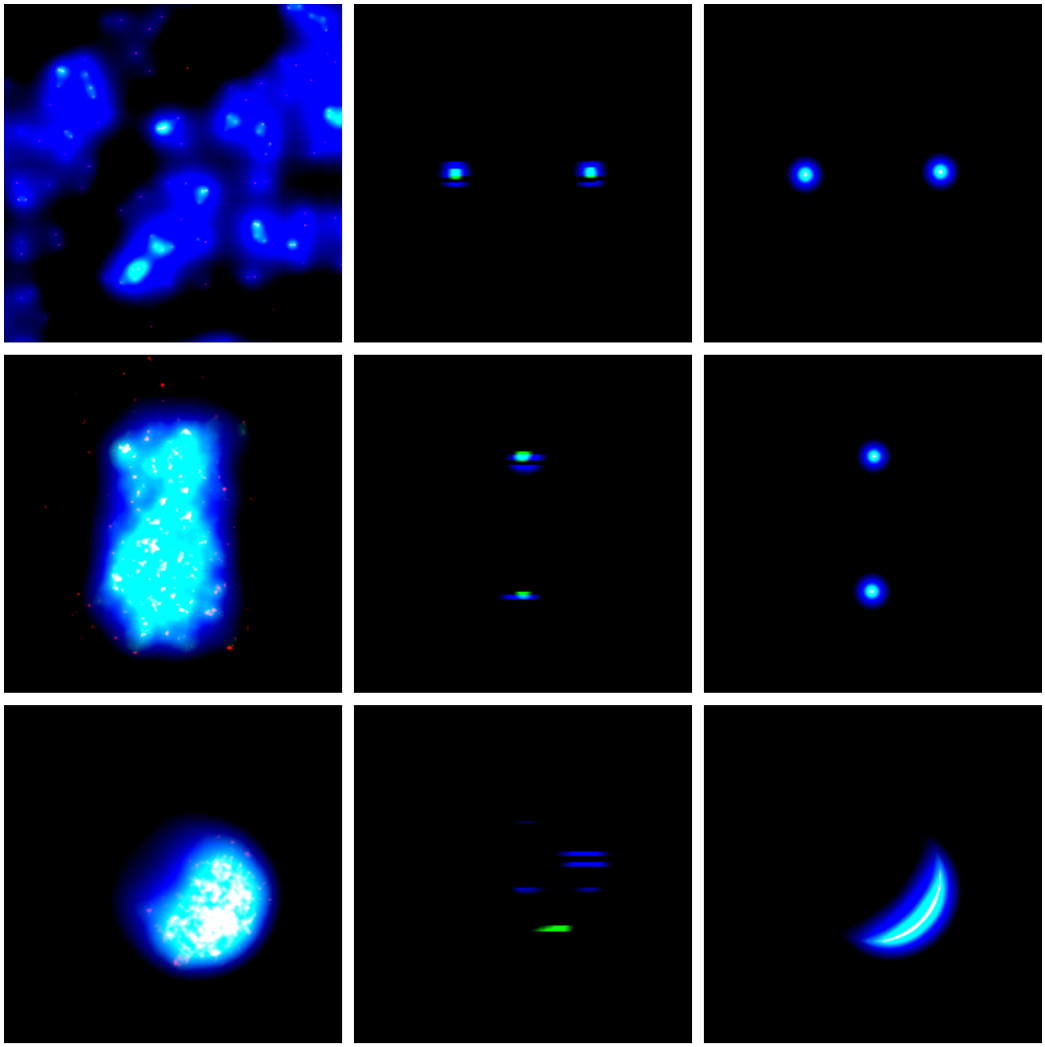


Figure 2.3: Denoised outputs of DenoMAE on unlabeled constellation diagrams at different SNRs during pretraining.

2.5.1 Pretraining Performance

The pretraining effectiveness of DenoMAE was evaluated using 10,000 unlabeled signal samples across five modalities. Visual inspection of the denoised constellation diagram

Table 2.1: Performance Comparison Across Modalities in Downstream Task

Constellation		Signal		Noise	Test Accuracy (%)
Clean	Noisy	Clean	Noisy		
✓					81.30
✓	✓				81.90
✓	✓	✓			83.20
✓	✓	✓	✓		83.30
✓	✓	✓	✓	✓	83.50

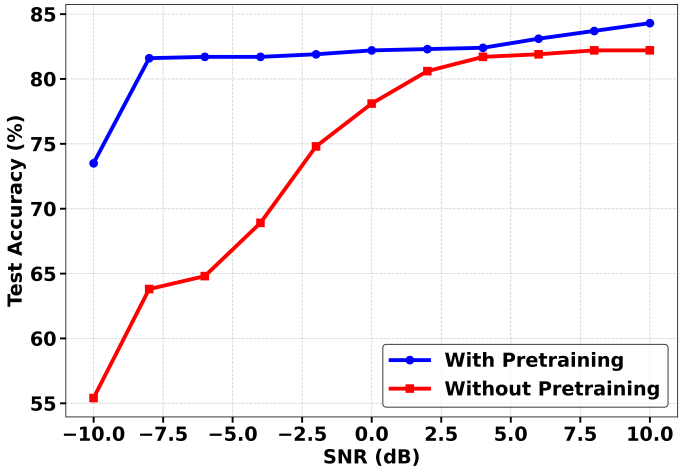


Figure 2.4: DenoMAE fine-tuned classification accuracy for constellation diagrams at different SNRs.

outputs (Figure 2.3) demonstrates strong reconstruction capabilities, particularly for samples in the first two rows where the model successfully removes noise while preserving

Table 2.2: Comparison of DenoMAE with other modulation classification methods

Methods	**Number of samples**		**SNR dB**	**Number of classes**	**Test accuracy (%)**
	Pretraining	**Fine-tuning**			
AlexNet [31]	800,000 (train)	8,000 (test)	2	8	82.8
NMformer	106,800	3,000	0.5 - 4.5	10	71.6
CNN-AMC [32]	79,200	228,060	-6	4	50.40
DL-GRF [33]	2,000 (train)	200 (test)	0	4	59
DenoMAE (**Ours**)	10,000	1,000	0.5	10	83.5
DenoMAE (**Ours**)	10,000	1,000	-10	10	77.50

signal characteristics. While reconstruction quality slightly diminishes for samples in the last row, the overall shape and pattern retention remain sufficient for downstream tasks.

2.5.2 Impact of SNR

In fine-tuning, we used 1000 labeled constellation images with 10 classes. The impact of SNR on classification performance reveals DenoMAE's robust noise resilience. As shown in Figure 2.4, the pre-trained model achieves 77.50% accuracy even at -10 dB SNR, with performance steadily improving to 84.30% at 10 dB. The pretraining advantage is most pronounced in challenging low-SNR conditions, demonstrating a 22.1% improvement over the non-pre-trained baseline at -10 dB. This performance gap gradually narrows at higher SNRs, suggesting that pretraining is particularly valuable for learning noise-robust features.

2.5.3 Extrapolation on out-of-bound SNRs

DenoMAE demonstrates impressive extrapolation capabilities when evaluated on SNR conditions significantly outside its training range. Although trained only on SNR levels above -10 dB, the model continues to deliver meaningful reconstruction performance at challenging SNR levels between -11 dB and -20 dB. As shown in Figure 2.2, DenoMAE effectively denoises highly noisy input constellation diagrams, making the first two samples visually recognizable. The final sample, while impacted by extreme noise, still approximates the outer shape of the ground truth.

2.5.4 Ablation Study on Modality Impact

Our ablation study on modality impact, detailed in Table 2.1, reveals the effectiveness of the multimodal approach. The full model with five modalities (noisy image, noisy signal, noiseless image, noiseless signal, and noise) achieves 83.50% accuracy. Performance

degrades gracefully as modalities are removed, with even the single-modality version maintaining 81.30% accuracy. This suggests that while additional modalities contribute to performance gains, the model maintains robust performance even with reduced modality input, indicating efficient cross-modal learning capabilities.

2.5.5 Comparative Analysis

Table 2.2 places DenoMAE next to published modulation classification results as literature context. It is not a controlled comparison: the entries differ in SNR, number of classes, and training and evaluation protocols, so its accuracy column cannot rank the methods. What it does show is data scale: DenoMAE reaches 83.5% at 0.5 dB and 77.50% at -10 dB on 10 classes using 10,000 unlabeled pretraining samples and 1,000 labeled fine-tuning samples, against hundreds of thousands of samples for some of the listed methods (for example, 80 times fewer than the 800,000 training samples of AlexNet). The matched comparison on identical data, SNR range, and classes is given in Section 2.6.1.

2.5.6 Cross-Framework Validation

The representation from this chapter is reused elsewhere in the dissertation, and its behavior there is part of the evidence for RQ1. In Chapter 3, the vision-based approach is tested on real RF captures, where pretrained vision architectures (YOLO, ResNet) operate on spectrograms. In Chapter 4, DenoMAE2.0 is evaluated as an alternative retrieval backbone for DiSC-AMC; it is strong in distribution but is outperformed by a generic DINOv2 encoder on RadioML (Section 4.6.9). Chapter 5 tests whether adding Meta-Task to a fine-tuned DenoMAE2.0 helps on RadioML, and it does not (Section 5.5). Section 2.7 collects these limits.

2.6 DenoMAE2.0: Adding Local Patch Classification

DenoMAE's pretext task is purely generative: reconstruct the masked patches. DenoMAE2.0 [1] adds a discriminative objective alongside the reconstruction. As Figure 2.5 shows, the encoder's output for each visible (unmasked) patch is also passed to a small MLP head that classifies the patch by its position in the image grid, and the two losses are combined,

$$\mathcal{L} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}}. \tag{2.1}$$

The motivation is that reconstruction alone favors coarse, global structure, while a local classification target pushes the encoder to keep fine-grained, position-specific patterns, which should help separate confusable classes at low SNR. In this dissertation DenoMAE2.0 is the signal encoder in its strongest form and is evaluated as an alternative retrieval backbone inside DiSC-AMC (Chapter 4).

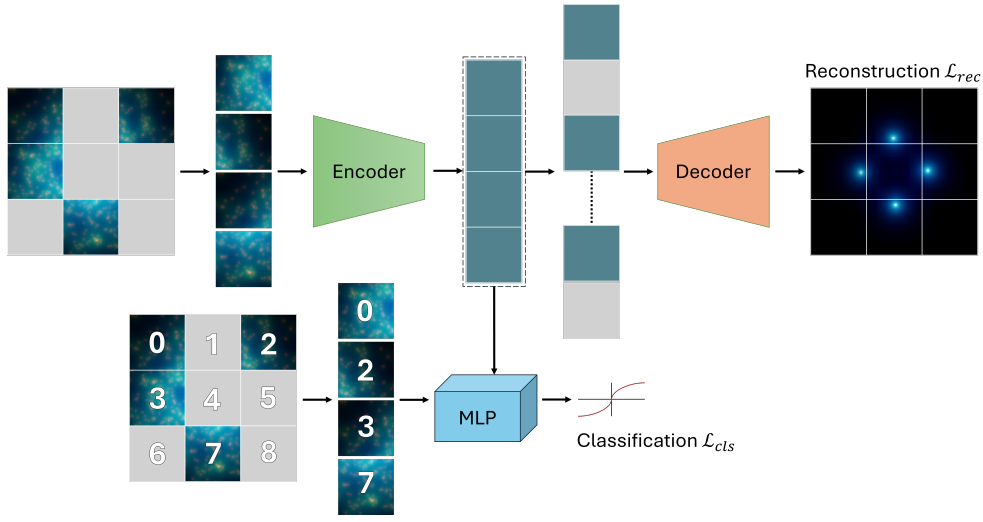


Figure 2.5: DenoMAE2.0 architecture overview. Masked autoencoding is extended with a classification head that treats each visible patch as a distinct class based on its spatial position; the reconstruction loss and the classification loss are combined during pretraining.

2.6.1 Matched Comparison on the Same Data

Table 2.3 compares DenoMAE and DenoMAE2.0 with other pretraining methods on the same dataset, the same 10 classes, and the same SNR range, which is the comparison that Table 2.2 cannot provide. The DenoMAE entry equals its constellation-only configuration in Table 2.1.

Table 2.3: Fine-tuned test accuracy on the same 10-class dataset and SNR range. Entries other than DenoMAE are as reported in the DenoMAE2.0 paper [1].

Method	Accuracy (%)
ViT (supervised, no pretraining)	79.9
MAE	80.1
BEiT	80.4
MoCov3	81.0
DeiT	81.2
DenoMAE	81.3
DenoMAE2.0	**82.4**

DenoMAE2.0 leads at 82.4%, 2.5 points above the supervised ViT and about one point above the other self-supervised methods. The bound on this result is that pretraining clearly helps, while the gaps among the self-supervised methods are small and reported without confidence intervals, so they should be read descriptively. The larger effect appears at low SNR, where DenoMAE is 22.1 points above the same model without pretraining at -10 dB (Section 2.7).

2.7 Where the Representation Holds and Where It Breaks

Moderate noise. The representation holds. DenoMAE reaches 77.5% at -10 dB with pretraining, 22.1 points above the no-pretraining baseline, and the advantage shrinks toward higher SNR (Figure 2.4).

Which modality matters. In the modality ablation (Table 2.1), adding the noise modality alone moves accuracy from 83.3% to 83.5%, a gain of 0.2 points, whereas adding the clean signal modality moves it from 81.9% to 83.2%. The signal modalities contribute most, and the noise modality's measured contribution in this ablation is small.

Extreme noise. At -30 dB, constellation diagrams become featureless and accuracy falls to roughly 5–10%. Every image-based pipeline in this dissertation collapses here, including the retrieval used by DiSC-AMC (Chapter 4), so this is a limit of the image-based representation, not of one method.

A new dataset. Features learned on synthetic constellations transfer weakly to RadioML. On RadioML at $+20$ dB, DenoMAE2.0 reaches 33.8% against 21.9% for DenoMAE, and accuracy at -10 dB is only 7.1% against a chance level of 4.2%. Inside DiSC-AMC the same encoder is also fragile to a modest SNR shift: its accuracy drops from 82.4% in distribution to 63.25% at -11 to -15 dB (Table 4.4), and on RadioML it is outperformed by a generic DINOv2 encoder (Section 4.6.9).

Difficult finding. The domain-specific representation that wins on matched in-distribution data is the weaker one once the data changes. RQ2 and RQ3 return to this: reasoning amplifies a representation only where its neighbors carry class information, and a learned regularizer does not repair a representation that did not transfer (Section 6.3.1).

2.8 Conclusion

This chapter presented DenoMAE and DenoMAE2.0, co-authored self-supervised representations that treat wireless signals as images and, in DenoMAE, noise as an explicit modality. The answer to RQ1 is yes, within bounds. Pretraining gives label-efficient features: 82.4% on matched 10-class data for DenoMAE2.0, and a 22.1-point gain from

pretraining at -10 dB. The bounds are equally part of the result: the gaps among self-supervised methods are small, the noise modality adds little in the ablation, accuracy collapses at -30 dB, and the features transfer weakly to RadioML.

Connection to the Dissertation Framework: DenoMAE2.0 supplies the encoder that the reasoning module retrieves over and the adaptation module regularizes. Its in-distribution strength and its out-of-distribution weakness together set the ceiling for both: DiSC-AMC's reasoning cannot recover class signal the encoder does not carry, and Meta-Task's regularizer does not repair it. This is why the dissertation's cross-dataset evidence returns to representation generality as the open problem (Section 6.3.1). The real-RF evidence for vision-based learning comes from CageDroneRF (Chapter 3).

Chapter 3

CDRF: A Large-Scale RF Benchmark and Toolkit for Drone Detection, Classification, and Open-Set Recognition

3.1 Introduction

The development of wireless foundation models requires validating a fundamental hypothesis: can representations learned by vision models transfer effectively to the wireless domain? Chapter 2 proposes that multimodal masked autoencoders can learn universal wireless signal representations by treating signals as images (spectrograms, constellation diagrams). Before pursuing this costly self-supervised learning approach, we must first establish that vision architectures (YOLO for detection, ResNet for classification) can indeed extract meaningful features from RF signals represented as spectrograms. This chapter addresses Research Question 1 (RQ1) through a practical validation: the CageDroneRF (CDRF) benchmark demonstrates that pre-trained computer vision models successfully

transfer to wireless perception tasks, providing empirical evidence for the cross-modal transfer principle underlying our modular framework.

Beyond this theoretical validation, CDRF serves a critical practical role: providing a realistic, stress-tested benchmark for evaluating foundation model components. While many wireless datasets capture signals under idealized laboratory conditions, operational deployment demands robustness to low signal-to-noise ratios (SNR), spectrum interference, frequency offsets, and domain shifts (indoor↔outdoor). CDRF's raw-signal augmentation pipeline and diverse collection methodology create exactly these challenging conditions, enabling us to assess whether foundation model approaches generalize beyond clean benchmarks to real-world scenarios.

The rapid proliferation of Unmanned Aerial Vehicles (UAVs) provides a compelling application domain for this validation. UAVs enable transformative applications in logistics, agriculture, and emergency response [34], while introducing risks to safety, privacy, and critical infrastructure [35]. Traditional sensing modalities (radar, vision, acoustics) exhibit notable limitations: environmental sensitivity, limited small-target range, line-of-sight dependence, and high deployment costs. Radio-Frequency (RF) sensing offers complementary advantages: continuous command-and-control links enable day/night, non-line-of-sight detection at low cost, with RF captures carrying rich device, protocol, and operator signatures [36].

Despite these advantages, progress in Machine Learning (ML) for RF-based drone perception has lagged behind vision and language domains, precisely the gap motivating this dissertation's framework. A central barrier is the absence of broad, standardized benchmarks that test models under realistic stress conditions. Existing datasets are often narrow in class diversity, limited in volume, collected in clean environments with restricted SNR ranges and weak interference diversity. Models trained on such data overfit to idealized conditions, achieving near-perfect scores on "easy" benchmarks yet degrading severely in

operational deployments with Bluetooth/Wi-Fi interference, spectrum crowding, and low SNR, exactly the conditions that foundation models must handle [37].

We introduce CageDroneRF (CDRF), a benchmark built to close this gap with a dataset-and-toolkit co-design. CDRF comprises real-world captures from controlled RF-cage facilities and open-campus settings, paired with a principled raw-signal augmentation pipeline that programmatically varies SNR, injects interfering emitters, and applies frequency shifts while preserving label consistency. Crucially, augmentations operate on complex baseband I/Q, before time-frequency conversion, so synthesized conditions faithfully reflect RF phenomena rather than image-level artifacts. For detection, frequency shifts are accompanied by exact recalculation of You-Only-Look-Once (YOLO [38])-format annotations with correct wrap-around behavior on the spectrogram's frequency axis. The data processing stack converts raw captures to spectrograms via Short-Time Fourier Transform (STFT) and exposes all parameters (e.g., sampling rate, FFT size, segment length) for reproducible ablations and reprocessing.

Beyond the dataset, CDRF provides interoperable utilities for dataset creation, cleaning, and evaluation. The data module loads .dat files via memory mapping, slices long recordings into time windows, generates spectrograms, appends per-sample metadata (including time bounds), and optionally adds controlled AWGN, Rician, or Rayleigh fading noise for precise SNR targets. The YOLO toolkit includes raw-IQ augmentation and automatic bounding-box recomputation; a dataset cleaner aligns third-party releases (e.g., Roboflow-style [39]) into a consistent hierarchy; and a lightweight patch exposes per-detection class probabilities from YOLO for richer calibration and analysis. For classification, CDRF ships PyTorch [40] baselines (binary and multi-class) built on ResNet-18 [41] with ready-to-run dataloaders for both spectrogram images and array/pickle formats, standardizing training, metrics, and confusion matrices. All tooling is dataset-agnostic, enabling applications to use existing public datasets to facilitate cross-benchmark comparability.

Collectively, CDRF aims to shift the field from narrow, clean benchmarks toward realistic, stress-tested evaluation under varied SNRs, interference, and frequency offsets, conditions that operational systems routinely encounter. By releasing both data and the underlying signal-first augmentation/evaluation stack, we seek to catalyze reproducible research on RF-based detection, identification, and related perception tasks. We summarize our key contributions as follows:

- **CDRF benchmark dataset:** Real-world RF captures from cage environments with standardized spectrogram generation and rich per-sample metadata.
- **Raw-signal augmentation pipeline:** Controlled SNR injection, interfering-signal mixing, and frequency shifts applied at I/Q level, with exact YOLO label recomputation (including wrap-around).
- **SNR-structured datasets:** Programmatic creation of SNR-stratified splits (including noise-only backgrounds) for stress testing and realistic robustness evaluation.
- **Interoperable tooling:** Open-source utilities for dataset creation, metadata generation, Roboflow-style cleaning, and spectrogram rendering; tools are dataset-agnostic for use on existing releases.
- **Detection analysis enhancements:** Patch exposing per-detection class probabilities from YOLO to support calibration studies and open-set analyses.
- **Baselines and evaluation:** Ready-to-run PyTorch baselines for binary and multi-class classification with standardized metrics and confusion matrices, enabling reproducible benchmarking across tasks.
- **Open set recognition:** Implementation of open-set recognition capabilities to handle previously unseen classes during inference.

Within the broader dissertation framework, CDRF serves as a critical validation domain demonstrating cross-modal knowledge transfer. By converting RF signals to spectrogram images, this work validates that vision models (YOLO, ResNet) can transfer learned representations to wireless perception tasks, a key principle underlying the multimodal approach in DenoMAE (Chapter 2). This empirical validation provides the foundation for the dissertation's modular, cost-effective approach: rather than training wireless foundation models from scratch, we can strategically leverage pre-trained vision architectures (proven here), integrate LLM reasoning (Chapter 4), and enable rapid adaptation via meta-learning (Chapter 5). CDRF provides the realistic, stress-tested benchmark needed to evaluate whether these foundation model components generalize beyond clean laboratory conditions to operational deployment scenarios.

The remainder of this paper is organized as follows. Section 3.2 reviews related work in RF-based drone detection, covering various sensing modalities and existing public datasets. Section 3.3 details the design and collection of the CDRF dataset, and Section 3.4 describes our data preprocessing, augmentation, and annotation pipeline. Section 3.5 presents the experimental setup and benchmark results for several machine learning tasks, including drone detection, single-label, open-set, and hierarchical classification. Finally, Section 3.7 concludes the paper and discusses future research directions.

3.2 Related Work

This section briefly positions CDRF within the existing landscape and highlights the specific gaps it addresses.

Despite growing public datasets, persistent challenges remain: lack of diversity in device classes and environments, restricted raw I/Q access, poorly characterized SNR distributions, scarcity of real-world negative samples, and absence of integrated augmentation

and evaluation frameworks. Existing datasets often achieve high in-benchmark accuracy but exhibit poor real-world generalization due to clean collection conditions [42, 43].

CDRF addresses these gaps by providing: (1) real-world cage and outdoor captures with controlled SNR stratification, (2) raw-signal augmentation pipeline operating at I/Q level with exact label recomputation, (3) dataset-agnostic tooling for spectrogram generation and evaluation, and (4) enhanced detection analysis exposing per-class probabilities for calibration and open-set studies.

3.3 CDRF: Design and Collection

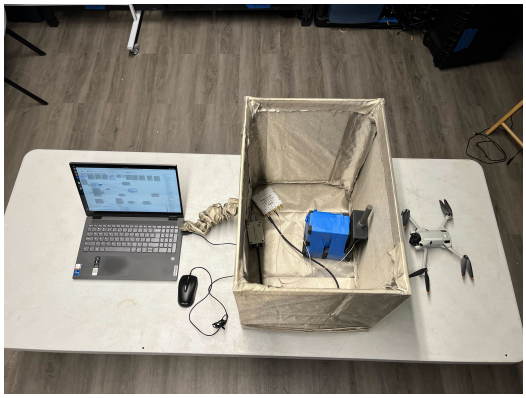
In the CDRF dataset, we aim to cover the RF signals emitted from the most common UAV-related devices in the market, including the downlink video signals and uplink Remote Controller (RC) signals. We start building the dataset by recording the UAV signals in a rather clean RF environment, in order to emphasize the characteristics of the signals of interest, and also relieve the impact of interfering RF sources or environmental variations. The clean data can also help on reducing the efforts on data-preprocessing and any following data augmentation processes.

3.3.1 Data Acquisition Platform

We built the data collection platform using a Software-Defined Radio (SDR) based system. In order to create an RF-isolation platform, we built a customized Faraday cage and placed both the SDR card and receiving antenna inside the cage to mitigate the interfering RF signals. The hardware setups and platform configurations are shown in Fig. 3.1.

We use a Lenovo IdeaPad Flex 5 as the data sink. The machine is equipped with an 8-core Intel Core i7-1165G7 CPU and 16 GB of memory. The operating system running on

(a) Recording configuration.



(b) Recording equipments.

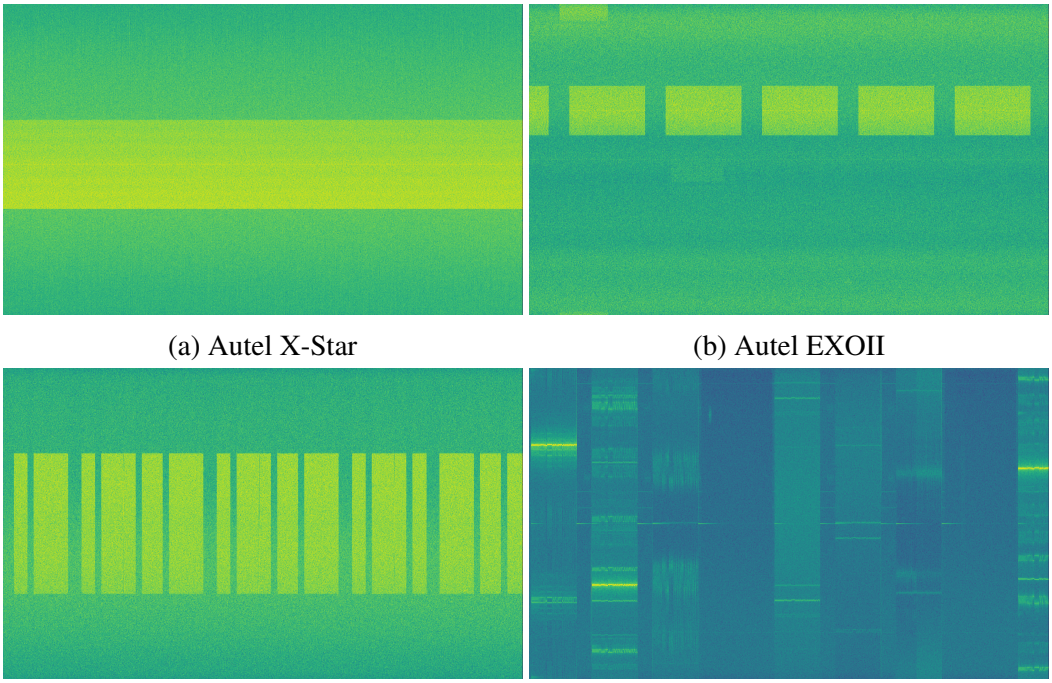
Figure 3.1: Equipment used to capture the data: portable RF shielded enclosure, SDR Card, a laptop, and a drone.

the machine is Ubuntu 24.04.2 LTS. The data collection flow and scripts are built through GNU Radio v3.10 to bridge the Universal Software Radio Peripheral (USRP) B200-mini for RF signal receiving and processing. The SDR device is then connected to a dual-band omnidirectional antenna, which is attached to a customized mount unit to ensure the antenna is static during data collection.

To ensure comprehensive coverage of drone transmissions, we employ sweep scanning across relevant frequency bands during data collection. This approach allows us to detect and record signals from devices operating at variable frequencies.

3.3.2 Data Collection Methodology

To initiate the data collection, we set up the platform as described in Fig. 3.1 (a), then turn on the UAV and the binding controller. While leaving the antenna inside the cage, we place the UAV outside and at a very close distance to the antenna. This placement ensures that the UAV signals delivered to the antenna are strong enough, while other inferencing RF sources in the surrounding areas are at further distances, and thus their signals will be mostly blocked by the Faraday cage. In addition, the RC of the UAV is kept in another



(a) Autel X-Star (b) Autel EXOII

(c) DJI Mavic 2 Pro (d) RadioMaster TX16S

Figure 3.2: Representative spectrograms from the CDRF dataset, illustrating the diversity of signal types and device behaviors captured during controlled data collection.

room next to the one where the data is collected. Our goal is to minimize the RC signals captured by the collection platform in this phase, so that the dataset can support further analysis on UAV signals and RC signals separately. This can also provide more flexibility for further data augmentation.

Next, we adjust the parameters on both the SDR devices and the UAVs to ensure they are properly aligned for optimal data recording quality. This includes settings such as the gain of the SDR device, the center frequency, and the bandwidth of the UAV's video transmission. In this work, we set the gain of the SDR device to 50 dB for indoor use and 76 dB for outdoor use, with a sampling rate of 20 MHz for both environments. For the video signal from UAVs, we select one of the channels in each of the frequency bands supported by the UAV, e.g., 900MHz, 2.4GHz, or 5.8GHz. Generally, we choose the channels that are

not commonly occupied by other non-UAV wireless traffic in order to ensure the quality of the collected data. After the UAV transmission channel is defined, we tune the center frequency of the SDR device to align it to the UAV channel. This ensures that any UAV signals with a bandwidth less than or equal to 20MHz can be fully captured. In addition, this frequency alignment can help simplify the data preprocessing step for any research focusing on frequency domain representation of the signals, because the resulting transformation from the raw data, e.g., spectrogram, will contain a centralized UAV signal.

After that, we run the data recording script from the laptop for a predefined time duration, and the raw data is generated and stored in a {drone_manufacturer}_{drone_model}_{bandwidth}_{drone_center_freq}_{drone_operation_mode}.dat file along with all the predefined labels. In CDRF, we record the UAV manufacturer, model, center frequency of UAV signals, and bandwidth for each recording in the corresponding filename or directory name.

Finally, we implement additional software scripts, including a signal playback module in GNU Radio, to load the raw data file, to review the validity of the recorded raw data, and to ensure the signals recorded match the predefined parameters. We also estimate the SNR and the channel condition to make sure the recorded signals are in a rather clean state. Fig. 3.2 shows some representative spectrograms from the CDRF dataset, illustrating device behaviors captured during controlled data collection.

3.3.3 Collection of Outdoor Data

To complement our controlled indoor recordings, we conducted extensive outdoor data collection sessions. This effort was aimed at enhancing the diversity and realism of the CDRF dataset by capturing UAV signals in complex RF environments, which better reflect real-world operational conditions with various signal sources and environmental factors.

For each outdoor session, we meticulously recorded a comprehensive set of metadata to ensure the data's utility and reproducibility. The raw I/Q data files are organized into a structured directory hierarchy where each directory name systematically encodes the key parameters of the recording session. The naming convention is as follows:

`{device}_{status}_{env}_{sdr_gain}_{splitter}_{duration_recording}_{distance}_{altitude}_{center_freq}_{drone_c_freq}_{bw}_{snr}_{sampling_rate}_{record_dir}.dat`

The fields in the directory name are defined as:

- `device`: The model of the drone or device being recorded.
- `status`: The operational state of the drone (e.g., hovering, flying, on the ground).
- `env`: The environment where the recording took place.
- `sdr_gain`: The receiver gain of the SDR in dB.
- `splitter`: A flag indicating if a signal splitter was used.
- `duration_recording`: The total duration of the recording in seconds.
- `distance`: The horizontal distance from the receiver to the drone in meters.
- `altitude`: The altitude of the drone in meters.
- `center_freq`: The center frequency of the SDR receiver in MHz.
- `drone_c_freq`: The drone's transmission center frequency in MHz.
- `bw`: The signal bandwidth in MHz.
- `snr`: The estimated SNR in dB.

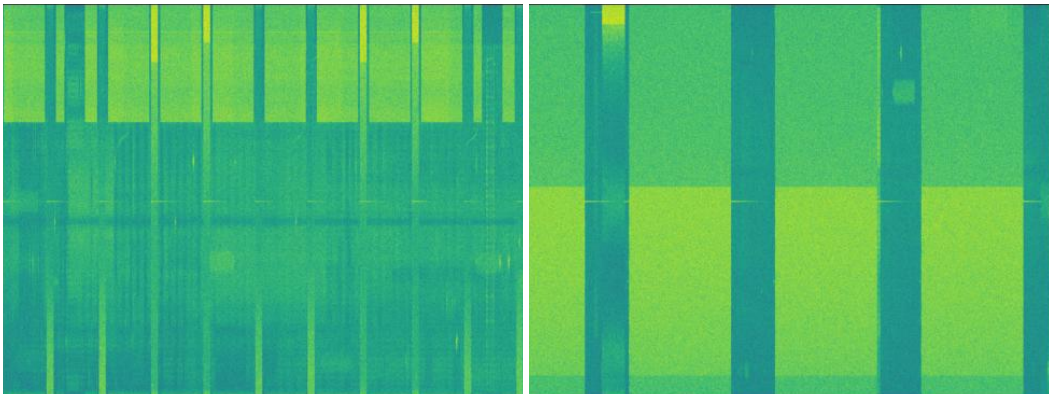
- `sampling_rate`: The sampling rate of the SDR in MHz.
- `record_dir`: The name of the directory where the recording is stored.

Fig. 3.3 presents a selection of representative spectrograms from our outdoor data collection, featuring signals from various drones such as the Autel EXOII, DJI Inspire1, DJI Mavic3, and DJI Phantom3 Advanced. In stark contrast to the clean, isolated signals captured indoors (as seen in Fig. 3.2), these outdoor spectrograms exhibit significantly higher levels of background noise and interference. This is visually evident from the brighter and more cluttered appearance of the spectrogram backgrounds, a direct result of the complex real-world RF environments in which the data was collected. The inclusion of this noisier, more challenging data is crucial for developing and validating drone detection models that are robust enough for practical, real-world deployment.

3.3.4 Dataset Composition and Signal Characteristics

The CDRF dataset is designed to provide a comprehensive and diverse collection of RF signals for drone detection and identification research. The dataset is built upon a large corpus of raw signal captures, totaling approximately 500+ GB of raw data.

At the core of our dataset is a significant diversity of drone models and signal types. The collection includes signals from 39 unique classes, encompassing 23 distinct commercial and hobbyist UAV models, their remote controllers, and operational variants (e.g., armed vs. unarmed states). This variety, spanning major manufacturers such as DJI, Autel, and Parrot, ensures that models trained on CDRF are exposed to a wide range of communication protocols, a substantial leap from the handful of models found in many previous datasets. In addition to drone signals, we include a substantial number of "no-drone" recordings, capturing ambient RF noise and other common interferers like Wi-Fi signals. This enables the training of robust binary classifiers that can distinguish between the presence



(a) Autel EXOII (b) DJI Inspire1

(c) DJI Mavic3 (d) DJI Phantom3 Advanced

Figure 3.3: Representative spectrograms from the outdoor dataset, showcasing the variability in signal characteristics due to environmental factors.

and absence of drone activity, a critical requirement for real-world deployment. To further challenge detection and identification algorithms, the dataset also contains recordings of multiple drones operating simultaneously in the same airspace.

All raw data is stored using the In-phase and Quadrature (I/Q) sampling method. Each sample is represented as an interleaved binary bitstream of two 32-bit floating-point numbers, capturing the full complex signal. This raw format is crucial as it allows researchers to develop and test novel signal processing and feature extraction techniques that operate directly on the baseband signal, prior to any transformations like the STFT.

The signals captured in CDRF exhibit a variety of characteristics typical of modern drone communication systems. Many of the drones utilize Frequency-Hopping Spread Spectrum (FHSS) for their command and control links. These signals are defined by key parameters such as their hopping frequency, duration, duty cycle, and the period of their hopping pattern. In contrast, signals used for video transmission typically have wider bandwidth and longer duration than control signals. We define the collection of these characteristics as the RF Drone Fingerprint, which serves as a rich input for deep learning models to differentiate between drone types.

Furthermore, our data captures the inherent behavioral variability of drone signals. For instance, the frequency hopping patterns of a drone during its initial pairing process can differ significantly from those during normal flight operations. The dataset also captures more subtle phenomena, such as the variable time duration of video transmission signals from certain drone models, a characteristic that can serve as another important feature for robust classification.

3.4 Data Preprocessing, Augmentation, and Annotation

We release both a raw-signal processing pipeline and augmentation utilities designed to create realistic, label-consistent training data from complex I/Q captures. All components are open-source and parameterized for reproducibility.

3.4.1 Spectrogram Generation

Raw complex I/Q streams are transformed into time–frequency representations via the STFT. Given a complex signal $x[n]$, window $w[n]$, window length N, and hop size H,

the STFT is [44]

$$X(m,k) \;=\; \sum_{n=0}^{N-1} x[n+mH]\; w[n]\; e^{-j2\pi kn/N},$$

where m is the discrete time index and k is the discrete frequency index.

We compute the power spectrogram $S(m,k) = |X(m,k)|^2$ and render in dB as

$$S_{\mathrm{dB}}(m,k) \;=\; 10\log_{10}\big(S(m,k)+\epsilon\big),$$

where ϵ represents a very small, positive constant used to avoid taking the logarithm of zero. After that map to RGB using a perceptual colormap. By default, we use `scipy.signal.stft` [45] with a Hann window, $N =$ `FFT_SIZE` $= 1024$, overlap $= 128$ samples, and return two-sided spectra with DC centered via `fftshift`. Default sampling is `SAMPLING_RATE` $=$ 20 MHz and we generate `NUM_FFT_SPEC` $= 1500$ time bins per sample. Spectrograms are exported as PNG images alongside per-sample metadata.

3.4.2 Frequency Resolution and Colormap Selection

The window size N and hop ($N -$ overlap) set the time–frequency trade-off; larger N improves frequency resolution at the cost of temporal precision. All parameters are exposed to support ablations. We provide multiple colormaps (`viridis`, `plasma`, `inferno`, `magma`, `cividis`, `gray`, `hot`) to test color sensitivity; the default pipeline normalizes spectrograms to $[0,1]$ before colorization and composites to RGB for consistent rendering.

3.4.3 Data Normalization

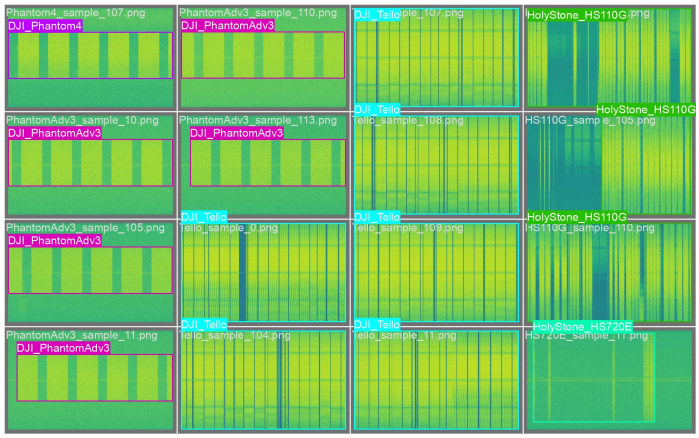
We include signal-level and spectrogram-level normalization operators:

- *Power normalization* of complex I/Q to unit average power: $\tilde{x} = x/\sqrt{\mathbb{E}|x|^2+\epsilon}$.

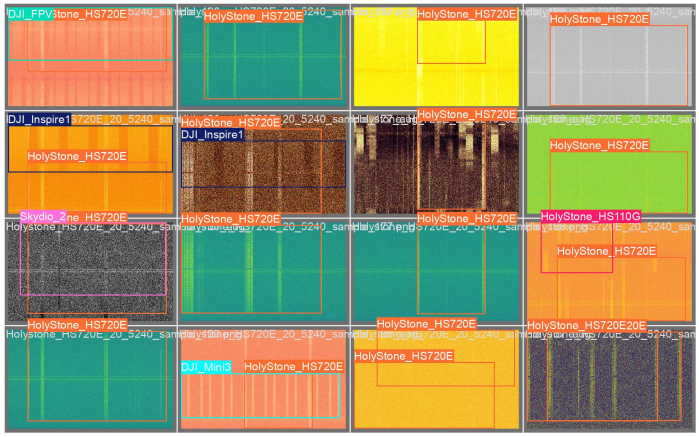
- *Z-normalization* of real and imaginary parts independently to zero mean, unit variance.

- *Spectrogram normalization* options: per-frequency z-score, per-time z-score, or global min–max to $[0, 1]$ prior to colorization.

These are selectable in the augmentation scripts and can be toggled per experiment.



(a) Annotated Data.



(b) Augmented Data.

Figure 3.4: Examples of annotated and augmented data.

3.4.4 Custom Augmentation Tools

Our augmentation operates at the *raw-signal* (I/Q) level to better preserve RF physics and then projects to spectrograms, with exact label propagation to detection annotations.

3.4.4.1 SNR Conditioning via AWGN

To synthesize controlled SNRs, we add complex AWGN $n \sim \mathcal{CN}(0, \sigma^2)$ to x where the noise power σ^2 is chosen for target $\mathrm{SNR}_{\mathrm{dB}}$:

$$P_s = \mathbb{E}|x|^2, \quad \mathrm{SNR}_{\mathrm{lin}} = 10^{\mathrm{SNR}_{\mathrm{dB}}/10}, \quad \sigma^2 = P_s/\mathrm{SNR}_{\mathrm{lin}}.$$

We optionally export noise-only spectrograms for analysis and create SNR-stratified dataset variants (e.g., per-SNR subdirectories), enabling robustness sweeps.

3.4.4.2 Frequency Shifting

We simulate carrier offsets and front-end errors by multiplying with a complex exponential

$$x_{\Delta f}[n] \;=\; x[n]\; e^{j2\pi \Delta f\, n/F_s},$$

which induces a vertical translation on the spectrogram and where F_s is sampling rate. This is applied before STFT and is label-consistent with our detection annotation recomputation (below).

3.4.4.3 Interferer Mixing

To emulate spectrum crowding, we mix two normalized signals at a controllable ratio $\alpha \in [0, 1]$:

$$x_{\mathrm{mix}} \;=\; \mathrm{norm}\left(x_1 \;+\; \alpha\, x_2\right),$$

optionally with independent frequency shifts on each source. This produces realistic co-channel scenarios directly in the I/Q domain.

3.4.4.4 Spectrogram Rendering Variants

Augmented samples are rendered under randomized, but reproducible, choices of colormap and normalization policy to increase visual diversity while keeping frequency content intact.

3.4.4.5 Background Noise and Non-Target Signals

Our dataset includes a comprehensive corpus of non-target signals, captured from diverse environments where no drones were active. To facilitate common machine learning workflows, we provide a suite of tools to extract these "no-drone" segments and generate unified metadata files for mixed-signal experiments. This curated collection of real-world negative samples is crucial for training well-calibrated binary detectors and is explicitly designed to support advanced strategies like hard-negative mining.

3.4.5 Annotation Strategy

We adopt YOLO-format annotations on spectrogram images with a *whole-signal* policy: contiguous RF emission bursts from a device are annotated as a single object to capture both spectral morphology and temporal occupancy. Fig. 3.4 shows examples of annotated data and augmented data with annotations.

In general, a spectrogram of a UAV signal segment consists of a sequence of ON-OFF states, which represents the transmission pattern of the UAV in both time and frequency domain. In order to lead YOLO model to better capture the signal characteristics in both domains, we define the annotation policy as that, the bounding box for each type of signals

in a spectogram image will start and end with an ON state of the signal. The OFF states at the beginning and end of a spectrogram are neglected. This can help the YOLO model to track the start and end of a transmission, and thus to output more accurate bounding boxes. Meanwhile, if the RF emission bursts captured by a spectrogram are with limited time duration, i.e., $< 10\%$ of the time duration of the spectrogram, or there is only a single ON state in the whole spectrogram, no bounding box annotation will be given, and the spectrogram image will be labeled as background in those cases.

As for the RC signals, due to its frequency-hopping property, in CDRF we aim at providing two types of YOLO-formated annotation to support the flexibility of future research or implementation in potentially different purposes.

- We provide one bounding box covering all the RC signal bursts captured in a spectrogram. Similar to the UAV signal labeling, we ensure that the edges of the bound box align with the start and end of the sequence of RC signal bursts

- CDRF also contains annotation for RC signal bursts transmitted at the same channel. Since the frequency-hopping behavior is not guaranteed to occur or being captured in every single spectrogram in a continuous raw data recording, we believe that this type of annotation can contribute the learning of YOLO models and improve the detection performance when deploying it in reality.

Moreover, due to the fact that most of the RC signals are narrowband and in short duration for each ON state, we do not provide any bounding box annotation to the spectrogram that contains only one single ON state. The reason behind this design choice for annotation strategy is that the bound box covering only single ON state does not provide any information for the model to learing about its pattern across either time or frequency domain. Similar statement on UAV video signal annotation is also demonstrated in [46].

Also, the detection or classification based on a single ON state is generally neither expected nor considered reliable in real-world UAV detector deployment scenarios.

3.4.5.1 Label-Preserving Transforms with Wrap-Around

Frequency shifts correspond to vertical translations in normalized image coordinates by $\Delta y = \Delta f / F_s$. We recompute bounding boxes analytically after augmentation and correctly handle wrap-around at the top/bottom edges: if a box straddles the boundary after shift, it is split into two valid boxes whose heights sum to the original. Boxes are also filtered by a configurable minimum height to avoid degenerate labels after extreme shifts.

3.4.5.2 Dataset Hygiene and Third-Party Integration

We include utilities to normalize third-party releases (e.g., Roboflow-style layouts) into a consistent hierarchy (`images/ labels/` with class folders and {train,val,test} splits) and to preserve class mappings. This enables direct augmentation and training across heterogeneous sources.

3.4.6 Reproducibility and Metadata

Each sample carries rich metadata to ensure reproducibility: original file path, device/model label, sampling rate, FFT size, number of STFT frames, per-sample time bounds (start/end in seconds), center frequency, and output paths for spectrograms. All processing parameters (e.g., window size, overlap, SNR target, shift amounts, mixing weights) are saved or deterministically re-creatable from configuration, facilitating exact regeneration of training splits and ablation studies.

3.5 Machine Learning Tasks and Baselines

This section details the machine learning tasks supported by our dataset, the baseline models implemented, and their performance. We address drone detection, single-label classification, multi-label classification, open-set recognition, and hierarchical classification.

3.5.1 Drone Detection (YOLO)

3.5.1.1 Task Definition

Given a time–frequency spectrogram, the goal is to detect and localize RF emissions attributable to drones. The detector predicts a set of axis-aligned bounding boxes with class labels in YOLO normalized coordinates, where each object corresponds to a contiguous emission burst (whole-signal policy) on the spectrogram.

3.5.1.2 Baseline Model Architecture and Training

We use a single-stage object detector, `YOLOv11n` [47] (Ultralytics [48] v8.3.156), trained on spectrograms produced by our pipeline (section 3.4). We evaluate two scenarios: (i) Clean, in which spectrograms are generated without augmentation; and (ii) Augmented, in which raw complex I/Q are transformed before STFT via frequency shifting, interferer mixing, and SNR conditioning. Models are initialized from COCO-pretrained weights [49, 48, 47], trained with deterministic seeds, and validated at each epoch. Labels follow the whole-signal policy, and for frequency shifts we recompute YOLO annotations exactly with wrap-around handling on the frequency axis to preserve label consistency. Unless otherwise specified, Ultralytics defaults are used for input size, data augmentation, and non-max suppression.

3.5.1.3 Evaluation Metrics

We report standard detection metrics: class-aggregated Precision and Recall; mean Average Precision at IoU 0.5 (mAP@0.5); and COCO-style mAP averaged over IoU thresholds from 0.5 to 0.95 in steps of 0.05 (mAP@[.5:.95]). For qualitative error analysis, we also include normalized confusion matrices computed from matched detections.

3.5.1.4 Experimental Results and Baseline Performance

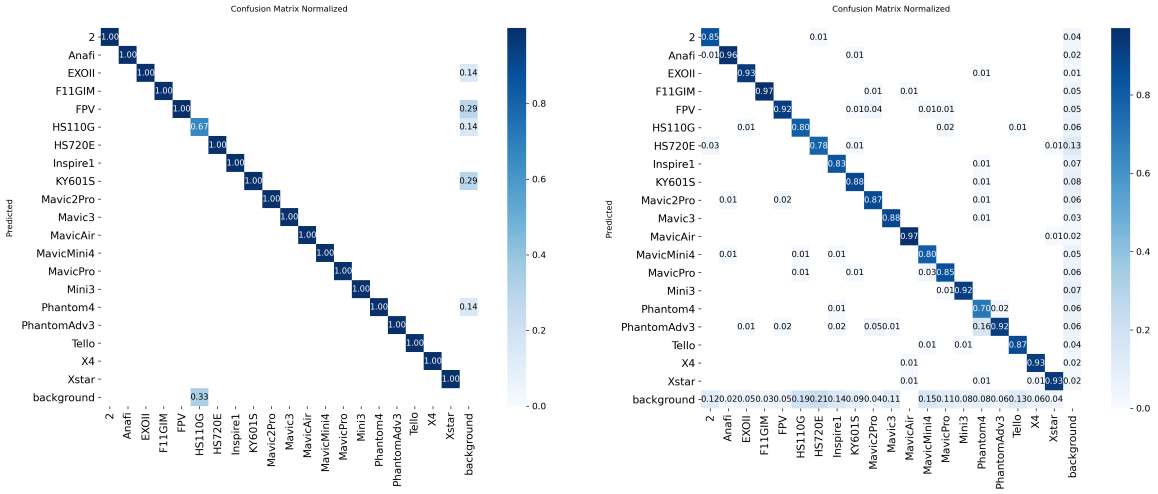
Table 3.1: YOLO detection results.

Run	Epochs	Precision	Recall	mAP@0.5	mAP@[.5:.95]
Clean	100	0.932	0.982	0.986	0.948
Augmented	50	0.940	0.855	0.935	0.838

Table 3.1 summarizes the best validation checkpoints: The model trained without any augmentations (referred to as "clean") achieves strong performance (Precision $= 0.932$, Recall $= 0.982$, mAP@0.5 $= 0.986$, mAP@[.5:.95] $= 0.948$), with the top mAP@[.5:.95] observed near epoch 78. Under augmentation, precision is maintained and slightly improved (Precision $= 0.940$) while recall drops (Recall $= 0.855$), yielding lower top-line mAP (mAP@0.5 $= 0.935$, mAP@[.5:.95] $= 0.838$; best near epoch 49). This reflects the expected precision–recall trade-off when training on harder, low-SNR and interferer-rich conditions: the detector becomes more conservative (fewer false positives) at the expense of increased misses (more false negatives).

Despite this shift, the normalized confusion matrices in Fig. 3.5 remain strongly diagonal for both regimes, indicating that whole-signal annotations preserve class separability even under frequency shifts and spectrum crowding. The dominant degradation under augmentation is recall (missed detections) rather than systematic label confusion, suggesting

room for robustness gains via longer training, class-balanced sampling, or stronger backbones without altering the annotation policy.



(a) Clean: normalized confusion.

(b) Augmented: normalized confusion.

Figure 3.5: Class-wise performance and error structure on validation.

3.5.2 Hierarchical Classification

3.5.2.1 Task Definition

To provide more robust and interpretable classifications, we explored a hierarchical approach that classifies signals in a coarse-to-fine manner. In Fig. 3.6, we provide the full structural tree of the hierarchical classification. Our hierarchy is structured into three levels: Modulation (3 classes), Protocol (5 classes), and finally the specific drone Model (27 classes).

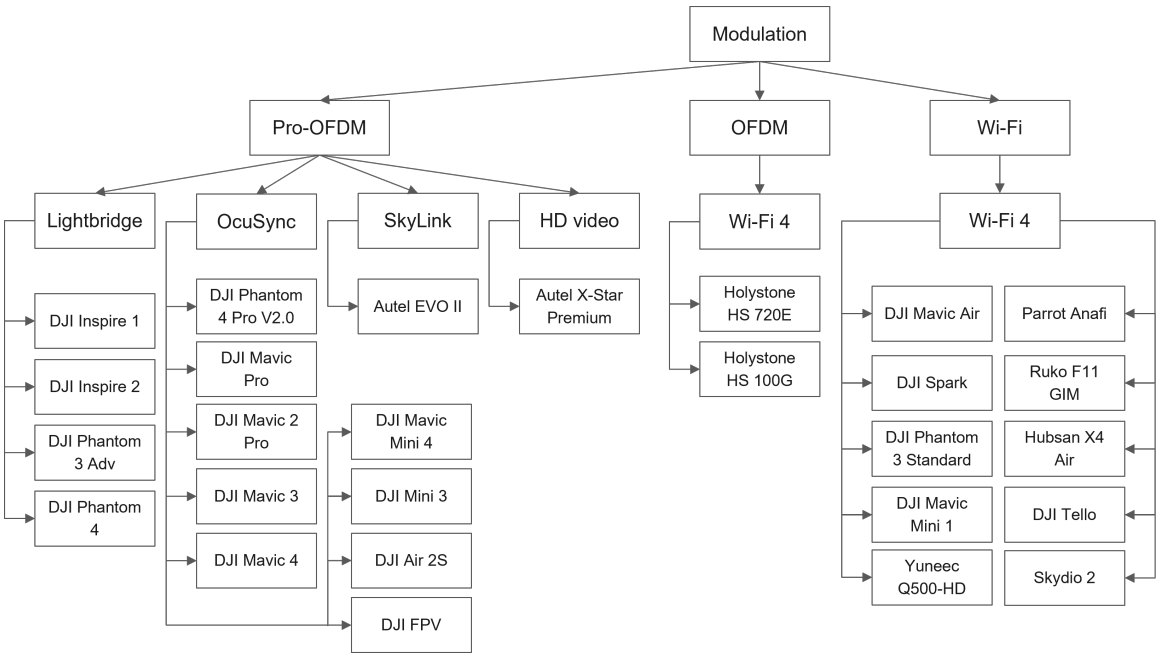


Figure 3.6: Hierarchical classification tree structure of CageDroneRF.

3.5.2.2 Baseline Model Architecture and Training

We implemented a multi-task learning model using a shared ResNet-18 [41] backbone to extract features from the input spectrograms. The output of the feature extractor is fed into three separate classification heads, each dedicated to one level of the hierarchy. The model is trained end-to-end using a "HierarchicalLoss", which is a weighted sum of the cross-entropy losses from each head.

To observe the effectiveness of our indoor data and how it generalizes to outdoor scenarios, we trained our hierarchical classifier on the indoor subset of CDRF and evaluated it on the outdoor subset. Furthermore, we examined how including outdoor data in the training set affects performance by training another model on the combined indoor and outdoor data. In both cases, the amount of indoor training data was kept constant to ensure a fair comparison. The models face challenging scenarios, including varying SNR levels, signal bandwidths, outdoor drones flying at different altitudes, distances, and the presence

of various interferers.

3.5.2.3 Experimental Results and Baseline Performance

The results, shown in Fig. 3.7, compare the generalization performance of two training configurations when tested on outdoor data across three classification levels: Modulation, Protocol, and Model.

When the model is trained on indoor data only, it achieves moderate accuracy on lower-level features (Modulation: 69.95%, Protocol: 66.11%) but struggles significantly with identifying the specific drone model (42.07%). This indicates poor domain generalization, as the model fails to adapt from indoor to outdoor environments.

In contrast, training on a combined set of indoor and outdoor data yields substantial improvements across all hierarchical levels. Accuracy increases to 89.42% for Modulation, 87.34% for Protocol, and 69.63% for Model. The most significant gain of +27.56% is observed in model-level classification, the task most sensitive to environmental shifts. This demonstrates that training on diverse, mixed-domain data is crucial for learning robust, invariant features.

3.5.3 Open-Set Recognition

3.5.3.1 Task Definition

Open-Set Recognition (OSR) extends classification to simultaneously: (1) accurately classify known drone types from the training set, and (2) identify and reject unknown samples as “novel” or “outlier” classes, preventing forced misclassification into known categories.

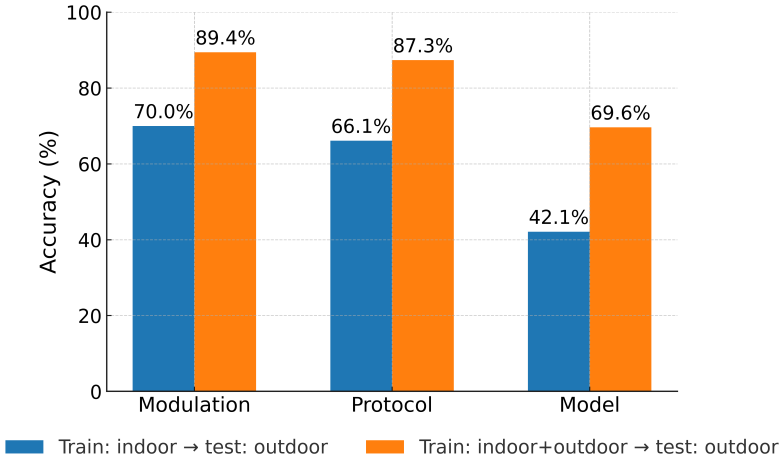


Figure 3.7: Hierarchical classification accuracy on different training setups.

3.5.3.2 Baseline Model Architecture and Training

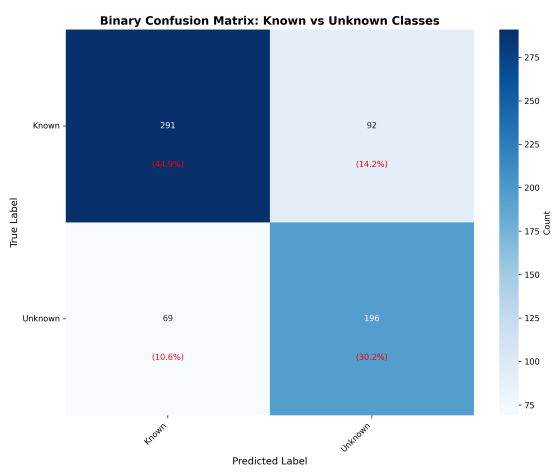
In our experiments, we utilized MetaMax [50], a state-of-the-art OSR method that combines deep feature learning with statistical modeling of known class distributions. For our OSR experiments, we curated a subset of CDRF comprising 14 distinct drone types totaling 2,180 spectrogram images. The dataset was strategically partitioned into known and unknown classes to simulate realistic deployment scenarios. The known class set contains 10 drone types with 1,915 images, and the unknown class set comprises 4 drone types with 265 images. The data was split into training (60%) and validation (20%) samples from known classes only, with the remaining samples reserved for testing.

3.5.3.3 Evaluation Metrics

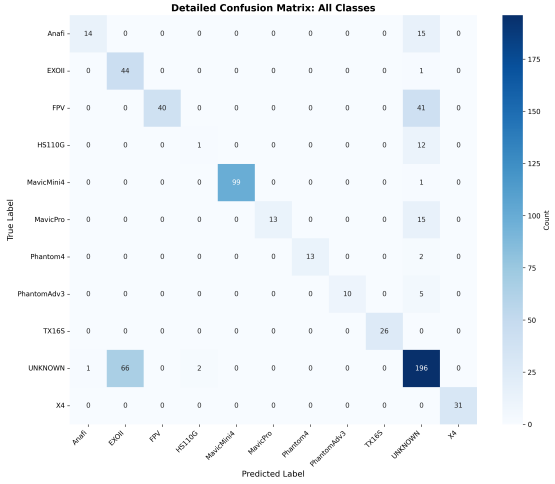
Performance metrics included: (1) closed-set accuracy on known classes, (2) unknown detection rate, and (3) overall accuracy.

3.5.3.4 Experimental Results and Baseline Performance

We evaluated MetaMax on our curated CDRF subset, achieving an overall accuracy of 75.15%, with known class accuracy of 75.98% and an unknown detection rate of 73.96%. The detailed confusion matrix (Fig. 3.8b) reveals strong performance correlation with class size. Among the 265 unknown test samples, 196 (73.96%) were correctly identified as novel. The binary confusion matrix for known vs. unknown discrimination is shown in Fig. 3.8a. These results demonstrate both the promise and challenges of OSR for RF drone detection.



(a) Binary confusion matrix for known vs unknown class discrimination.



(b) Detailed confusion matrix showing per-class performance for open-set recognition.

Figure 3.8: Open-set recognition confusion matrices: (a) known vs unknown discrimination; (b) per-class detailed performance.

3.5.4 Drone Classification (Single-Label)

3.5.4.1 Task Definition

The single-label drone classification task involves assigning each detected RF signal to exactly one drone type from a predefined set of classes.

3.5.4.2 Baseline Model Architecture and Training

We implemented a ResNet-18 convolutional neural network as the backbone, leveraging its proven effectiveness for spectrogram-based RF signal classification. The model was trained on spectrogram images generated from the CDRF dataset, using standard data augmentation and normalization techniques to improve generalization. We had a total number of 20 different drone types in our dataset. The ResNet-18 model was trained with a cross-entropy loss and optimized using Adam.

3.5.4.3 Experimental Results and Baseline Performance

After training, the model achieved a test accuracy of 53.49%, as shown in the confusion matrix (Fig. 3.9). The confusion matrix reveals that certain drone types are classified with high accuracy, while others exhibit significant misclassifications, especially for classes that seen less training data, indicating areas for potential improvement. The results highlight the challenges of single-label classification in RF drone signals, particularly in distinguishing between similar models.

3.6 Discussion and Future Work

Cross-Modal Transfer Validation: CDRF's results validate the central hypothesis underlying the dissertation's framework: vision models successfully transfer to wireless domains

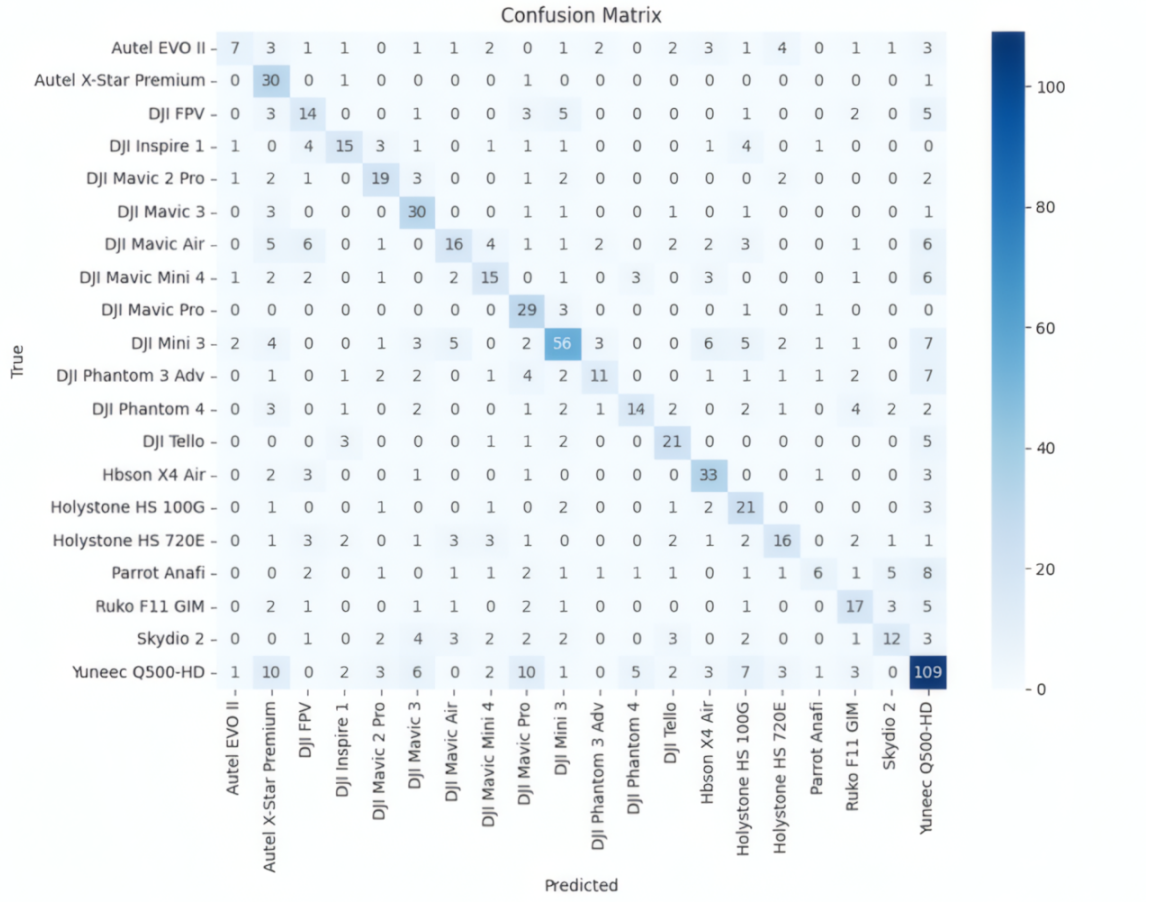


Figure 3.9: Confusion matrix for single-label drone classification.

when signals are represented as images. The YOLOv11n detector achieves mAP@0.5 = 0.986 on clean spectrograms and 0.935 under augmentation, demonstrating that object detection architectures pretrained on natural images (COCO) extract meaningful features from RF time-frequency representations without wireless-specific modifications. Similarly, ResNet-18 baselines achieve 53.49% single-label accuracy across 20 drone types and 75.98% known-class accuracy in open-set scenarios, confirming that convolutional architectures transfer across modalities. This empirical evidence directly supports Research Question 1 (RQ1) and justifies DenoMAE's approach of leveraging vision transformers for

wireless representation learning (Chapter 2).

Robustness Gap and Foundation Model Implications: The performance degradation under augmented conditions reveals critical challenges for foundation models. The recall drop from 0.982 to 0.855 (YOLOv11n) reflects the precision-recall tradeoff when training on low-SNR, high-interference conditions: the detector becomes more conservative to avoid false positives but misses more true detections. This robustness gap motivates the dissertation's modular framework: self-supervised pretraining (DenoMAE) on diverse unlabeled signals should learn noise-invariant representations; LLM-based reasoning (Plug-and-Play) should provide complementary decision-making that doesn't degrade under noise; and meta-learning (Meta-Task) should enable rapid adaptation to deployment-specific SNR distributions with minimal labeled data. The hierarchical classification results (indoor-only: 42.07% model accuracy; mixed training: 69.63%, +27.56%) demonstrate that domain shift, a core challenge for foundation models, can be addressed through diverse pretraining data, validating the framework's emphasis on broad data coverage.

Open-Set Recognition and Real-World Deployment: The 73.96% unknown detection rate establishes that models can learn to reject novel threats, but the class-size correlation in performance (Fig. 3.8b) highlights a persistent challenge: data imbalance. This finding connects directly to Meta-Task (Chapter 5), which addresses few-shot learning scenarios where new drone types emerge with only 5-10 labeled examples. The open-set capability validated here provides a safety mechanism for foundation model deployment: rather than forcing every input into a known category, models can flag uncertainty and request human verification.

Confounded Comparisons and Split Provenance: Two methodological caveats bound the results above. First, the indoor-only versus indoor+outdoor hierarchical comparison (42.07% vs. 69.63% model-level accuracy) changes both domain coverage and training population size simultaneously, since the indoor-only condition was not also evaluated with

a matched, enlarged indoor-only sample; the reported gain should therefore be attributed to the combination of more data and more diverse data, not cleanly isolated to domain diversity alone. Second, this chapter does not ship a recording/session/device-level split manifest proving that augmented or time-windowed descendants of the same raw recording never cross a train/validation/test boundary. Repeated windows or synthetic descendants from one raw recording are statistically related, so a large number of generated images is not automatically a large number of independent acquisition conditions; without a published lineage manifest, the independence of the reported splits cannot currently be audited externally. This is an absence of documented evidence rather than a demonstrated leakage finding, and releasing such a manifest is a priority for future releases of CDRF.

To ensure the long-term relevance and utility of CDRF, we are committed to its continuous expansion. Future work will focus on several key areas:

- **Dataset Expansion:** We will regularly update the dataset with new and emerging drone models, as well as more diverse background and interference signals captured from a wider range of environments (e.g., urban, industrial, and rural). We also plan to integrate CDRF with other public datasets to create an even more comprehensive and diverse benchmark.

- **Advanced Model Architectures:** We will explore the application of more sophisticated deep learning models, such as Transformers and graph neural networks, which may be better suited to capturing the complex temporal and spectral dependencies in RF signals.

- **Enhanced Augmentation Techniques:** We will continue to develop more advanced data augmentation techniques that can more realistically simulate a wider range of real-world channel effects, such as multipath fading and Doppler shifts.

- **Focus on Advanced Tasks:** We will further investigate the more challenging tasks of multi-label and open-set classification, which are critical for real-world applications where multiple drones may be present, and novel threats are a constant concern.

3.7 Conclusion

This chapter presented CDRF, a comprehensive benchmark that validates a fundamental premise of the dissertation's framework: vision models can successfully transfer to wireless perception tasks. By demonstrating that YOLO detection (mAP@0.5 = 0.986 clean, 0.935 augmented) and ResNet classification operate effectively on RF spectrograms, CDRF provides empirical evidence for the cross-modal transfer principle underlying Research Question 1 (RQ1). This validation justifies the multimodal approach pursued in Deno-MAE (Chapter 2), where masked autoencoders learn universal wireless representations by treating signals as images.

Beyond theoretical validation, CDRF establishes a realistic stress-test benchmark for foundation model components. The performance degradation under augmented conditions, particularly the recall drop from 0.982 to 0.855 in YOLO detection, reveals the robustness challenges that operational systems face. The hierarchical classification results (+27.56% model-level gain with mixed indoor/outdoor training) demonstrate the domain shift problem that foundation models must address through diverse pretraining and effective adaptation mechanisms. The open-set recognition capability (73.96% unknown detection) validates that models can learn to reject novel threats, a critical requirement for real-world deployment.

Framework Integration: Within the dissertation's modular system, CDRF serves three essential roles. First, it validates the feasibility of vision-to-wireless transfer, establishing that pre-trained computer vision architectures provide a strong foundation for

wireless perception. Second, it provides a diverse, challenging benchmark (39 classes, 500GB raw data, controlled SNR stratification) for evaluating whether foundation model approaches, namely self-supervised learning (DenoMAE), LLM-based reasoning (Plug-and-Play), and meta-learning adaptation (Meta-Task), generalize beyond clean laboratory conditions. Third, its raw-signal augmentation pipeline and dataset-agnostic tooling enable systematic robustness evaluation across the SNR, interference, and domain shift conditions that real deployments encounter.

The success of vision models on CDRF's drone detection tasks demonstrates that cross-modal knowledge transfer is not merely theoretical but practically achievable. This empirical validation provides confidence for pursuing the more sophisticated approaches in subsequent chapters: DenoMAE's self-supervised representation learning can leverage vision transformer architectures because those architectures demonstrably work for wireless signals; the Plug-and-Play framework's LLM integration becomes more plausible given that one domain (vision) has already successfully transferred to wireless; and Meta-Task's few-shot adaptation addresses the remaining challenge of rapidly specializing foundation model capabilities to new wireless scenarios with minimal data.

By open-sourcing both dataset and toolkit, CDRF aims to foster reproducible research and collaborative progress toward robust, field-ready wireless AI systems. Future work will expand the dataset with emerging drone models, explore advanced architectures (transformers, graph neural networks), and investigate how CDRF can serve as a pretraining corpus or evaluation benchmark for the integrated foundation model framework proposed in this dissertation.

Chapter 4

LLM-Based Reasoning for Open-Set Modulation Classification: Plug-and-Play and DiSC-AMC

4.1 Introduction

Automatic Modulation Classification (AMC) is a critical technology in modern wireless communications, underpinning applications in cognitive radio, spectrum sensing, and interference management. By accurately identifying modulation schemes, AMC facilitates efficient spectrum utilization and enhances the adaptability and reliability of communication networks. However, despite its significance, AMC remains a challenging problem due to the complex interplay of signals with ambient noise and other channel impairments.

Early approaches to AMC predominantly relied on traditional machine learning techniques and handcrafted feature extraction, but these methods have gradually been superseded by deep learning models. For instance, Convolutional Neural Networks (CNNs) have

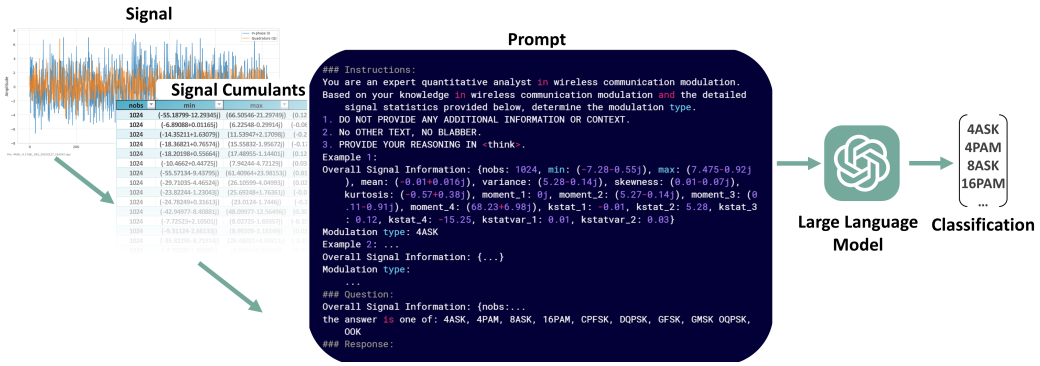


Figure 4.1: Overview of our in-context AMC framework. The raw signal is first summarized into high-order statistics and cumulants, which are then embedded in a text prompt. This prompt is fed into a large language model to perform modulation classification from a predefined set of modulation types (e.g., 4ASK, 4PAM, 8ASK, 16PAM).

been extensively applied in this domain. Peng et al. [28] transformed raw modulated signals into constellation diagrams to serve as inputs for CNN architectures such as AlexNet, while subsequent studies introduced variants like MCNet [51] that leveraged multiple convolutional blocks with skip connections and asymmetric kernels to capture spatio-temporal correlations. In another investigation, a CNN framework was explored using multiple signal representations, such as constellation diagrams, ambiguity functions, and eye diagrams, with empirical results indicating that combining these representations could improve classification accuracy [52].

More recently, attention-based models, particularly transformers, have garnered significant interest for AMC. Transformers, introduced by Vaswani et al. [53], are adept at processing sequential data by dynamically focusing on the most relevant parts of the input. Cai et al. [54] demonstrated that a transformer network (TRN)-based approach can capture long-range dependencies more effectively than conventional CNNs, which is especially beneficial in low SNR scenarios. Building on this idea, Kong et al. [10] further improved transformer-based AMC by incorporating convolutional embedding and attention pooling techniques, and Qu et al. [55] combined transformers with Long Short-Term

Memory (LSTM) networks to address long-range temporal dependencies and enhance robustness against noise through novel data augmentation strategies.

Despite these advancements, many existing methods are developed under idealized, noise-free conditions [31, 56]. Such conditions are rarely encountered in real-world environments where signals are typically contaminated with noise. Moreover, several approaches struggle with fine-tuning downstream tasks when only limited labeled data is available [57, 58].

Nmformer [59] employs a Vision Transformer (ViT) trained on a large labeled dataset to achieve strong performance when retrained on low-SNR signals. However, the model struggles to attain high accuracy due to the absence of unsupervised pretraining. In contrast, DenoMAE [26] and DenoMAE2.0 [1] aim to achieve competitive performance with only a few thousand samples during pretraining and significantly fewer labeled samples in the downstream task. Nevertheless, the limited data constrains the model's ability to achieve substantial performance improvements. This underscores the need for a more advanced framework that effectively leverages large-scale data to enhance accuracy.

Addressing these limitations, recent work has begun to explore more sophisticated frameworks that can balance performance across varying SNR conditions. Gao et al. in the MoE-AMC study [9] proposed a Mixture-of-Experts (MoE) approach that integrates a ResNet-based model for high SNR signals with a transformer-based model for low SNR signals, achieving a more balanced and robust classification performance. Complementing this, the review by Jafarigol et al. [60] provided a comprehensive analysis of contemporary AI/ML-based AMC models, highlighting both recent advances and persistent challenges, particularly the difficulty of maintaining accuracy in noisy environments and the need for efficient models that operate under constrained computational resources.

In this work, we take a novel direction by leveraging a pretrained LLM-based approach

for automatic modulation classification. Our method utilizes traditional modulation statistics and cumulants, capitalizing on the LLM's inherent familiarity with classical signal processing techniques. This capability enables our framework to perform competitive one-shot modulation classification without the need for exhaustive training across all SNRs, modulation types, or channel conditions, and without the computational overhead of extra preprocessing steps such as denoising. By transforming quantitative signal features into natural language descriptions, our approach bridges the gap between classical signal processing and modern AI, achieving robust performance even in challenging, noisy scenarios.

Our contributions include:

- A novel in-context approach for AMC using LLMs that simplifies the processing pipeline.
- A one-shot classification method that eliminates the need for extensive training data.
- Competitive performance under both noiseless and noisy conditions.

Framework Position: Within the dissertation's modular framework for Wireless Physical-Layer Foundation Models (WPFMs), this Plug-and-Play approach serves as the *intelligent reasoning engine*, the second core component. While DenoMAE (Chapter 2) learns rich signal representations through self-supervised learning and Meta-Task (Chapter 5) enables rapid adaptation with minimal data, Plug-and-Play demonstrates a fundamentally different pathway: leveraging pre-trained LLMs to perform training-free wireless classification by converting statistical signal features into natural language prompts. This validates the dissertation's central thesis that wireless foundation models need not be built through massive isolated training but can strategically approximate and integrate world knowledge. By achieving competitive performance (72.4% accuracy) without wireless-specific training, this work proves that sophisticated reasoning capabilities can be transferred from language

models to the wireless domain, providing interpretability, open-set classification, and cost-effective deployment. The modular design enables future integration where DenoMAE's learned features could replace or augment the statistical summaries, while Meta-Task could enable few-shot adaptation of prompt strategies.

Beyond improving AMC performance, our approach paves the way for developing foundation models for wireless communications. Such foundation models can provide a robust unified solution across diverse scenarios, reducing the need for channel-specific models and lowering the associated development and deployment costs. By enabling models that generalize well without extensive retraining, our method has the potential to streamline the design of wireless communication systems, making them more adaptable, scalable, and cost-effective.

Overall, this work not only simplifies modulation classification but also sets the stage for further research in robust, scalable, and versatile signal processing using large language models.

4.2 Theoretical Background

Plug-and-Play rests on classical statistical features of the received signal. The central moments of a signal summarize the shape of its amplitude distribution, and k-statistics are the minimum-variance unbiased estimators of its cumulants. Cumulants of order higher than two vanish for Gaussian noise, so they suppress additive Gaussian noise while retaining structure that depends on the modulation, and fourth-order cumulants in particular take different values for BPSK, QPSK, and QAM constellations [61]. These features are compact and interpretable, which is why classical feature-based AMC used them, and they can be written as short numeric descriptions that a language model can read as text.

4.3 Method

Modular Design: This method operates as a standalone component using statistical signal features, demonstrating training-free classification. Future work could integrate DenoMAE's learned representations (Chapter 2) as enhanced feature inputs while maintaining the framework's modularity.

In this section, we present a novel approach for modulation classification using higher-order statistics and prompt-based LLM generation through three key stages:

(Stage 1) extracting comprehensive statistical summaries (e.g., moments, variance, kurtosis); **(Stage 2)** linearizing these features into a text format [62] as

$$\text{linearize}(x) = [c_k : x_k]_{k=1}^{K}, \tag{4.1}$$

where c_k is the k^{th} feature name and x_k its value; and **(Stage 3)** constructing a structured prompt combining instructions (I), exemplar context (C), the linearized statistics, and modulation directives (S):

$$d = \text{LLM}\big(I, C,\ \text{linearize}(x),\ S\big). \tag{4.2}$$

This process generates concise signal descriptions to guide the LLM in selecting the appropriate modulation type (Figure 4.1).

4.3.1 Signal Summarization

We compute key descriptive statistics and cumulant-related metrics to obtain a robust statistical characterization of each signal. These statistical summaries serve as the foundation for both individual signal queries and for constructing exemplar contexts, reference points

with known modulation labels that help the model infer the underlying patterns associated with each modulation type. Algorithm 1 details the procedure used to convert a raw signal vector x into a succinct string d that encapsulates its statistical properties.

Algorithm 1: Signal Summarization

Input: x (Vector of signal values)
Output: d (A string summarizing cumulants and descriptive statistics)

```
stats ← {
  "nobs": x.nobs,
  "min": x.min,
  "max": x.max,
  "mean": x.mean,
  "variance": x.variance,
  "skewness": x.skewness,
  "kurtosis": x.kurtosis
};
# Compute k-th moment and cumulant
for i ← 0 to 7 do
    stats["moment_i"] ← moment(signal, order = i);
for i ← 1 to 4 do
    stats["kstat_i"] ← kstat(signal, i);
for i ← 1 to 2 do
    stats["kstatvar_i"] ← kstatvar(signal, i);
d ← dict_to_string(stats);
return d;
```

4.3.2 Example Context Construction

To provide context for the language model, we generate statistical summaries for a set of exemplar signals, one for each modulation type, using the same procedure outlined in Algorithm 1. These summaries, paired with their known modulation labels, are formatted into

reference examples. Including these exemplars in the prompt furnishes the model with concrete illustrations of the statistical characteristics defining each modulation class, thereby enhancing classification accuracy, especially for signals affected by noise or channel impairments.

4.3.3 Prompt Formulation

The final prompt is assembled by combining three components:

- **Instruction Block** (I): A concise set of instructions that mandates the selection of a modulation type from a predefined list (e.g., "4ASK," "4PAM," "8ASK," etc.) and emphasizes that the output must consist of a single valid answer.
- **Exemplar Summaries** (C): The formatted examples derived from the exemplar signals, which display both the overall statistical summaries and their corresponding ground-truth modulation types.
- **Query Section** (linearize$(x) + S$): The statistical summary for the test signal, concatenated with a directive suffix (S) that reinforces the modulation classes.

This carefully designed prompt minimizes ambiguity and ensures that the language model's output adheres strictly to the allowed modulation types and an example of a prompt is shown in Table 4.1.

4.3.4 Generation and Post-Processing

Once the prompt is constructed, it is tokenized and submitted to the pre-trained causal language model (no fine-tuning is used in Plug-and-Play). To encourage deterministic responses, the model is configured with constrained generation parameters (e.g., a low temperature and a restricted maximum token count). After the model generates its output,

post-processing is performed to remove extraneous characters and extract the predicted modulation type by matching the response against the predefined list of valid classes. Finally, the predicted modulation type is compared with the ground truth label to compute an accuracy metric.

4.4 Experiments and Results

Dataset Context: We evaluate on synthetically generated signals to provide controlled proof-of-concept validation for the LLM reasoning component. This complements the diverse datasets used for DenoMAE (Chapter 2) and Meta-Task (Chapter 5), collectively demonstrating framework versatility across multiple evaluation scenarios.

We evaluate our context-aware automatic modulation classification framework on a synthetically generated dataset. The dataset comprises signals generated under two distinct conditions: (1) ideal, completely noiseless samples and (2) samples subjected to randomly generated noise with SNRs ranging from -10 dB to 10 dB. Since the ideal, noiseless samples contain no noise, they are significantly easier to visually distinguish and classify. In contrast, as noise levels increase, classification becomes progressively more challenging. Moreover, at lower SNRs, particularly in the negative dB range, the signals become extremely difficult to visually differentiate and accurately classify.

Our dataset comprises signals from 10 diverse modulation schemes (4ASK, 4PAM, 8ASK, 16PAM, CPFSK, DQPSK, GFSK, GMSK, OOK, OQPSK), with SNR values systematically varied from -10dB to 10dB to simulate realistic channel conditions. We evaluate our framework based on classification accuracy across different noise conditions, comparing various prompt engineering techniques and model architectures to assess both performance and generalization capability.

4.4.1 Dataset Generation

We generated our testing dataset to include 2,000 signals featuring various modulations. For the downstream classification task, we used 20 samples as context for both noisy and noiseless data. These samples are evenly distributed among the 10 modulation classes, with SNR values ranging from -10 dB to 10 dB, where applicable. This setup allows us to assess the performance of our framework under ideal conditions as well as in the presence of channel-induced noise.

4.4.2 Experimental Setup

For each data sample, the raw complex-valued signal $x_{i,t}$ is processed to extract its I/Q components, and a comprehensive statistical summary is computed as described in Section 4.3. This summary is then transformed into a natural language prompt using our linearization technique. The prompt, which includes instructions I, context-setting prefix C, the linearized statistical summary, and a directive suffix S, is fed into the pre-trained language model configured with deterministic decoding parameters. The model's output, restricted to a predefined list of modulation types, is compared against the ground truth to compute classification accuracy.

4.4.3 Overall Classification Performance

Table 4.2 reports classification accuracy across prompt contexts, SNR conditions, and model sizes (cleaned accuracy excludes invalid or null responses). Our results highlight the impact of context and scale on AMC performance.

First, incorporating exemplar context into the prompt ($I+C+S$) substantially improves performance compared to using only the instruction and suffix ($I + S$). For example, with the DeepSeek-R1-Distill-Qwen-7B model, the inclusion of exemplar context increases the

cleaned accuracy from 14.35% under noiseless conditions to 54.52%, while under noisy conditions accuracy improves from 13.66% to 27.81%.

Second, the effect of model size is evident. The DeepSeek-R1-Distill-Qwen-32B model, which has significantly more parameters than the 7B variant, achieves a cleaned accuracy of 61.44% (noiseless) and 53.52% (noisy) with the $I + C + S$ prompt, an improvement of over 40% relative to its $I + S$ configuration. This indicates that larger models, with their enhanced representational capacity, can better exploit the rich contextual information provided by exemplar summaries.

The proprietary o3-mini model achieves the highest accuracy (69.92% overall, 72.4% cleaned under noise). Because it is available only through a hosted proprietary API, however, it cannot be deployed on-device and its cost and latency are outside the user's control, motivating future work on distillation and quantization.

Moreover, performance degrades across all models as noise increases, suggesting that explicit SNR cues could further improve robustness.

4.4.4 Validity, Coverage, and the Correct Denominator

Table 4.2 reports two different accuracy columns, and the distinction between them is not cosmetic. For n queries, let n_c be the number of correct predictions and n_v the number of *valid* predictions: outputs that identify one of the predefined modulation labels at all. Define

$$A = \frac{n_c}{n}, \qquad V = \frac{n_v}{n}, \qquad A_v = \frac{n_c}{n_v}, \tag{4.3}$$

the *all-query accuracy*, the *valid-output coverage*, and the *valid-only ("cleaned") accuracy*, respectively. When invalid outputs are never counted correct, these satisfy $A = V \cdot A_v$: all-query accuracy is the product of how often the model returns a usable answer and how often that answer is right. Reporting only A_v conditions away a real failure mode (the

model refusing or failing to produce a parseable label); reporting only A conflates two distinct error sources (wrong decisions versus unusable output). Both columns are necessary to understand a generative classifier's behavior, which is why Table 4.2 keeps them separate rather than reporting a single summary number.

This distinction changes the reading of the 7B noisy-condition result. Valid-only accuracy rises from 13.66% to 27.81% when exemplar context is added, which in isolation reads as a clear improvement. But all-query accuracy over the same comparison *falls*, from 11.8% to 5.2%. Using the identity above, the implied valid-output coverage is

$$V_{\text{no context}} \approx \frac{11.8}{13.66} \approx 0.864, \qquad V_{\text{context}} \approx \frac{5.2}{27.81} \approx 0.187, \tag{4.4}$$

a collapse from roughly 86% to 19% coverage. Adding exemplar context to a small model's prompt here does not reliably improve the system a downstream receiver would actually experience: it makes the model noticeably better *when it answers*, but it answers usably far less often, so a system that must respond to every query would obtain fewer correct, usable decisions after the change, not more. This is a genuine limitation of the small-model, raw-floating-point configuration, not an artifact of how the metric is reported, and it is the central motivation for DiSC-AMC's redesign (Section 4.5): a shorter, more structured prompt is partly a validity intervention, not only an accuracy intervention.

A second caution applies to the o3-mini comparison. Its 69.92% all-query / 72.4% valid-only accuracy is the strongest result in Table 4.2, but the comparison against the 7B and 32B open-weight models is not a controlled test of parameter count alone: model architecture, pretraining data and recipe, inference procedure, and API-level access all differ simultaneously with model identity. The result supports the narrower claim that a sufficiently capable proprietary model can perform well on this task with the PnP prompt; it does not isolate scale as the causal factor, and later results (Table 4.7) show a 7B model

with a better-designed prompt matching this accuracy at a fraction of the token cost, which is itself evidence that prompt construction, not raw scale, is what dominates.

4.5 From Plug-and-Play to DiSC-AMC: The Token-Efficiency Bottleneck

This paper presents a novel in-context approach to automatic modulation classification that takes advantage of higher-order statistics and cumulative estimation through prompt-based classification using a pre-trained large language model. By transforming quantitative signal features into natural language descriptions, our framework effectively captures the intrinsic characteristics of complex-valued signals across diverse scenarios without relying on explicit channel or SNR inputs. Experimental results on synthetically generated datasets demonstrate that incorporating exemplar context into the prompt significantly boosts classification accuracy, and that stronger proprietary models such as o3-mini can further enhance performance, particularly under noisy conditions.

Plug-and-Play establishes feasibility: a pre-trained LLM, given no wireless-specific training, can classify modulation schemes from statistical descriptions alone. However, the proof-of-concept exposes a severe efficiency bottleneck. Serializing raw, high-precision floating-point statistics into the prompt consumes a massive token budget (2.9K tokens per query) and overwhelms small models with numerical noise rather than actionable patterns: the 7B-parameter DeepSeek-R1-Distill-Qwen model reaches only 5.2% accuracy (27.81% cleaned) on noisy data at this prompt length, and competitive accuracy (69.92%) is only achieved by routing through the proprietary o3-mini model. This renders the approach impractical for resource-constrained, real-time, or edge deployment, precisely the deployment regime a cost-effective wireless foundation model must target.

The remainder of this chapter introduces **DiSC-AMC** (Discretized Self-supervised Candidate Retrieval AMC) [12], the dissertation's central answer to RQ2. DiSC-AMC keeps Plug-and-Play's core insight, that statistical signal summaries can be linearized into LLM prompts, but replaces raw floating-point serialization with a token- and parameter-efficient pipeline built on three mechanisms: (i) **feature discretization**, which maps continuous statistics to compact symbolic tokens; (ii) **context retrieval**, which uses a self-supervised visual encoder to select query-specific exemplars rather than a static, generic set; and (iii) **candidate retrieval**, which prunes the label space to a small, plausible multiple-choice set before the LLM ever reasons about the query. Together these mechanisms cut prompt length by over 50% and let a 7B open-weight model match or beat the proprietary o3-mini baseline, while a fine-tuned 7B model matches or surpasses dedicated supervised AMC encoders in- and out-of-distribution.

4.6 DiSC-AMC: Discretized Self-Supervised Candidate Retrieval

4.6.1 Problem Formulation

Let $x \in \mathbb{C}^N$ denote an observed baseband signal comprising N complex I/Q samples received under an unknown modulation scheme $m \in \mathcal{M}$, where $\mathcal{M} = \{m_1, m_2, \ldots, m_C\}$ is the set of C candidate modulation classes. The goal of AMC is to learn a mapping $f : \mathbb{C}^N \to \mathcal{M}$ that assigns x to its true class. DiSC-AMC decomposes f into a chain of transformations,

$$f = f_{\text{LLM}} \circ f_{\text{prompt}} \circ f_{\text{disc}} \circ f_{\text{feat}}, \tag{4.5}$$

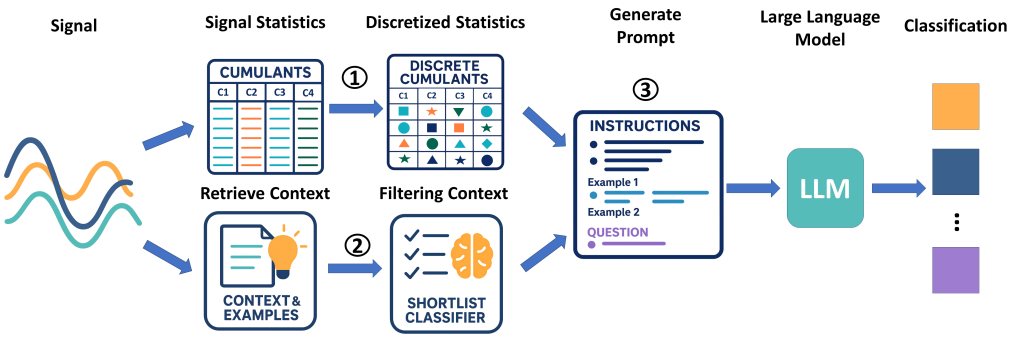


Figure 4.2: Overview of the DiSC-AMC pipeline. (1) Raw I/Q signals are rendered as constellation diagrams and encoded into latent embeddings via a self-supervised visual encoder (DINOv2 ViT-B). (2) The continuous feature vector is quantized into ordinal bins and mapped to compact alphabetic symbols, achieving 3–5$\times$ token compression over raw floating-point serialization. (3) A two-stage retrieval module shortlists the k_{top} most probable classes via centroid or FAISS proximity and retrieves the k_{NN} most similar labeled exemplars, forming a query-specific, pruned context. (4) The discretized query, retrieved exemplars, and candidate options are assembled into a $\sim$0.5K–1.3K token prompt. (5) A QLoRA fine-tuned DeepSeek-R1-Distill-Qwen-7B model performs chain-of-thought reasoning over the symbolic features before predicting the modulation class.

where each component (feature extraction, discretization, prompt assembly, and LLM classification) is detailed below and illustrated in Figure 4.2.

4.6.2 Stage 1: Signal Representation and Feature Extraction

DiSC-AMC supports two parallel feature extraction pathways. From the raw I/Q signal x, a *statistical feature* vector is computed,

$$s = [\hat{\mu}_2, \ldots, \hat{\mu}_{10},\ \hat{\kappa}_1, \ldots, \hat{\kappa}_4,\ \hat{\gamma}_1, \hat{\gamma}_2,\ \hat{\sigma}_{\kappa_1}, \hat{\sigma}_{\kappa_2},\ \text{SNR}]^\top \in \mathbb{R}^{D_s}, \tag{4.6}$$

where $\hat{\mu}_k$ are sample central moments of order k, $\hat{\kappa}_j$ are k-statistics (minimum-variance unbiased cumulant estimators), $\hat{\gamma}_1, \hat{\gamma}_2$ are skewness and kurtosis, and SNR is the signal-to-noise ratio label provided with each sample (the generation SNR, not an estimate from the

signal). Apart from this SNR label, which Plug-and-Play did not use, this is the statistical summary used earlier in this chapter (Section 4.2), with higher-order cumulants providing theoretical separation among BPSK, QPSK, and QAM families.

Alternatively, the signal is rendered as a constellation-diagram image and passed through a self-supervised visual *encoder embedding* pathway: $z = g_\theta(\mathrm{img}(x)) \in \mathbb{R}^{D_z}$, reduced via PCA to $D_e \ll D_z$ components, $e = \mathrm{PCA}_{D_e}(z) \in \mathbb{R}^{D_e}$. This produces a compact, model-agnostic feature vector capturing learned visual patterns from the constellation geometry, the same DenoMAE-style representation philosophy developed in Chapter 2, here instantiated with a DINOv2 ViT-B encoder (frozen or fine-tuned depending on the scenario). Either feature vector (s or e) is standardized via a `StandardScaler` fit on the training set.

4.6.3 Stage 2: Discretization Mechanism

The central innovation of DiSC-AMC maps any numerical signal representation to compact symbolic tokens, normalizing feature scales and compelling the model to focus on qualitative patterns rather than decimal precision. Given a feature vector $v \in \mathbb{R}^D$ (either s or e), each dimension v_j is independently quantized into one of B ordinal bins via a `KBinsDiscretizer` with uniform strategy,

$$q_j = \left\lfloor \frac{v_j - v_j^{\min}}{v_j^{\max} - v_j^{\min}} \cdot B \right\rfloor, \qquad q_j \in \{0, 1, \ldots, B-1\}, \tag{4.7}$$

with bin edges fit on training data and applied deterministically at test time. Each integer bin index q_j is encoded as a base-26 alphabetic symbol, $\phi(q_j) = \mathrm{base26}(q_j) \in \{A, B, \ldots, Z, AA, \ldots\}$, so the full signal representation becomes a sequence of symbolic tokens,

$$f_{\mathrm{disc}}(v) = [\mathrm{feat}_1 : \phi(q_1),\ \mathrm{feat}_2 : \phi(q_2),\ \ldots,\ \mathrm{feat}_D : \phi(q_D)]\,. \tag{4.8}$$

Discretization offers three benefits: (i) it compresses each feature from a multi-digit floating-point string to one or two characters, cutting prompt token count by roughly 3–$5\times$; (ii) it abstracts away measurement noise, making the representation robust to small perturbations; and (iii) ordinal letter codes align naturally with the symbolic reasoning strengths of LLMs, rather than their comparatively weak numerical precision.

4.6.4 Stage 3: Dynamic Context Construction

Rather than a large, fixed exemplar set as in Plug-and-Play, DiSC-AMC builds a compact, query-specific context via two retrieval mechanisms, both driven by the self-supervised visual encoder.

Context (few-shot) retrieval. A FAISS index [63] is built over the scaled feature vectors of all training signals. At inference, for a query feature vector v_q, the k_{NN} nearest training signals are retrieved via inverse-squared-distance weighting, grouped by class label, and added to the prompt as signal-specific, diverse few-shot demonstrations. When a minimum-class-diversity constraint is specified, the search radius expands (up to $3k_{\mathrm{NN}}$ candidates) with per-class brute-force scans to fill any missing classes.

Candidate retrieval. To further constrain the LLM's decision space, the full label set $\mathcal{M}$ is narrowed to the k_{top} most plausible classes using the same encoder, via one of four interchangeable strategies: (a) *Centroid*: Euclidean distance from z to per-class prototype centroids; (b) *FAISS kNN*: inverse-distance-weighted voting among nearest neighbors; (c) *DNN head*: softmax probabilities from a trained classification head; (d) *Random Forest*: class vote counts from an RF classifier on encoder features. Applied to constellation images of the 10-class synthetic benchmark, candidate retrieval at $k_{\mathrm{top}} = 5$ achieves **99.83% recall** (the correct class is included in the shortlist in nearly every query), reducing the LLM's task from C-way to k_{top}-way classification and preventing hallucinated out-of-vocabulary

labels. Shortlist recall is an upper bound on final accuracy, not an accuracy figure itself: writing $R = \Pr(y \in \mathcal{C}(x))$ for recall, final accuracy satisfies $A = R \cdot \Pr(\hat{y} = y \mid y \in \mathcal{C}(x)) \leq R$. A 99.83% shortlist recall therefore only bounds the opportunity available to the downstream LLM decision; it does not by itself guarantee, or report, how often that decision is correct.

4.6.5 Stage 4: LLM Inference and Prompt Design

The final prompt assembles three components: (1) an *instruction template* (role description and response-format guide, optionally annotated with the candidate-retrieval method and feature type); (2) *few-shot context* (the dynamically retrieved exemplars, each a discretized feature string paired with its ground-truth label); and (3) the *query* (the discretized feature string of the test signal, followed by its candidate class options). The LLM is instructed to reason step-by-step inside `<think>` tags before outputting a single label, encouraging chain-of-thought deliberation over the symbolic features rather than the raw query.

4.6.6 Experimental Setup

Datasets. *SynthMod-10* follows the Plug-and-Play generation protocol (Section 4.4): 10 digital modulation classes (4ASK, 4PAM, 8ASK, 16PAM, CPFSK, DQPSK, GFSK, GMSK, OOK, OQPSK), 10,000 training signals, 200 in-distribution test queries (-10 to $+10$ dB), and 200 out-of-distribution (OOD) queries (-15 to -11 dB, never seen in training). *RadioML 2018.01a* [64] is a 24-class benchmark with real channel impairments (Rician fading, AWGN, frequency/phase drift), entirely disjoint from SynthMod-10, evaluated at five SNR levels ($-20, -10, 0, +10, +20$ dB) with 2,400 training and 240 test samples per SNR level.

Baselines. Training-free baselines use the Plug-and-Play prompt format on DeepSeek-R1-Distill-Qwen (7B/32B); supervised/OOD baselines use DenoMAE [26], DenoMAE2.0 [1], and NMformer [59] as fixed, non-LLM classification heads.

Default configuration. Unless noted otherwise, results use DeepSeek-R1-Distill-Qwen-7B fine-tuned via QLoRA on the training set, 5 discretization bins, FAISS $k_{\mathrm{NN}} = 10$ context retrieval, and $k_{\mathrm{top}} = 5$ candidate retrieval (FAISS voting over the $k = 50$ nearest training signals) over DINOv2 encoder embeddings.

Evaluation protocol. Five scenarios isolate specific capabilities: (I) in-distribution classification on SynthMod-10; (II) near-OOD robustness on SynthMod-10; (III) cross-dataset transfer to RadioML with a fine-tuned encoder; (IV) cross-dataset transfer to RadioML with an unadapted encoder; (V) token and parameter efficiency. Ablations then dissect pipeline architecture choices and prompt-design tradeoffs.

4.6.7 Results

Scenario I: In-distribution classification. Table 4.3 compares DiSC-AMC against supervised baselines on SynthMod-10. DiSC-AMC, using a 7B open-weight LLM with FAISS-retrieved exemplars over DINOv2 embeddings, achieves **83.00%**, numerically above every supervised baseline including DenoMAE2.0 (82.40%, the strongest encoder-only method from Chapter 2). This margin should be read descriptively rather than as an established statistical superiority: the test set is 200 queries, a binomial 95% confidence half-width at this accuracy and sample size is approximately 5 points, and no repeated-seed or matched-split uncertainty estimate was computed for either row. The 0.6-point gap over DenoMAE2.0 is well within that uncertainty band; the result supports "DiSC-AMC is competitive with the strongest supervised baseline," not "DiSC-AMC is proven superior to it."

Scenario II: Near-OOD robustness. Shifting the test set to an unseen SNR range (-15

to -11 dB) outside the training distribution, DiSC-AMC maintains **82.50%** OOD accuracy (Table 4.4), a drop of only 0.5 points from in-distribution, far exceeding standalone encoder baselines.

Scenarios III & IV: Cross-dataset transfer to RadioML. These scenarios test whether DiSC-AMC's reasoning amplifies an encoder backbone's own classification head when transferred to the entirely disjoint 24-class RadioML benchmark, with the encoder either fine-tuned (III) or left unadapted (IV). In both scenarios, the classification head and, in Scenario III, the encoder backbone are trained on RadioML's own 12,000 labeled training samples (Section 4.6.6); this is *supervised target-domain adaptation*, not label-free transfer: RadioML labels are used in both scenarios, and the distinction between them is whether the encoder's weights are updated (III) or held fixed while only a head is fit (IV), not whether target labels are used at all. With this scoping, Table 4.5 shows that with a fine-tuned DINOv2 backbone, DiSC-AMC reaches **76.25%** at $+20$ dB versus just 9.58% for the fine-tuned encoder's own head, an **8**$\times$ improvement. Table 4.6 shows the same effect holds with the encoder weights left unadapted (only the head is trained on RadioML labels): DiSC-AMC with an unadapted DINOv2 reaches 45.42% at $+20$ dB versus 7.08% for the head alone. In both scenarios, standalone encoder heads plateau below 15% across all SNR levels, confirming that reasoning, not the encoder alone, drives cross-dataset generalization. Notably, DenoMAE2.0 as the DiSC-AMC backbone *underperforms* DINOv2 on RadioML at higher SNRs (Table 4.5, $+10$/$+20$ dB), indicating that DenoMAE2.0's synthetic-data-trained features do not retrieve informative neighbors on this out-of-family dataset even though they excel in-distribution (Scenario I).

Scenario V: Token and parameter efficiency. Table 4.7 isolates the cost of reasoning. The Plug-and-Play baseline requires a 2.9K-token prompt and reaches only 5.20% on a 7B model; even the proprietary o3-mini reaches 69.92%. DiSC-AMC's training-free 7B model, without retrieved context exemplars, reaches **71.9%** valid-output accuracy (143

of 199 valid answers; 71.5% over all 200 queries) at just **0.5K tokens**, surpassing the proprietary o3-mini baseline at roughly one-sixth the prompt length.

4.6.8 Ablation Studies

Using DeepSeek-R1-Distill-Qwen-7B and GLM-4.6V-Flash in a training-free setting, four independent design axes are varied: candidate-retrieval strategy, feature representation, context retrieval (RAG), and distribution shift.

Candidate retrieval: FAISS vs. centroid. FAISS kNN retrieval substantially outperforms centroid-based selection: 66.1% vs. 32.8% in-distribution, and 62.6% vs. 42.5% under moderate OOD shift. Centroid-selected exemplars cluster around class means and lack query-specific diversity; a key factor is *SNR matching*, because signal statistics vary substantially with SNR, exemplars from a mismatched noise level confuse rather than guide the LLM, and FAISS retrieval naturally preserves SNR proximity where centroid selection does not.

Feature representation: embeddings vs. statistics. Replacing discretized statistics with self-supervised encoder embeddings as the retrieval feature consistently improves accuracy: the best embedding-based configuration reaches 71.5% in-distribution versus 69.0% for the best statistics-only configuration; the encoder captures visual invariances of constellation diagrams that cumulant statistics do not.

Context retrieval (RAG) vs. fixed exemplars. RAG-augmented and fixed-exemplar pipelines achieve comparable mean accuracy in-distribution (47.2% vs. 51.8%); RAG's dynamic retrieval catches up to the fixed set at 52.6% under OOD shift, indicating that *which* exemplars are retrieved (FAISS vs. centroid) matters far more than *how many* are added per query.

Out-of-distribution generalization and the noise floor. Accuracy degrades gracefully

under moderate OOD shift (-11 to -15 dB): the best FAISS configuration retains 68.5%, only 3 points below its 71.5% in-distribution score. Under extreme shift (-30 dB), however, all configurations collapse to near-random levels ($\sim$5–10%). At this noise level constellation diagrams become effectively featureless, so the self-supervised encoder's embeddings lose discriminative structure and both retrieval and classification degrade to chance. This is a fundamental limitation of any retrieval pipeline built on visual signal representations, and directly echoes the low-SNR collapse observed for DenoMAE in Chapter 2.

Prompt design tradeoffs. Table 4.8 isolates prompt format in the training-free setting. The 7B DeepSeek model under the raw Plug-and-Play prompt achieves only 9%, confirming that floating-point serialization overwhelms small models; Gemini-2.5-Flash with a 1.3K-token DiSC-AMC prompt reaches 45.5%, exceeding a 32B DeepSeek PnP + candidates baseline (32.5%) at under half the token cost, and Gemini-2.5-Pro at 0.9K tokens reaches 51.0%, the best configuration in this table. Increasing the candidate-set size k_{top} from 4 to 5 yields a marginal gain (44.5%$\rightarrow$45.5%) for a small token cost; expanding to $k_{\text{top}} = 10$ (10 bins) expands the prompt to 2.9K tokens while accuracy *drops* to 29.5%, since candidate-retrieval recall already reaches 99.83% at $k_{\text{top}} = 5$. Discretization granularity shows a similar "less is more" pattern: Gemini-2.5-Flash peaks at 5 bins (45.5%) and degrades monotonically with finer bins, while the more capable Gemini-2.5-Pro peaks later, at 10 bins (47.5%), before declining. Coarse symbolic codes align better with LLM reasoning over noisy physical-layer signals than high-precision numerical detail, for most models.

4.6.9 RQ1 Meets RQ2: Does the Encoder Matter?

Scenarios III and IV (Tables 4.5–4.6) are deliberately designed to probe the seam between RQ1 (representation) and RQ2 (reasoning): DiSC-AMC's retrieval stage depends entirely

on an RQ1 encoder, so its RadioML performance is a direct test of whether reasoning can compensate for representation quality, or whether it is bottlenecked by it.

With a *fine-tuned* DINOv2 backbone, DiSC-AMC reaches 76.3% at $+20$ dB versus 9.6% for the DINOv2 head alone, a $\mathbf{8}\times$ improvement from reasoning over the same features. Strikingly, this amplification holds even with an *unadapted* DINOv2 encoder, where only the classification head was trained: DiSC-AMC still reaches 45.4% versus 7.1% for the head alone, a $\mathbf{6}\times$ gain, showing that reasoning amplifies a general-purpose visual encoder without any target-domain fine-tuning, provided its embeddings carry some class-relevant structure.

The counter-case is equally informative. With DenoMAE2.0 as the backbone, the stronger encoder on SynthMod-10 (82.4% vs. DINOv2's 65.3%, Table 4.3), DiSC-AMC plateaus at or below 25% across every SNR on RadioML (Table 4.5), and in the unadapted setting its own head alone *beats* DiSC-AMC at $+10$ and $+20$ dB (Table 4.6): reasoning provides no gain, and can even be outperformed by the plain head. The difference is not reasoning capacity but retrieval quality: DenoMAE2.0 was trained exclusively on the synthetic SynthMod-10 distribution, so its nearest neighbors on the out-of-family RadioML dataset are not meaningfully related to the query's true class, and the LLM is reasoning over uninformative context. DINOv2, pretrained on broad natural-image self-supervision rather than narrow synthetic constellations, retrieves more generically relevant neighbors even without wireless-specific adaptation.

Difficult finding: reasoning only helps when the encoder's nearest neighbors are meaningful. This is the dissertation's clearest demonstration that RQ2's gains are conditional on RQ1: DiSC-AMC is not an encoder-agnostic amplifier, but a mechanism that converts representation quality into classification accuracy, and inherits representation failures just as readily as it inherits representation strengths.

4.6.10 Testing a Non-Generative Classifier: Jev

The DiSC-AMC pipeline was designed around chain-of-thought generation: the LLM reasons inside `<think>` tags before committing to a label. A natural question is whether the same discretized, retrieval-augmented prompt transfers to a fundamentally different decision mechanism: a non-generative classifier that scores a fixed set of options rather than generating free text. We tested this using **Jev**, which consumes three inputs: a *state* (the full DiSC-AMC prompt: role, binned statistics, retrieved exemplars, and the query), a set of *options* (the k_{top} shortlisted candidate classes), and *criteria* (one fixed natural-language description per option, e.g., for 16QAM: "16 symbols on a 4×4 square grid, varying both amplitude and phase"); it returns a single selected option with a probability, with no reasoning trace.

Table 4.9 compares Jev against the paper's generative LLMs (Gemini-2.5-Pro on the Table 4.8 prompt; DeepSeek-R1-Distill-Qwen-7B on RadioML) under three identical shortlist conditions.

On the centroid-shortlist condition, Jev modestly *outperforms* Gemini-2.5-Pro on the identical Table 4.8 prompt (56.5% tailored vs. 51.0%, a $+5.5$-point gain), showing the DiSC-AMC prompt format transfers to a non-generative scoring mechanism. On the FAISS-shortlist condition the two are closer (65.0% vs. 71.5%). But on RadioML, where 13 of the 24 classes differ only in modulation *order* (e.g., 16/32/64QAM, several APSK variants), Jev collapses to 32.5–33.3% while the reasoning LLM reaches 76.3%, more than double. **Finding:** the DiSC prompt transfers to a non-generative model for coarse class discrimination, but fine-grained order discrimination, which requires reasoning about relative structure rather than matching a fixed description, needs a model that can actually reason over the symbolic evidence rather than score against static criteria.

This comparison, however, is not a clean controlled experiment, and the limitations are

reported explicitly rather than left implicit in the headline numbers:

- **Prompt not designed for Jev:** tailoring only removed the `<think>`-tag and output-format instructions meant for generative decoding; no Jev-specific prompt was searched for.
- **Criteria cannot carry query-specific evidence:** a fixed per-class blurb (e.g., restating that 16QAM, 32QAM, and 64QAM differ only in symbol count) cannot encode the query's actual discretized statistics, which live only in the state.
- **Default runs had empty criteria:** a configuration bug sent null descriptions in the untailored condition, so those runs saw class names only, not even the static blurbs.
- **Two changes confounded at once:** the tailored condition simultaneously strips the generative-decoding rules and adds criteria, so their separate contributions are not isolated.
- **Small, single runs:** each condition used 200–240 queries, one trial, one model version; tailoring moved results by -1.5 to $+3.0$ points (roughly 6 queries), which is within the noise floor of a single run at this sample size.
- **Options are pre-filtered by the shortlist:** a class the candidate-retrieval shortlist drops is never available to Jev; 31% of plain-FAISS-shortlisted queries had only one option, trivializing the decision for those queries.

Takeaway: the DiSC-AMC prompt format is not tied to generative decoding, but a fair comparison between reasoning and non-reasoning classifiers on this task requires query-specific criteria and repeated runs at larger sample sizes; this is work left open in Section 6.7.

4.6.11 Qualitative Reasoning: Grounded Inference and Confident Confabulation

Aggregate accuracy numbers obscure *how* the fine-tuned model reaches its answers. Inspecting individual chain-of-thought traces reveals that the 83% fine-tuned model's correct and incorrect predictions differ qualitatively, not just in outcome.

On a correctly classified 8ASK query (true label 8ASK, options {8ASK, 4PAM, 4ASK}, 5.73 dB), the model grounds its answer in a specific, checkable feature-to-hypothesis link: "The feature pc_0 is E, which is the highest bin. The rest of the features (pc_1 to pc_9) are mostly C and D. [...] For 8ASK, we'd expect a wider range of amplitude values, leading to more varied bins. [...] For 4PAM, there are only four amplitude levels, so the features should reflect a more limited range." This reasoning ties the spread of the binned features directly to the number of amplitude levels each candidate modulation implies, a correct and verifiable inference.

On an incorrectly classified query (true label CPFSK, predicted GFSK, options {GFSK, CPFSK}, -2.70 dB), the model's own trace admits the features are uninformative, then answers anyway: "GFSK (Gaussian Filtered-Smoothed FM) [...] CPFSK (Chirp-Frequency-Smoothed Keyed FM) [...] I notice that the features don't strongly indicate a particular scheme. [...] Given the options and the features, the most probable classification is GFSK." Both expansions are also wrong (GFSK is Gaussian *Frequency* Shift Keying; CPFSK is *Continuous*-Phase FSK), suggesting the model's uncertainty is genuine but its explanation does not surface that uncertainty to a reader; it produces a confident-sounding rationale around what amounts to a guess. **Finding:** even the strongest (83%) configuration fails on the hardest class pair in the benchmark, two continuous-phase FSK variants that are physically similar, and its failure mode is not an obviously garbled trace but a fluent one built on an incorrect premise.

Comparing the fine-tuned (83.0%) and training-free (71.0%) configurations on identical retrieval (same FAISS shortlist, same DINOv2 embeddings, same 10 retrieved exemplars, same 5 bins, differing only in whether QLoRA fine-tuning was applied; 166 vs. 142 of 200 correct) shows the same pattern at smaller scale. A correct training-free prediction (true and predicted GFSK, 1.99 dB) reasons from one weak but ultimately correct heuristic after dismissing the rest: "The other features don't clearly point towards one over the other, but the SNR is a strong indicator. Therefore, I'm leaning towards GFSK." An incorrect training-free prediction (true GFSK, predicted CPFSK, -6.85 dB) again builds a plausible-sounding cumulant argument on an incorrect premise: "GFSK is Gaussian Filtered SSB Modulation. [...] I'm leaning towards CPFSK because I remember that CPFSK signals have non-zero even-order cumulants". In fact, GFSK is frequency-shift keying, not single-sideband modulation. Fine-tuning does not change the *style* of reasoning observed in these traces; it changes how often the underlying domain knowledge the model draws on is correct, which is consistent with QLoRA correcting factual associations (e.g., modulation-family definitions) rather than altering the reasoning strategy itself.

These qualitative traces temper the chapter's quantitative claims: DiSC-AMC's interpretability advantage over end-to-end classifiers is real: every prediction ships with an inspectable rationale. But that rationale is not a reliable confidence signal. A fluent explanation can accompany a wrong answer as readily as a right one, which matters for any deployment that would use the chain-of-thought text itself (rather than just the final label) to decide how much to trust a given classification.

4.6.12 Discussion: Answering RQ2

DiSC-AMC provides the dissertation's strongest evidence for Research Question 2 (RQ2): *Can RF statistics and relevant examples be given to an LLM efficiently enough for useful,*

open-set AMC? Two complementary results support a confident **yes, within bounds**. First, a QLoRA fine-tuned 7B model reaches 83.00% in-distribution and 82.50% OOD accuracy on SynthMod-10, matching or surpassing every supervised baseline evaluated, including DenoMAE2.0, demonstrating that symbolic, retrieval-augmented reasoning is not merely a cheaper substitute for supervised classification but can be competitive with it. Second, a training-free 7B model reaches 71.9% accuracy at 0.5K tokens, exceeding the proprietary o3-mini model given 2.9K tokens of raw statistics (69.92%), demonstrating that *prompt construction, not model scale, is the dominant driver of LLM-based AMC accuracy*.

The bounds on this answer come from three directions, each explored in the preceding subsections. On cross-dataset transfer to RadioML, DiSC-AMC amplifies a frozen or fine-tuned encoder's own head by up to $8\times$ (Section 4.6.9), but only when the encoder's nearest neighbors remain informative: DenoMAE2.0's synthetic-trained embeddings, which excel in-distribution, retrieve poorly on RadioML, so the composition's ceiling is ultimately set by the quality of the RQ1 representation it retrieves over (Chapter 2, Chapter 3): reasoning cannot manufacture class signal an encoder's embeddings do not carry. On decision mechanism, the DiSC-AMC prompt transfers to a non-generative classifier (Jev) for coarse, well-separated classes, but a reasoning LLM is needed for fine-grained order discrimination on RadioML's APSK/QAM families (Section 4.6.10). On interpretability, qualitative inspection (Section 4.6.11) shows that chain-of-thought explanations are fluent regardless of correctness, so the rationale text is not itself a reliable confidence signal, even from the best-performing configuration. The one hard noise limit is physical rather than algorithmic: below roughly -30 dB, constellation-based features of any kind collapse to chance, for both retrieval and final classification.

Connection to the Dissertation Framework: Together, Plug-and-Play and DiSC-AMC demonstrate that the *intelligent reasoning engine* of this dissertation's modular framework can be built economically: training-free when deployment cost dominates, or lightly

fine-tuned when accuracy must match dedicated supervised encoders, in both cases at a small fraction of the parameter count of frontier LLMs. This validates the dissertation's central thesis that wireless foundation-model capabilities need not be built through massive isolated training, but can be assembled by strategically integrating pre-trained vision (RQ1, Chapters 2–3) and language (RQ2) world knowledge. DiSC-AMC's retrieval stage is itself a direct composition of RQ1 and RQ2: it is the self-supervised encoder from Chapter 2 that supplies the exemplars the LLM reasons over, making DiSC-AMC the dissertation's concrete demonstration that the framework's components compose rather than merely coexist. Section 5.5 (Chapter 5) returns to this composition from the RQ3 (adaptation) side, testing whether Meta-Task's learnable regularization can further improve the DiSC-AMC pipeline on RadioML.

Overall, this chapter represents a substantive step toward practical, open-set wireless reasoning with LLMs. The transition from Plug-and-Play to DiSC-AMC shows concretely how an initial proof-of-concept, once its cost bottleneck is diagnosed, can be redesigned into a deployment-realistic pipeline, while being explicit about where that pipeline's limits still lie.

Instruction Block	
Prompt Instructions	You are an expert quantitative analyst in wireless communication modulation. Based on your knowledge in wireless communication modulation and the detailed signal statistics provided below, determine the modulation type. 1. DO NOT PROVIDE ANY ADDITIONAL INFORMATION OR CONTEXT. 2. No OTHER TEXT, NO BLABBER.
Exemplar Summaries	
Example 1	{nobs: 1024, min: -7.3, max: 7.5, mean: -0.0, ... }, Answer 1: 4ASK
Example 2	{nobs: 1024, min: -6.6, max: 7.5, mean: -0.1, ... }, Answer 2: 4PAM
Example 3	{nobs: 1024, min: -19.8, max: 15.2, mean: 0.2, ... }, Answer 3: 8ASK
... (Additional examples not shown)	
Query Section	
Question Sample	{nobs: 1024, min: -33.502, max: 32.946, mean: -0.629, ... }
Answer Choices	`4ASK, 4PAM, 8ASK, ...`

Table 4.1: Prompt Structure for Modulation Classification: Instruction Block, Exemplar Summaries, and Query Section.

Table 4.2: Modulation Classification Performance Across Models, Prompt Contexts, and Model Sizes

Model	Parameters	Prompt	SNR (dB)	Accuracy (%)	Cleaned Accuracy (%)
Nmformer[59]	-	-	[0.5 4.5]	71.6	-
DenoMAE[26]	-	-	[0 10]	81.3	-
DenoMAE2.0[1]	-	-	[0 10]	**82.4**	-
DeepSeek-R1-Distill-Qwen	7B	$I + S$	100	12.6	14.35
DeepSeek-R1-Distill-Qwen	7B	$I + S$	[-10 10]	11.8	13.66
DeepSeek-R1-Distill-Qwen	7B	$I + C + S$	100	19.3	54.52
DeepSeek-R1-Distill-Qwen	7B	$I + C + S$	[-10 10]	5.2	27.81
DeepSeek-R1-Distill-Qwen	32B	$I + S$	100	9.1	10.83
DeepSeek-R1-Distill-Qwen	32B	$I + S$	[-10 10]	7.9	9.58
DeepSeek-R1-Distill-Qwen	32B	$I + C + S$	100	**58.8**	**61.44**
DeepSeek-R1-Distill-Qwen	32B	$I + C + S$	[-10 10]	**47.8**	**53.52**
OpenAI's o3-mini	-	$I + C + S$	[-10 10]	**69.92**	**72.4**

$I + S$ refers to the scenario where the language model receives only a concise block of instructions to select the modulation type from a predefined list without any additional exemplar context. In contrast, *Example* $I + C + S$ incorporates exemplar summaries derived from known modulation signals into the prompt. *Cleaned Accuracy* refers to the accuracy obtained after removing responses that did not identify any modulations from the predictions. Parameter counts of proprietary models are not publicly disclosed.

Table 4.3: Scenario I: In-distribution classification on SynthMod-10.

Model	Params	Acc. (%)
DINOv2	86M	65.33
NMformer [59]	86M	71.60
ViT [1]	86M	79.90
DEiT [1]	86M	81.20
MoCov3 [1]	86M	81.00
BEiT [1]	86M	80.40
MAE [1]	86M	80.10
DenoMAE [26]	86M	81.30
DenoMAE2.0 [1]	86M	82.40
DiSC-AMC (ours)	**7B**	**83.00**

Table 4.4: Scenario II: Near-OOD robustness on SynthMod-10.

Model	Params	OOD Acc. (%)
DINOv2	86M	63.90
DenoMAE2.0 [1]	86M	63.25
DiSC-AMC (ours)	**7B**	**82.50**

Table 4.5: Scenario III: Cross-dataset transfer with a fine-tuned encoder (RadioML 2018.01a, 24 classes).

	Fine-tuned encoder (%)		DiSC-AMC (%)	
SNR (dB)	DINOv2	DenoMAE2.0	DINOv2	DenoMAE2.0
-20	3.33	4.17	19.58	17.92
-10	3.33	5.42	17.50	19.17
0	6.67	8.75	40.83	25.00
$+10$	8.75	13.33	72.50	24.17
$+20$	9.58	11.25	**76.25**	21.25

Table 4.6: Scenario IV: Cross-dataset transfer with an unadapted encoder (RadioML 2018.01a, 24 classes).

	Encoder only (%)		DiSC-AMC (%)	
SNR (dB)	DINOv2	DenoMAE2.0	DINOv2	DenoMAE2.0
-20	3.75	3.33	10.83	15.42
-10	5.00	2.92	11.67	5.83
0	6.67	9.58	28.33	12.50
$+10$	15.00	14.58	36.25	5.83
$+20$	7.08	12.92	**45.42**	6.25

Table 4.7: Scenario V: Token and parameter efficiency. Parameter counts of proprietary models are not publicly disclosed.

Model	Params	#Tokens	Acc. (%)
DeepSeek-R1 (PnP)	7B	2.9K	5.20
DeepSeek-R1 (PnP)	32B	2.9K	47.80
o3-mini (PnP)	–	2.9K	69.92
DeepSeek-R1 (DiSC-AMC, training-free)	**7B**	**0.5K**	**71.9**

Table 4.8: Effect of prompt format on training-free LLMs (centroid candidate retrieval, no RAG). Parameter counts of proprietary models are not publicly disclosed.

Method	Model	Params	#Tokens	Acc. (%)
PnP	DeepSeek-R1	7B	2.9K	5.20
PnP	DeepSeek-R1	32B	2.9K	47.80
PnP	o3-mini	–	2.9K	69.92
PnP + candidates	DeepSeek-R1	7B	2.9K	9.00
PnP + candidates	DeepSeek-R1	32B	2.9K	32.50
PnP + candidates	Gemini-2.5-Flash	–	2.9K	29.50
PnP + candidates	Gemini-2.5-Pro	–	2.9K	42.50
DiSC-AMC	DeepSeek-R1	7B	1.3K	33.50
DiSC-AMC	DeepSeek-R1	32B	1.3K	39.00
DiSC-AMC	Gemini-2.5-Flash	–	1.3K	45.50
DiSC-AMC	Gemini-2.5-Pro	–	0.9K	**51.00**

Table 4.9: Jev vs. the paper's generative LLM on identical shortlists and candidate sets.

Condition	Paper LLM	Jev (default)	Jev (tailored)
Centroid shortlist, 10 classes	51.0%	53.5%	56.5%
FAISS shortlist, 10 classes	71.5%	65.0%	65.0%
RadioML $+20$ dB, 24 classes	76.3%	33.3%	32.5%

Chapter 5

Meta-Task: A Method-Agnostic Framework for Learning to Regularize in Few-Shot Learning

5.1 Introduction

Few-Shot Learning (FSL) has emerged as a promising solution to the problem of data scarcity in machine learning, particularly in computer vision, medical applications, and personalized health systems[65, 66, 67]. By leveraging meta-learning, FSL enables models to learn how to learn[68] with the primary objective of extracting meaningful, transferable representations from sparse datasets, reducing the need for extensive labeled data, and significantly lowering data collection and annotation costs [69, 70, 71]. In wireless communications, where foundation models must rapidly adapt to diverse modulation schemes and channel conditions, FSL offers a practical solution. Rather than exhaustive retraining for each deployment scenario, meta-learning enables adaptation with minimal labeled data,

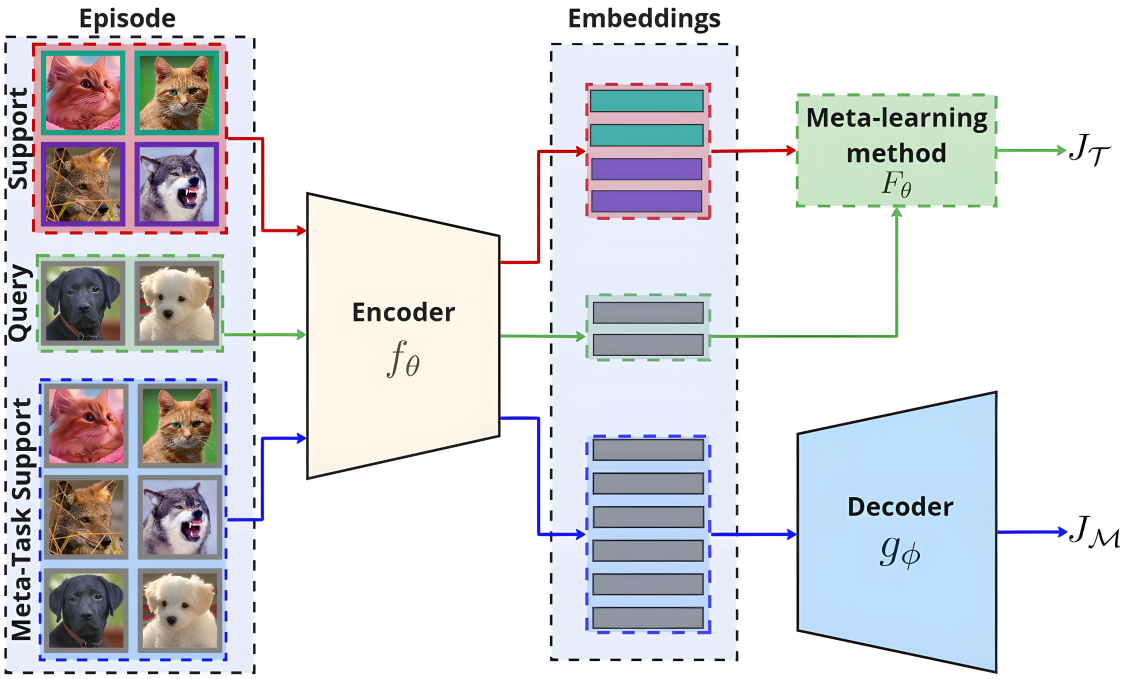


Figure 5.1: Detailed architecture of the Task-Decoder within the Meta-Task framework. The Meta-Task support set consists of unlabeled copies of support and query samples, which are fed into an encoder f_θ to produce feature embeddings. These embeddings serve two key functions: (1) they are passed to a decoder g_ϕ to reconstruct the input images, aiding in the learning of rich, meaningful representations, and (2) they are used to generate the meta-learning method (F_θ) for query image classification. The model optimizes two objectives: classification loss ($J_\mathcal{T}$) and reconstruction loss ($J_\mathcal{M}$), each with a dedicated backpropagation step. The overall training loss is $J_\mathcal{T} + \lambda J_\mathcal{M}$, balancing both classification accuracy and representation learning. For a broader application, the decoder g_ϕ could be replaced by one or more neural networks that solve various unsupervised tasks.

a critical capability for cost-effective foundation model deployment.

Advances in meta-learning have led to techniques such as Matching Networks [72], Relation Networks [73], Model-Agnostic Meta-Learning (MAML)[74], Prototypical Networks[75], and Reptile [76]. Despite performance gains, FSL models remain prone to overfitting, limiting generalization.

Regularization plays a critical role in mitigating overfitting and enhancing generalization in FSL. Conventional methods such as weight decay[77], dropout[78], and noise injection[79, 80, 81] fail to address FSL-specific challenges. Meta-learning requires both adaptation-generalization and meta-generalization[82, 83, 84, 85, 86, 87, 88], yet existing techniques remain static, inflexible, and reliant on manually designed loss functions or auxiliary tasks that do not adapt to the unique structure of each task. This limitation motivates the need for a more dynamic, learnable regularization mechanism.

To address this limitation, we introduce Meta-Task, a method-agnostic framework that reformulates regularization as a learnable task within the FSL optimization process. Meta-Task dynamically learns task-specific regularization strategies through a parameterized neural layer ϕ. This adaptive mechanism enables our framework to seamlessly integrate with existing FSL pipelines, ensuring flexibility across different task distributions and learning paradigms. Meta-Task remains compatible with existing auxiliary tasks and regularization losses, leveraging prior techniques while introducing a dynamic, learnable component. This enhances overfitting mitigation without extensive hyperparameter tuning or task-specific modifications, making it scalable and practical for real-world FSL applications.

To operationalize Meta-Task regularization, we propose using an autoencoder-based Task-Decoder (Figure 5.1). The Task-Decoder serves as a flexible and effective regularizer, refining task-specific representations by constraining the embedding vector to reconstruct the original input accurately. We choose autoencoders because they are data-agnostic, easy to implement, architecturally flexible, and allow tunable parameterization for optimization. This adaptability makes them well-suited for various FSL applications, as they can seamlessly integrate into different model architectures.

Our method leverages the latent variable z, inherently present in most FSL settings due to the N-Way, K-Shot paradigm. The standard inner-outer loop optimization structure

ensures the existence of an intermediate latent representation within the support-query split or gradient-based adaptation process. This guarantees compatibility across various FSL architectures, allowing Meta-Task to enhance regularization effectiveness dynamically.

The main contributions of this paper are as follows. (1) Meta-Task Framework: We propose Meta-Task, a novel, method-agnostic framework that introduces auxiliary tasks to regularize individual tasks in FSL. (2) Task-Decoder (Figure 5.1): We introduce the Task-Decoder, utilizing autoencoders as a new regularization task for meta-learners, leveraging unsupervised learning to refine hidden representations. (3) Improved Performance: Our method demonstrates faster convergence, higher accuracy, and better generalization than existing techniques, all without extensive hyperparameter tuning. (4) Versatility and Applicability: The framework is method-agnostic and can be integrated with various FSL settings, making it suitable for real-world, resource-constrained scenarios.

Framework Position: Within the dissertation's modular framework, Meta-Task provides the rapid adaptation mechanism essential for wireless foundation models. While DenoMAE (Chapter 2) learns robust representations and the Plug-and-Play LLM (Chapter 4) enables training-free reasoning, Meta-Task addresses the critical challenge of adapting to new wireless scenarios with only 5-10 labeled examples. This few-shot capability is vital for cost-effective deployment across diverse channel conditions, modulation types, and operational environments. These are scenarios where acquiring extensive labeled data is prohibitively expensive. Meta-Task's method-agnostic design ensures compatibility with various architectures, making it a practical adaptation layer for wireless foundation models.

The remainder of the paper is organized as follows: Section 5.2 reviews related work to contextualize the problem. Section 5.3 details our proposed approach, and Section 5.4 presents experimental results. Finally, conclusions are drawn in Section 5.7.

5.2 Related Work

Here we position Meta-Task within this landscape.

Existing regularization approaches include Explicit Regularization (ER), Episode Augmentation Regularization (EAR), Task Augmentation Regularization (TAR), and Adversarial Task Upsampling (ATU) [82, 83, 89, 86, 90]. While effective, these methods often require extensive pre-training, synthetic data generation, or task-specific modifications.

Meta-Task distinguishes itself by introducing a learnable, task-specific regularization strategy through a parameterized neural layer ϕ that dynamically adapts without modifying the underlying FSL architecture. Unlike iMAML [82] and MR-MAML [83] (which focus on meta-initialization and gradient regularization) or ProtoDiff [91] (which requires computationally intensive diffusion models), Meta-Task learns robust embeddings during meta-training through auxiliary tasks, making it method-agnostic, computationally efficient, and suitable for resource-constrained environments.

Table 5.1: Comparison of meta-learning regularization methods.

Method	**Category**	**Method-Agnostic**	**Complexity**
iMAML [82]	ER	No	Medium
MR-MAML [83]	ER	No	Medium
Minimax-Meta [85]	ER	No	Low
DAC-MR [92]	EAR	Yes	Medium
MetaMix [93]	EAR	Yes	Medium
ProtoDiff [91]	TAR	No	High
ATU [90]	TAR	Yes	High
LEO [94]	ER	No	High
Task-Decoder (Ours)	**MTR**	**Yes**	**Medium**

The table compares various meta-learning regularization techniques by category, method-agnostic capability, and computational complexity. Categories include Explicit Regularization (ER), Episode Augmentation Regularization (EAR), Task Augmentation Regularization (TAR), and our proposed Meta-Task Regularization (MTR).

5.3 Meta-Task Framework

While our framework is domain-agnostic and demonstrated on image classification benchmarks, it is particularly motivated by wireless foundation models where rapid adaptation to new scenarios (channel conditions, modulation types) with minimal labeled data is essential for economical deployment. The method operates on learned representations regardless of whether they originate from images, wireless signals represented as images (spectrograms, constellation diagrams), or other modalities.

In this section, we introduce our proposed Meta-Task Framework, a novel approach that enhances generalization by integrating learnable Meta-Tasks as a form of regularization. By treating regularization itself as a learnable task, our framework encourages the model to develop richer and more robust feature representations. The framework is method-agnostic and can be seamlessly integrated with existing FSL models, such as Prototypical Networks and MAML. In the following subsections, we detail the components of the framework, including the Task-Decoder, the problem formulation, and the theoretical justification for its effectiveness.

Our framework extends MAML [74] by introducing regularization as a *learnable Meta-Task*, enabling unsupervised task integration during meta-training without requiring additional data. The following subsections detail the implementation and theoretical foundations of our contributions:

- **Task-Decoder:** We introduce an unsupervised autoencoder architecture that implements our learnable regularization concept by reconstructing input images from their embeddings, proving its effectiveness in enriching feature representations.
- **Problem Formulation:** We present the mathematical framework that enables the integration of unsupervised auxiliary tasks with existing FSL methods, demonstrating how our approach remains method-agnostic through separable loss functions.

- **Theoretical Analysis:** We provide theoretical justification for our framework's convergence properties and establish how the joint optimization of primary and Meta-Tasks enhances generalization capabilities.

5.3.1 Task-Decoder

At the core of our Meta-Task Framework is the Task-Decoder (Figure 5.1), an unsupervised learning component designed to enhance the embedding space by reconstructing input images from their embeddings. The Task-Decoder employs an autoencoder architecture, where the encoder f_θ maps the input images to a latent embedding space, and the decoder g_ϕ reconstructs the images from these embeddings.

By encouraging the embeddings to retain sufficient information for accurate reconstruction, the Task-Decoder prevents the encoder from focusing solely on discriminative features that may not generalize well. This leads to richer and more robust feature representations, ultimately improving the model's performance on unseen tasks.

The Task-Decoder operates in parallel with the primary few-shot learning model, processing the same inputs and contributing to the overall learning objective through its reconstruction loss.

5.3.2 Problem Formulation

Let the Meta-Task $\mathcal{M}$ denote the Task-Decoder, and our meta-learning method F be Prototypical Networks. We define the encoder network f_θ parameterized by θ, and g_ϕ represent the decoder network parameterized by ϕ.

For a given query sample $x \in \mathcal{Q}$ and its true class y belonging to task $\mathcal{T}$, the Prototypical Network F_θ computes class prototypes using support set $\mathcal{S}$ and classifies x based on the closest prototype to the embedding f_θ.

Primary Task Loss: The probability of x belonging to class n is:

$$p_\theta(y = n \mid x) = \frac{\exp\left(-d\left(f_\theta(x), c_n\right)\right)}{\sum_{k=1}^{N} \exp\left(-d\left(f_\theta(x), c_k\right)\right)}, \tag{5.1}$$

where N is the number of tasks in the episode, c_n is the prototype of class n and $d(\cdot, \cdot)$ denotes the Euclidean distance. The classification loss is then:

$$J_{\mathcal{T}}(x, \theta) = -\log p_\theta(y = n \mid x). \tag{5.2}$$

Meta-Task Loss: The Task-Decoder aims to reconstruct x from f_θ, with the reconstruction loss defined as:

$$J_{\mathcal{M}}(x, \theta, \phi) = \left\| g_\phi(f_\theta(x)) - x \right\|^2. \tag{5.3}$$

Combined Loss Function: The total loss for a sample x is a weighted sum of the primary task loss and the Meta-Task loss:

$$J(x, \theta) = J_{\mathcal{T}}(x, \theta) + \lambda J_{\mathcal{M}}(x, \theta, \phi), \tag{5.4}$$

where λ is a hyperparameter balancing the two losses or the learning rate for the Meta-Task update.

Optimization: The model parameters θ and ϕ are optimized jointly over all tasks and samples. For a given task $\mathcal{T}$, the overall loss is calculated by summing the combined losses over all query samples:

$$J_{\mathcal{T}}(\theta) = \sum_{x \in \mathcal{T}} J_{\mathcal{T}}(x, \theta). \tag{5.5}$$

During meta-training, we sample a batch of tasks $\{\mathcal{T}_i\}_{i=1}^{L}$. The total loss across all tasks

in an episode is:

$$J(\theta) = \sum_{i=1}^{L} \left(J_{\mathcal{T}_i}(\theta) + \lambda J_{\mathcal{M}_i}(\theta, \phi) \right). \tag{5.6}$$

By minimizing this combined loss, the model learns embeddings that are both discriminative for classification and informative for reconstruction, enhancing generalization across tasks. The full algorithm of the Task-Decoder integration is provided in Algorithm 2.

We define tasks and Meta-Tasks as separable if their loss functions can be computed independently and summed without including interaction terms. This structure allows us to optimize each component separately while still contributing to the overall objective. If the regularization term is separable (as in our case with the autoencoder), we can simplify (5.6) as

$$J(\theta) = \sum_{i=1}^{L} J_{\mathcal{T}_i}(\theta) + \lambda \sum_{i=1}^{L} J_{\mathcal{M}_i}(\theta, \phi). \tag{5.7}$$

If each task may have multiple or no Meta-Tasks attached, the overall loss function generalizes to:

$$J(\theta) = \sum_{i=1}^{L} J_{\mathcal{T}_i}(\theta) + \lambda \sum_{r=1}^{R} J_{\mathcal{M}_r}(\theta, \phi), \tag{5.8}$$

where R is the total number of regularizers applied to the model.

5.3.3 Justification

Our Meta-Task framework provides unique theoretical guarantees, distinguishing it from existing regularization approaches. In Equations (5.7) and (5.8), we demonstrated that incorporating auxiliary tasks into training effectively reformulates the FSL paradigm to a new FSL setting with a combined distribution of primary tasks and Meta-Tasks, denoted as $\mathcal{T}'$. This reformulation allows the overall training objective to be expressed as a sum over

$L + R$ tasks, preserving the structure necessary for standard FSL optimization techniques.

When using MAML as the FSL method, based on convergence guarantees proven in literature [95], our approach converges to an optimal set of parameters for $\mathcal{T}'$. Since the Meta-Tasks are carefully designed to enhance generalization, this optimal parameter set enables the model to generalize effectively across both primary tasks and Meta-Tasks. The convergence of ϕ to its global optima further ensures that the learned embeddings retain the essential information necessary for both classification and reconstruction. This joint optimization (θ) constructs a globally optimal representation space, improving overall model performance on unseen tasks by reducing generalization error.

A critical advantage of Meta-Task is the separability of the primary and auxiliary tasks. Since their loss functions can be computed independently, optimization can proceed efficiently, with each task contributing to the overall objective. This design simplifies the optimization process and supports convergence to a global optimum, as the auxiliary tasks act as an implicit regularizer, reducing overfitting by penalizing non-generalizable embeddings.

A rigorous mathematical proof is more challenging for FSL methods beyond MAML. However, as all FSL methods aim to optimize parameters that generalize across task distributions with limited examples, it is reasonable to infer that our approach improves generalization capabilities, as demonstrated in the results section. Additional experiments, including visualizations of reconstructed features in Section 5.7.3, empirically validate how Meta-Task enhances embedding quality and task generalization.

Remark 1. *The proposed regularization approach is method-agnostic and can be applied to any FSL learning method.*

Remark 2. *Equations* (5.7)*,* (5.8) *hold only if tasks and Meta-Tasks are separable or independent.*

Algorithm 2: Meta-Tasks: Task-Decoder

Require: $p(\mathcal{T})$: distribution over tasks;
Require: f_θ: embedding network;
Require: g_ϕ: decoder network;
Require: α, λ: learning rates;
Require: F: meta-learning method (e.g., Prototypical Networks);
Require: $J_\mathcal{T}(x, \theta)$: classification loss;
$J_\mathcal{M}(x, \theta, \phi) = \|g_\phi(f_\theta(x)) - x\|^2$: reconstruction loss

Randomly initialize θ and ϕ
while not done **do**
  Sample support set: $\mathcal{S} \sim p(\mathcal{T})$
  Sample query set: $\mathcal{Q} \sim p(\mathcal{T})$
  Compute reconstruction loss:
$$J_\mathcal{M}(\theta, \phi) \leftarrow \frac{1}{|\mathcal{S} \cup \mathcal{Q}|} \sum_{x \in \mathcal{S} \cup \mathcal{Q}} J_\mathcal{M}(x, \theta, \phi)$$
  Use the support set $\mathcal{S}$ and embedding network f_θ to get meta-learning method F_θ.
  Compute classification loss on query set:
$$J_\mathcal{T}(\theta) \leftarrow \frac{1}{|\mathcal{Q}|} \sum_{x \in \mathcal{Q}} J_\mathcal{T}(x, F_\theta)$$
  Update embedding network parameters:
$$\theta \leftarrow \theta - \alpha \nabla_\theta \left(J_\mathcal{T}(\theta) + \lambda J_\mathcal{M}(\theta, \phi) \right))$$
  Update decoder parameters:
$$\phi \leftarrow \phi - \lambda \nabla_\phi J_\mathcal{M}(\theta, \phi)$$
end while

Remark 3. *The use of autoencoders is because autoencoders can be used with any type of data, are easy to use, and are flexible in architecture.*

5.4 Experiments and Results

We validate Meta-Task on standard computer vision benchmarks to demonstrate its general applicability and enable comparison with existing FSL methods. This validation establishes the framework's effectiveness on vision few-shot tasks; Section 5.5 then tests directly whether the same mechanism transfers to a wireless AMC deployment.

In this section, we conduct comprehensive experiments to validate the effectiveness of

our proposed Meta-Task framework in enhancing generalization and mitigating overfitting in few-shot learning. Our objectives are to:

1. Demonstrate that incorporating Meta-Tasks as auxiliary regularization improves model generalization to both seen and unseen tasks.

2. Validate that leveraging both labeled and unlabeled data through an unsupervised autoencoder refines hidden representations.

3. Showcase the method-agnostic applicability of Meta-Task across various FSL settings and its practical efficiency in resource-constrained scenarios.

We begin by outlining the experimental setup, including datasets, model architectures, and implementation specifics. Following this, we present our results along with a comprehensive analysis of our findings, focusing on the impact of the Meta-Task framework on performance, convergence speed, and variance reduction.

5.4.1 Datasets

We evaluated the proposed Meta-Task framework on three standard benchmarks commonly used in few-shot learning:

MiniImageNet [96]: This dataset consists of 100 classes selected from ImageNet, with 600 images per class resized to 84×84 pixels. It is divided into 64 training, 16 validation, and 20 test classes, ensuring that test classes are entirely disjoint from training classes for robust evaluation of generalization capabilities.

TieredImageNet [97]: A larger and more challenging dataset comprising 608 classes organized into 34 high-level categories, providing semantic diversity. It includes 351 training, 97 validation, and 160 test classes, with images resized to 84×84 pixels. Its hierarchical structure makes it suitable for evaluating models' ability to generalize across different

levels of abstraction.

FC100 [2]: Derived from the CIFAR-100 dataset, FC100 contains 100 classes, each with 600 images resized to 32×32 pixels. The dataset is split into 60 training, 20 validation, and 20 test classes. It presents unique challenges due to low-resolution images and high intra-class variability, making it an excellent benchmark for assessing model robustness.

These datasets offer diverse and challenging scenarios for few-shot learning, enabling us to thoroughly evaluate the generalization capabilities of the Meta-Task framework.

5.4.2 Methods

To demonstrate the method-agnostic nature of our approach, we integrated the Meta-Task framework with four few-shot learning models:

Prototypical Network (PN) [75]: Utilizes a ResNet-18 [98] backbone for feature extraction. It computes class prototypes in the embedding space, classifying query samples based on their distances to these prototypes.

MAML [74]: Employs a four-layer ConvNet architecture aiming to find initial model parameters that can be quickly adapted to new tasks with minimal gradient updates.

MetaOptNet [99]: Uses a ResNet-12 backbone and incorporates differentiable convex optimization layers for efficient classification, seamlessly integrating with our Meta-Task framework.

P>M>F [100]: Uses DINO[101] framework as its backbone. Like PN, it computes class prototypes in the embedding space. It classifies query samples based on their distances to these prototypes but with the added benefit of leveraging pre-trained transformer-based feature extraction.

In our Meta-Task framework, we introduced a Task-Decoder employing an unsupervised autoencoder to reconstruct input images from their embeddings. The decoder mirrors

the encoder architecture used in the backbone models. This process leverages both labeled and unlabeled data (support and query sets), refining the hidden representations and mitigating overfitting.

For the autoencoder architecture, we experimented with different configurations to assess the impact of the Task-Decoder's depth on performance. Specifically, we used both shallow and deep autoencoders, with parameters ranging from 75K to 250K. The shallow autoencoder consists of two convolutional layers, while the deep autoencoder includes four convolutional layers. This allowed us to analyze how the complexity of the Task-Decoder affects the overall performance.

We trained all models using the Adam optimizer with learning rates set to 1×10^{-4} for the encoder and classifier, and 1×10^{-6} for the Task-Decoder. Each model was trained over 50,000 episodes using 5-way 1-shot and 5-way 5-shot scenarios. The query size was maintained at 15 for both training and evaluation phases. To enhance model robustness, we applied standard data augmentation techniques, including random cropping, horizontal flipping, and color jittering.

Furthermore, we investigated the effect of using multiple autoencoders per task by associating different autoencoders with specific layers of the encoder network. However, we found that using a single autoencoder provided a good trade-off between performance improvement and computational overhead.

5.4.3 Performance Evaluation

Our experimental results demonstrate that the Meta-Task framework consistently outperforms baseline models across all datasets and settings, achieving higher accuracy with significantly fewer training episodes. For example, as shown in Figure 5.2, TD enables higher test accuracy within 20k episodes, whereas PN Test fails to achieve the same even after 50k

Table 5.2: Comparison of few-shot classification accuracies (%) with standard deviation on MiniImageNet and TieredImageNet datasets.

		MiniImageNet		TieredImageNet	
Method	Backbone	5-way 1-shot(%)	5-way 5-shot(%)	5-way 1-shot(%)	5-way 5-shot(%)
PN	ResNet18	45	65.2	46.6	65.4
PN(TD)	ResNet18	**47.9**	**66.1**	**50**	**71.3**
MAML	CNN	47.54	62.23	-	-
MAML (TD)	Shallow CNN Decoder	**47.81**	**62.52**	-	-
MAML (TD)	CNN	**47.67**	**63.01**	-	-
Relation Networks[73]	64-96-128-256	50.44 ± 0.82	65.32 ± 0.70	54.48 ± 0.93	71.32 ± 0.78
R2D2[102]	96-192-384-512	51.2 ± 0.6	68.8 ± 0.1	-	-
Transductive Prop Nets[103]	96-192-384-512	55.51 ± 0.86	69.86 ± 0.65	59.91 ± 0.94	73.3 ± 0.75
SNAIL[104]	ResNet-12	55.71 ± 0.99	68.88 ± 0.92	-	-
AdaResNet[105]	ResNet-12	56.88 ± 0.62	71.94 ± 0.57	-	-
TADAM [2]	ResNet-12	58.5 ± 0.30	76.7 ± 0.30	-	-
MetaOptNet	ResNet-12	61.23 ± 0.29	77.14 ± 0.21	63.10 ± 0.34	78.58 ± 0.27
MetaOptNet (TD)	ResNet-12	**61.35 ± 0.29**	**77.3 ± 0.22**	**63.20 ± 0.34**	**78.66 ± 0.27**
P>M>F	IN1K,ViT-base	**72.80**	**90.70**	-	-
P>M>F(TD)	IN1K,ViT-base	**74.10**	**92.30**	-	-

Note: Results are reported for 5-way 1-shot and 5-way 5-shot. 'PN' and 'TD' refer to Prototypical Networks and Task-Decoder, respectively.

Table 5.3: Comparison of few-shot classification accuracies (%) with standard deviation on the FC100 dataset [2].

	FC100	
Method	5-way 1-shot(%)	5-way 5-shot(%)
PN	31.1	49.6
PN (TD)	**39**	**51.1**
TADAM[2]	40.1 ± 0.4	56.1 ± 0.4
MetaOptNet	40.19 ± 0.25	55.04 ± 0.26
MetaOptNet (TD)	**40.59 ± 0.26**	**55.38 ± 0.25**

Note: Results are reported for 5-way 1-shot and 5-way 5-shot settings. 'PN' and 'TD' refer to Prototypical Networks and Task-Decoder, respectively.

episodes. Similarly, in Table 5.2, TD P>M>F reaches 92.30% accuracy in 3k episodes compared to P>M>F's 90.70% after 4k episodes.

These results highlight the framework's efficiency in leveraging a small number of additional parameters to enhance generalization while reducing computational overhead. By introducing Meta-Tasks, the learning process is effectively regulated, eliminating the need for extensive parameter tuning or large-scale episodic training. This makes the Meta-Task framework both lightweight and practical for real-world applications. Further details on are provided in Table 5.2 and Table 5.3.

From a wireless foundation model perspective, these results validate the feasibility of rapid adaptation with minimal labeled data. The 5.9 percentage point improvement on TieredImageNet 5-way 5-shot (Table 5.2) demonstrates that foundation models need not require extensive labeled datasets for each deployment scenario. The 20k vs. 50k episode convergence reduction translates directly to training cost savings, which is critical for economical wireless AI deployment across diverse channel conditions and modulation types.

Prototypical Network Results

Table 5.2 presents our experimental results on MiniImageNet and TieredImageNet, where our proposed method, the Task-Decoder, demonstrates significant performance improvements over the baseline Prototypical Networks. Specifically, on TieredImageNet in the 5-way 5-shot setting, our approach achieved an increase from 65.4% to 71.3% in test set accuracy, reflecting a substantial 5.9% absolute improvement. This gain indicates superior meta-generalization capabilities of our method.

In the 5-way 1-shot scenario, the Task-Decoder improved accuracy from 46.6% to 50% on TieredImageNet, highlighting its efficacy in scenarios with extremely limited data. These consistent improvements across different datasets and settings demonstrate the robustness and effectiveness of our approach.

MAML Results

Table 5.2 also showcases the performance of the Task-Decoder integrated with MAML. Our method outperforms the baseline MAML by up to 0.27% in the 5-way 1-shot setting and up to 0.78% in the 5-way 5-shot setting on MiniImageNet. Although the absolute improvements are modest, they demonstrate the effectiveness of the Task-Decoder in enhancing the adaptability of MAML.

To examine the impact of the Task-Decoder's architecture (ϕ) on generalization, we trained MAML using two differentTask-Decoders: a shallow decoder with 75K parameters and a deeper decoder with 250K parameters. Interestingly, the shallow decoder excelled in the 5-way 1-shot task, suggesting that deeper models are not always superior when data is scarce. In contrast, the deeper decoder performed better with more shots, highlighting the importance of model depth in specific contexts. This indicates that the Task-Decoder and its Meta-Task extensions can be effective even with a shallow network, which is beneficial in resource-constrained environments.

It's important to note that MAML requires the computation of second derivatives during optimization. However, to remain consistent with our claim of a method-agnostic framework, we chose to use a first-order approximation for the Task-Decoder's optimization. This decision avoids the computational overhead associated with second-order derivatives but may introduce approximation errors. Another limitation may stem from the inherent difficulties of optimizing the learning rate in MAML, which is complicated by the need for careful tuning due to its sensitivity to hyperparameters. While using the same learning rate as the Prototypical Networks simplifies comparisons, it likely constrained MAML's overall performance. These factors suggest that further optimization and fine-tuning could enhance the benefits of integrating the Task-Decoder with MAML.

MetaOptNet Results

The results presented in Table 5.2 indicate that integrating the Task-Decoder with MetaOptNet [99] leads to modest but consistent performance improvements, with an average accuracy increase of 0.12% in the 5-way 1-shot setting and 0.16% in the 5-way 5-shot setting on MiniImageNet. While these gains are not as pronounced as those observed with other methods, they underscore the potential of the Task-Decoder to enhance performance even in well-optimized frameworks like MetaOptNet.

We attribute the smaller improvements to MetaOptNet's distinctive training setup, where variations in batch size and the number of shots significantly impact model performance. Due to hardware limitations, we reduced the batch size and the number of training shots, which may have constrained the performance gains. These findings suggest that further hyperparameter optimization, such as adjusting batch size or learning rate, could yield more substantial improvements.

P>M>F Results

Due to the pre-trained nature of the encoder, we implemented a custom scheduler to align TD's optimization. Additionally, hardware constraints necessitated deviations from the original hyperparameters, such as reducing image sizes and query shots, which may have influenced the observed improvements. Despite these constraints, TD achieved a notable 1.45% performance increase, demonstrating its flexibility and adaptability to transformer-based methods under constrained conditions.

5.4.4 Regularization Improvements

Figure 5.2 illustrates that our method achieves higher accuracy more quickly during all training, validation, and testing phases compared to the baseline, which demonstrates faster

convergence. Additionally, Figure 5.3 illustrates that the loss associated with our method decreases more rapidly and consistently remains lower throughout the training process across all training, validation, and test sets. The Task-Decoder achieves a mean loss of 1.23 with a variance of 0.001, whereas the baseline has a mean loss of 1.3 and a variance of 0.0002. This indicates more efficient learning and improved generalization.

To address concerns regarding implementation specifics, we provide detailed information about the hyperparameters, training times, and autoencoder architectures in Section 5.7.1. We also conducted ablation studies to analyze the effect of different autoencoder architectures and the use of multiple autoencoders per task. The results indicate that while deeper autoencoders can provide marginal performance gains, the computational overhead increases significantly. Therefore, a single, appropriately sized autoencoder strikes a good balance between performance and efficiency.

5.5 Wireless Validation: Meta-Task for AMC on RadioML

The vision benchmarks above establish that Meta-Task's learnable regularization helps few-shot generalization when the base learner has a weak inductive bias. This section asks the question those benchmarks cannot answer directly: *does the same auxiliary task help when the backbone is a wireless signal encoder, the deployment is cross-dataset, and classification runs through the DiSC-AMC pipeline of Chapter 4?*

5.5.1 Experimental Design

We evaluate on RadioML 2018.01a (24 classes, five SNR levels from -20 to $+20$ dB), using the full DiSC-AMC pipeline (DeepSeek-R1-Distill-Qwen-7B reasoning over discretized, retrieval-augmented exemplars; Section 4.6) as the fixed downstream endpoint, so that any effect of the auxiliary task is measured on the metric that matters for the dissertation's RQ2 contribution, not on an encoder's classification head in isolation. We compare three configurations that isolate the Task-Decoder's effect while holding the encoder backbone (DenoMAE2.0) and the reasoning pipeline fixed:

(A) DenoMAE2.0 baseline the encoder frozen, only a classification head trained, as published in Chapter 4's DiSC-AMC Scenario III results.

(B) DenoMAE2.0, fine-tuned the encoder updated end-to-end on RadioML, with no auxiliary task.

(C) DenoMAE2.0 + Meta-Task identical to (B), with the Task-Decoder's reconstruction loss added on a separate optimizer.

The main comparison is **C vs. B**: the two configurations share the same backbone and fine-tuning protocol, differing only in whether the auxiliary task is present.

Two protocol fixes were necessary for this comparison to be fair, both motivated by an earlier result that did not survive scrutiny. An initial run with a *frozen* encoder (Meta-Task's auxiliary loss added only to the classification head) showed a +4.6 percentage point gain at -20 dB. This gain disappeared once the comparison was corrected: **Fix 1** trains both the Task-Decoder and the vanilla fine-tuning baseline with real, matched end-to-end fine-tuning of the encoder (matching the protocol used for DINOv2 elsewhere in Chapter 4), rather than leaving the encoder frozen only for the Task-Decoder arm; **Fix 2** places the auxiliary reconstruction loss on its own optimizer with a small learning rate (10^{-6}), as in

the original Meta-Task implementation (Section 5.3), rather than folding it into the main loss via a scalar weight λ, which had been inflating its apparent effect.

5.5.2 Results

Table 5.4: Wireless validation: DiSC-AMC pipeline accuracy on RadioML 2018.01a (24 classes), averaged over five SNR levels (-20 to $+20$ dB).

Configuration	Mean Accuracy (%)
(A) DenoMAE2.0, frozen encoder	21.5
(B) DenoMAE2.0, fine-tuned, no auxiliary task	23.0
(C) DenoMAE2.0, fine-tuned + Meta-Task	22.2

Table 5.4 reports the result under the corrected protocol: **(C) 22.2% vs. (B) 23.0%**: Meta-Task's auxiliary task gives *no net gain* over plain fine-tuning on this cross-dataset wireless AMC task, and is marginally, not significantly, below it. The one effect that held up under both protocol fixes is the end-to-end fine-tuning itself: moving from a frozen encoder to a fine-tuned one (A$\rightarrow$B) improves mean accuracy by +1.5 points (21.5%$\rightarrow$23.0%), confirming that adapting the backbone to the target domain matters, even though adding Meta-Task's regularization on top of that adaptation does not help further. For reference, DiSC-AMC with a fine-tuned DINOv2 backbone (Chapter 4, Table 4.5) still reaches 76.25% at $+20$ dB on the same benchmark, an order of magnitude above any DenoMAE2.0 configuration here, which underscores that representation quality, not the regularizer, is the binding constraint on this dataset.

5.5.3 Discussion: A Bounded, Honest Null Result

We report this as a **robust null result**: both plausible objections to the initial positive finding (an unfair frozen-encoder comparison and an improperly weighted auxiliary loss)

were checked explicitly, and neither reversal flips the conclusion. This matters for two reasons. First, it demonstrates the value of adversarial self-checking of promising early results before they are reported as a contribution; the original +4.6-point frozen-encoder result would have overstated Meta-Task's wireless applicability had it not been re-examined under a fair protocol. Second, it sharpens the dissertation's account of *why* Meta-Task helps where it helps: Section 5.4 showed its largest gains on metric learners with a weak inductive bias (Prototypical Networks) applied to within-domain few-shot vision tasks; here, the task is instead cross-dataset transfer (SynthMod-10-trained representations evaluated on RadioML), where Section 4.6.12 already identified the self-supervised *representation* itself, not the classifier's regularization, as the bottleneck. A learnable auxiliary task cannot compensate for an encoder whose embeddings do not retrieve informative neighbors on the target dataset. This connects directly back to RQ1 (Chapters 2–3): on wireless cross-dataset transfer, representation quality dominates regularization.

5.6 A Second Wireless Test: Few-Shot Channel Estimation

The RadioML experiment above tests Meta-Task on one wireless task. An independent study, Roshan et al. [3] (second-author work of this dissertation, with R. Gh. Roshan first), asks the broader question that RQ3 poses for a different wireless task: can a model that has learned to adapt specialize to an unseen channel from only a few labeled examples? It does not use Meta-Task. It uses standard MAML and MultiGrade-MAML (MG-MAML), so it is evidence about learning-to-learn on wireless data, not about the Task-Decoder, and it is used here only as corroboration.

5.6.1 Setup

The task is pilot-aided channel estimation in a single-antenna OFDM system: reconstruct the full time-frequency channel response from a noisy least-squares (LS) estimate interpolated from sparse pilots, scored by mean squared error (MSE, lower is better). Source and target channels are simulated: five Tapped Delay Line profiles (TDL-A to TDL-E, at 0, 5, and 10 dB SNR, giving 15 channel conditions) and nine urban-microcell (UMi) conditions from the CeBed framework. Splits are made at the channel level, so each evaluated channel is unseen in training (10 of 15 TDL conditions and 4 of 9 UMi conditions are used for training). For each unseen channel, 5 labeled frames are used for fine-tuning and the remaining 226 frames for evaluation. Four estimators are compared under the same budget: LS; linear MMSE (LMMSE), which uses second-order channel and noise statistics; ChannelNet, a convolutional estimator trained without meta-learning; and the two meta-learners, MAML and MG-MAML.

5.6.2 Results

Table 5.5 combines the paper's NLOS, LOS, and domain-shift results. The UMi column is a domain shift, since the learned models were trained mostly on TDL channels.

Table 5.5: Few-shot (5-shot) channel estimation MSE; lower is better. TDL-A and TDL-C are rich-multipath NLOS channels, TDL-D and TDL-E are LOS-dominant, and UMi is a domain shift. Values are as reported by Roshan et al. [3].

Method	TDL-A (0 dB)	TDL-A (5 dB)	TDL-C (5 dB)	TDL-D (5 dB)	TDL-E (10 dB)	UMi (5 dB)
LS (input)	1.7326	0.5483	0.5490	0.5508	0.1759	0.1844
LMMSE	**0.0202**	**0.0065**	**0.0075**	**0.0073**	**0.0024**	**0.0022**
ChannelNet	0.1571	0.0742	0.0744	0.0432	0.0229	0.0258
MAML	1.4831	1.5827	1.1330	0.1926	0.2282	0.0563
MG-MAML	0.4366	0.4736	0.4170	0.0793	0.0482	0.1164

Four observations follow from the table. First, the classical LMMSE estimator has

the lowest error in every condition, because it uses explicit second-order statistics that act as a structural prior. Second, the learned estimators are sensitive to noise and channel complexity: in the NLOS conditions MAML's error is as large as, or larger than, that of the LS input it starts from (for example 1.5827 against 0.5483 on TDL-A at 5 dB), consistent with inner-loop gradients being estimated from five noisy frames. Third, the non-meta ChannelNet is better than both meta-learners in every condition shown, and MG-MAML, which stages the meta-updates for stability, is steadier than MAML on NLOS channels but is not robust to the UMi shift (0.1164, worse than ChannelNet's 0.0258). Fourth, increasing the number of adaptation frames helps the meta-learners but does not close the gap (Table 5.6): MAML and MG-MAML improve from 1 to 10 shots, with a non-monotonic step for MAML at 15, while ChannelNet is nearly flat, and none approaches LMMSE's 0.0022.

Table 5.6: Effect of the number of adaptation frames K on UMi channel estimation (MSE; lower is better), from Roshan et al. [3].

Model	1-shot	5-shot	10-shot	15-shot
ChannelNet	0.0291	0.0258	0.0263	0.0251
MAML	0.0902	0.0563	0.0508	0.0581
MG-MAML	0.1487	0.1164	0.0921	0.0870

5.6.3 What It Adds to RQ3

This study supports the RQ3 answer in three ways. It shows that the RadioML null result is not specific to Meta-Task, to classification, or to the DiSC-AMC pipeline: on a second wireless task, learning-to-learn did not reliably beat a plain learned estimator, and under noise and domain shift it was often worse. It suggests a mechanism that fits the RadioML finding, namely that adaptation from a handful of noisy examples amplifies variance, so

what limits adaptation is the quality and generality of the starting representation or prior, not the adaptation procedure. And it supplies a counterpart to the dissertation's arc from hand-crafted features to learned representations (Section 6.3.1): the estimator built on explicit channel statistics is the most robust under data paucity and shift, while learned priors are distribution-dependent.

The bounds are as important as the support. MAML and MG-MAML are not Meta-Task, and the task is regression on simulated channels rather than modulation classification, so the study does not test the Task-Decoder or the RadioML claim directly. The results are single reported values per condition without confidence intervals, and LMMSE benefits from channel statistics that the learned models must acquire from data. Whether Meta-Task's reconstruction objective, which is close in spirit to channel estimation viewed as image reconstruction, helps here is untested and is left to future work.

5.7 Conclusion

In this paper, we introduced Meta-Task, a method-agnostic framework that treats regularization as a learnable task for few-shot learning. By integrating an autoencoder-based Task-Decoder directly within the meta-learning optimization, our approach provides continuous, unsupervised regularization that prevents overfitting and improves generalization. Extensive experiments across multiple datasets (Mini-ImageNet, Tiered-ImageNet, FC100) and meta-learning methods (Prototypical Networks, MAML, MetaOptNet, P>M>F) demonstrate consistent improvements in accuracy, convergence speed, and cross-domain generalization.

Connection to the Dissertation Framework: Meta-Task provides a *partial* answer to Research Question 3 (RQ3): *Can a learnable auxiliary task improve generalization when only a few labeled examples exist?* On vision few-shot benchmarks, the answer is

yes: Prototypical Networks improve from 65.4% to 71.3% on Tiered-ImageNet 5-way 5-shot, reaching the baseline's best accuracy in under 20k episodes versus more than 50k without the Task-Decoder, with consistent (if uneven) gains across MAML, MetaOptNet, and P>M>F. On the wireless deployment tested directly in Section 5.5, however, Meta-Task gives no net gain over plain fine-tuning on cross-dataset RadioML AMC through the DiSC-AMC pipeline (22.2% vs. 23.0%). The lesson this bounded result supports is that *on wireless cross-dataset transfer, the representation is the bottleneck, not the regularizer*: Meta-Task's adaptation layer is most valuable when the base learner's inductive bias is weak and the domain is in-distribution, and is not a substitute for a better encoder (Chapters 2–3) when the encoder itself fails to transfer. This work complements DenoMAE's representation learning (Chapter 2), benefits from CDRF's diverse evaluation tasks (Chapter 3), and was tested directly as a fine-tuning addition to the Plug-and-Play/DiSC-AMC pipeline (Chapter 4). Meta-Task's method-agnostic design keeps it a practical adaptation layer for in-distribution few-shot scenarios within the modular wireless foundation model framework, even though it is not, by itself, a solution to cross-dataset representation gaps.

The Task-Decoder variants show consistently superior performance across metrics, particularly in test accuracy, while maintaining reasonable computational requirements. The deeper CNN architecture demonstrates the best overall performance, though at the cost of increased training time.

5.7.1 Experimental Details

We trained all models using the recommended hyperparameters for each method. For the Task-Decoder, we maintained consistent settings across experiments. Table 5.7 lists the hyperparameters used for training the Prototypical Networks and their Task-Decoder counterparts.

Table 5.7: Hyperparameters used to train Prototypical Networks for 5-shot and 1-shot experiments.

Hyperparameter	**5-shot**	**1-shot**
Epoch	5	5
Train Num Episodes	10,000	10,000
Test Num Episodes	10,000	10,000
Train Way	5	5
Train Shot	5	1
Train Query	15	15
Test Way	5	5
Test Shot	5	1
Test Query	15	15
Optimizer	Adam	Adam
Task-Decoder Learning Rate	1×10^{-6}	1×10^{-6}
Learning Rate	1×10^{-4}	1×10^{-4}

Due to hardware constraints, we adjusted certain parameters for MetaOptNet, such as reducing the batch size from 8 to 6 and decreasing the number of training shots from 15 to 10. These adjustments were necessary to fit the models into GPU memory but may have affected performance.

Due to hardware constraints, we adjusted certain parameters for P>M>F, such as reducing the image size from 128 to 92 and decreasing the number of training shots from 15 to 10. These adjustments were necessary to fit the models into GPU memory but may have affected performance.

5.7.2 Ablation Study: Additional Meta-Tasks

To further demonstrate the convergence of Meta-Tasks, we designed simple neural layers and implemented basic Meta-Tasks to compare their impact on performance. As shown in Tables 5.8 and 5.9, even these simple Meta-Tasks provide better regularization and improve

performance. However, designing more complex Meta-Tasks requires careful attention and hyperparameter tuning to be effective. For instance, the GAN PN Meta-Task in Table 5.8 negatively impacted training performance, while all other Meta-Tasks led to improvements. Moreover, Table 5.9 highlights that while not all Meta-Tasks yield dramatic performance gains, they generally contribute to performance improvements.

Table 5.8: Performance comparison of different Meta-Tasks with vanilla Prototypical Networks in a 5-Way 5-Shot 5-Query for 50,000 episodes.

Meta-Task	**Meta-Task Loss**	**Accuracy(%)**
Vanilla	-	64.70
Task-Decoder (TD)	Reconstruction Loss	**67.30**
RL PN	Reward Function	66.88
Rotating PN	Cross-Entropy	66.06
MIXWAYS PN	Cross-Entropy	67.19
GAN PN	Cross-Entropy	46.10

Table 5.9: Performance comparison of different Meta-Tasks with vanilla Prototypical Networks in a 5-Way 5-Shot 5-Query for 500 episodes.

Meta-Task	**Meta-Task Loss**	**Accuracy(%)**
Vanilla	-	24.00
TD	Reconstruction Loss	**36.20**
ArcFace	Cosine Distance	33.80
ArcMarginProduct	Cosisne Distance	32.20
AddMarginProduct	Cosine Distance	20.60
SphereProduct	Cosine Distance	23.80

5.7.3 Feature Space Analysis

In this section, we analyze the effect of few-shot learning on the feature space and the Task-Decoder. Figure 5.7 shows the image reconstructions before few-shot learning, while Figure 5.4 shows the reconstructions after. Similarly, Figure 5.8 presents the reconstructions

before few-shot learning, and Figure 5.9 depicts them after. Interestingly, the performance of the Task-Decoder changes following few-shot learning. Although the exact reasons for this behavior are unclear, several factors may contribute to this effect, including the imbalance in the number of parameters between the encoder (85M) and the decoder (3M), insufficient training of the decoder (only 25 episodes with approximately 5,000 images), or other underlying issues.

Additionally, Figure 5.5 and Figure 5.6 display the t-SNE projections of the first 10 classes of MiniImageNet with and without the Task-Decoder. Both visualizations exhibit improved clustering with the Task-Decoder, indicating better feature space representation. This suggests that the Task-Decoder enhances the performance of Prototypical Networks while improving generalization and regularization capabilities.

5.7.4 Extended Classification Results

5.7.4.1 Prototypical Networks

We evaluated test accuracies using standard classification metrics and computed the F1-score to compare the performance of the two models. Tables 5.10, 5.11, 5.12, 5.13 present a comparison between Prototypical Networks and their Task-Decoder counterparts, demonstrating that the Task-Decoder exhibits significantly superior adaptation-generalization compared to Prototypical Networks. The F1-scores further indicate superior meta-generalization to novel tasks by the Task-Decoder.

Table 5.10: 5-way 1-shot classification report of Precision (%), Recall (%), and F1-score (%) on MiniImageNet using **Prototypical Networks** after 50,000 episodes.

Class	Support	Precision (%)	Recall (%)	F1-score (%)
0	46635	36.64	29.59	32.74
1	44925	51.78	35.31	41.99
2	37020	42.86	39.67	41.20
3	30330	46.58	46.50	46.54
4	44670	34.14	34.82	34.48
5	29460	31.27	46.75	37.47
6	36465	38.62	68.32	49.35
7	44640	41.56	32.46	36.45
8	22770	36.59	27.04	31.10
9	14670	69.57	93.33	79.72
10	36945	34.01	34.48	34.24
11	38595	32.40	31.89	32.14
12	37125	38.08	32.01	34.78
13	59445	30.77	28.49	29.59
14	46245	43.04	46.19	44.56
15	44925	46.80	58.68	52.07
17	67725	52.55	58.48	55.35
18	29100	49.93	28.34	36.15
19	38310	62.36	52.30	56.89
Accuracy (%)		42.90		
Macro Average (%)		43.13	43.40	42.46
Weighted Average (%)		42.45	42.20	41.64

5.7.4.2 MAML

Table 5.14 provides a detailed comparison of the performance of MAML and Task-Decoder models on the MiniImageNet dataset. The table shows results for both 5-way 1-shot and 5-way 5-shot classification tasks across training, validation, and test sets. Notably, the Task-Decoder with a deeper CNN architecture consistently outperforms the standard MAML across all metrics, particularly in test accuracy, indicating its superior ability to generalize to unseen data. These results highlight the effectiveness of the Task-Decoder in improving

Table 5.11: 5-way 1-shot classification report of Precision (%), Recall (%), and F1-score (%) on MiniImageNet using **Prototypical Networks with Task-Decoder** after 50,000 episodes.

Class	**Support**	**Precision (%)**	**Recall (%)**	**F1-score (%)**
0	46635	40.08	31.99	35.58
1	44925	57.97	47.64	52.30
2	37020	53.45	43.70	48.09
3	30330	59.32	52.89	55.92
4	44670	40.26	37.58	38.87
5	29460	45.57	58.67	51.30
6	36465	45.14	73.41	55.90
7	44640	45.68	41.21	43.33
8	22770	53.30	36.17	43.09
9	14670	66.96	96.80	79.16
10	36945	39.39	38.42	38.90
11	38595	41.17	39.37	40.25
12	37125	44.74	39.99	42.23
13	59445	36.32	36.61	36.46
14	46245	45.68	46.86	46.26
15	44925	53.66	54.31	53.98
17	67725	53.84	67.44	59.88
18	29100	45.04	31.58	37.13
19	38310	56.64	56.94	56.79
Accuracy (%)			47.86	
Macro Average (%)		48.64	49.03	48.18
Weighted Average (%)		47.72	47.86	47.27

model performance on few-shot learning tasks.

5.7.4.3 Computational Analysis

To provide insights into the computational requirements of our approach, Table 5.15 presents the training times for various model configurations. It is important to note that the reported times correspond to training for exactly 50,000 episodes. However, in practice, the

Table 5.12: 5-way 5-shot classification report of Precision (%), Recall (%), and F1-score (%) on MiniImageNet using **P>M>F with Task Decoder** after 3,000 episodes.

Class	Support	Precision (%)	Recall (%)	F1-score (%)
0	2370	100.00	98.06	99.02
1	3420	95.95	93.45	94.68
2	4155	92.43	98.72	95.47
3	6270	96.52	88.60	92.39
4	3825	95.86	92.05	93.92
5	810	100.00	86.67	92.86
6	4320	88.03	98.87	93.13
7	795	100.00	100.00	100.00
8	3135	96.94	94.93	95.92
9	5535	89.10	86.97	88.02
10	2970	90.49	94.78	92.58
11	3720	93.28	94.81	94.04
12	3810	92.72	90.31	91.50
13	4710	95.02	87.90	91.32
14	3330	88.83	87.15	87.98
15	5145	93.43	96.40	94.89
16	4080	89.26	90.44	89.85
17	5010	95.29	93.71	94.50
18	3600	87.67	81.58	84.52
19	3990	81.23	95.89	87.95
Accuracy (%)	92.25			
Macro Average (%)	75000	93.10	92.57	92.73
Weighted Average (%)	75000	92.44	92.25	92.25

Task-Decoder (TD) often converges in significantly fewer episodes, further highlighting its computational efficiency.

The Task-Decoder variants show consistently superior performance across metrics, particularly in test accuracy, while maintaining reasonable computational requirements. The deeper CNN architecture demonstrates the best overall performance, though at the cost of increased training time.

Table 5.13: 5-way 5-shot classification report of Precision (%), Recall (%), and F1-score (%) on MiniImageNet using **P**>**M**>**F** after 4,000 episodes.

Class	**Support**	**Precision (%)**	**Recall (%)**	**F1-score (%)**
0	5700	98.35	96.18	97.25
1	5100	93.46	91.14	92.29
2	2460	97.81	96.18	96.99
3	2040	100.00	82.45	90.38
4	2685	95.23	89.16	92.09
5	4530	78.52	95.36	86.12
6	4965	95.35	95.91	95.63
7	1485	92.70	94.07	93.38
8	1635	100.00	63.55	77.71
9	3630	98.21	88.95	93.35
10	3105	92.10	93.88	92.98
11	2895	94.66	90.09	92.32
12	4785	92.77	92.54	92.66
13	4605	91.58	89.55	90.56
14	6090	85.71	89.36	87.50
15	2145	91.15	97.53	94.23
16	3900	88.29	94.74	91.40
17	4425	95.97	87.19	91.37
18	5040	80.83	79.48	80.15
19	3780	79.10	94.31	86.04
Accuracy (%)	90.74			
Macro Average (%)	75000	92.09	90.08	90.72
Weighted Average (%)	75000	91.24	90.74	90.75

Table 5.14: Comparison of few-shot classification accuracies (%) on MiniImageNet using MAML and Task-Decoder models. Results are reported for 5-way 1-shot and 5-way 5-shot settings, including Train, Validation, and Test accuracies.

Method		**5-way 1-shot (%)**			**5-way 5-shot (%)**		
Model	**Backbone**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**
MAML	CNN	57.58	46.36	47.54	76.64	61.11	62.23
MAML (Task-Decoder)	Shallow CNN	**58.10**	**45.24**	**47.81**	**76.82**	**61.04**	**62.52**
MAML (Task-Decoder)	CNN	**57.94**	**46.13**	**47.67**	**76.84**	**61.57**	**63.01**

Table 5.15: Comparison of training times (in hours) on MiniImageNet using MAML and Task-Decoder models for 5-way 1-shot and 5-way 5-shot experiments.

Model	**Backbone**	**5-way 1-shot**	**5-way 5-shot**
MAML	CNN	5.1h	7.2h
MAML (Task-Decoder)	Shallow CNN	**5.6h**	**7.9h**
MAML (Task-Decoder)	CNN	**6.4h**	**8.4h**

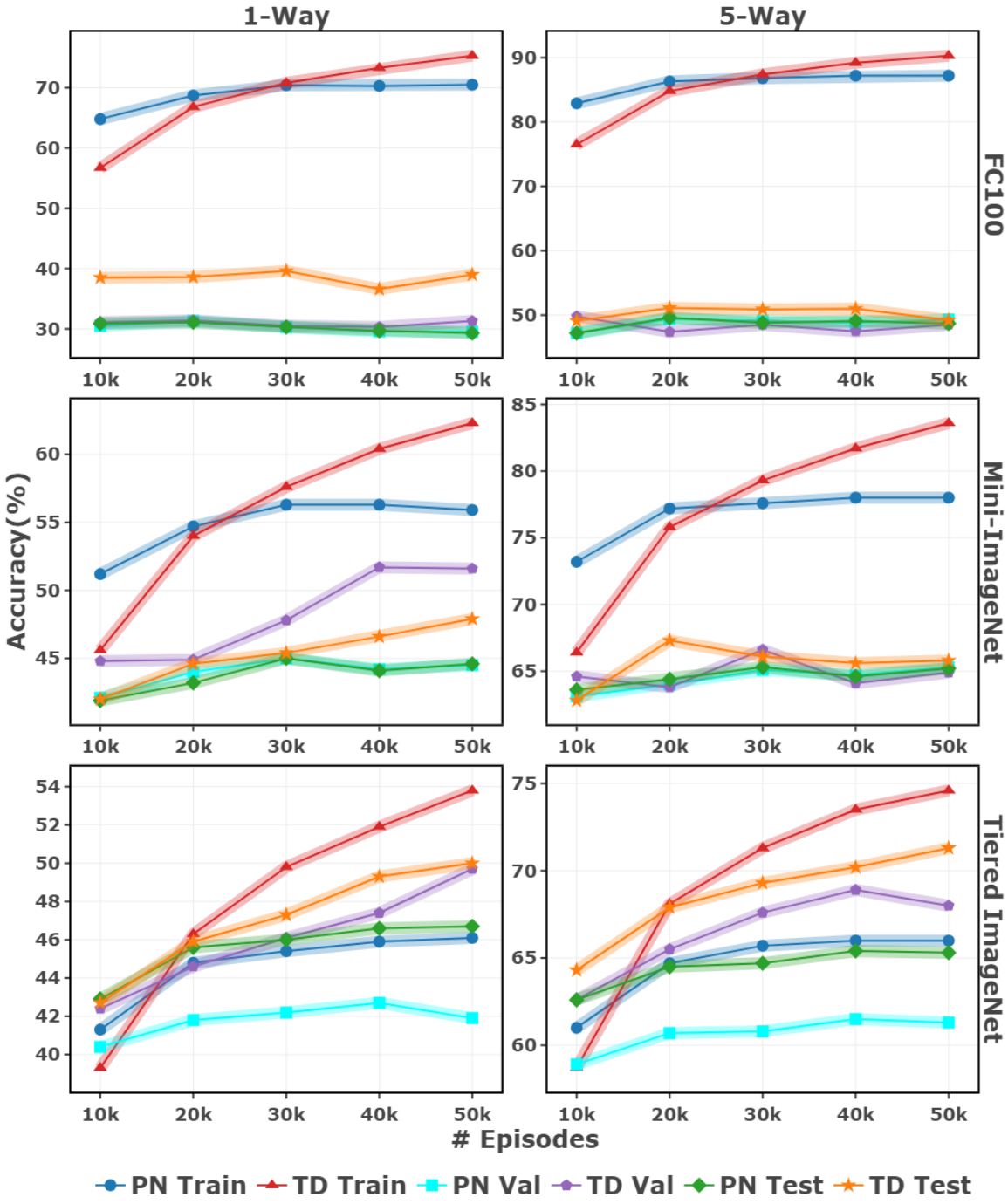


Figure 5.2: Accuracy curves for Prototypical Networks (PN) and Task-Decoder (TD) across different datasets, plotted against the number of episodes.

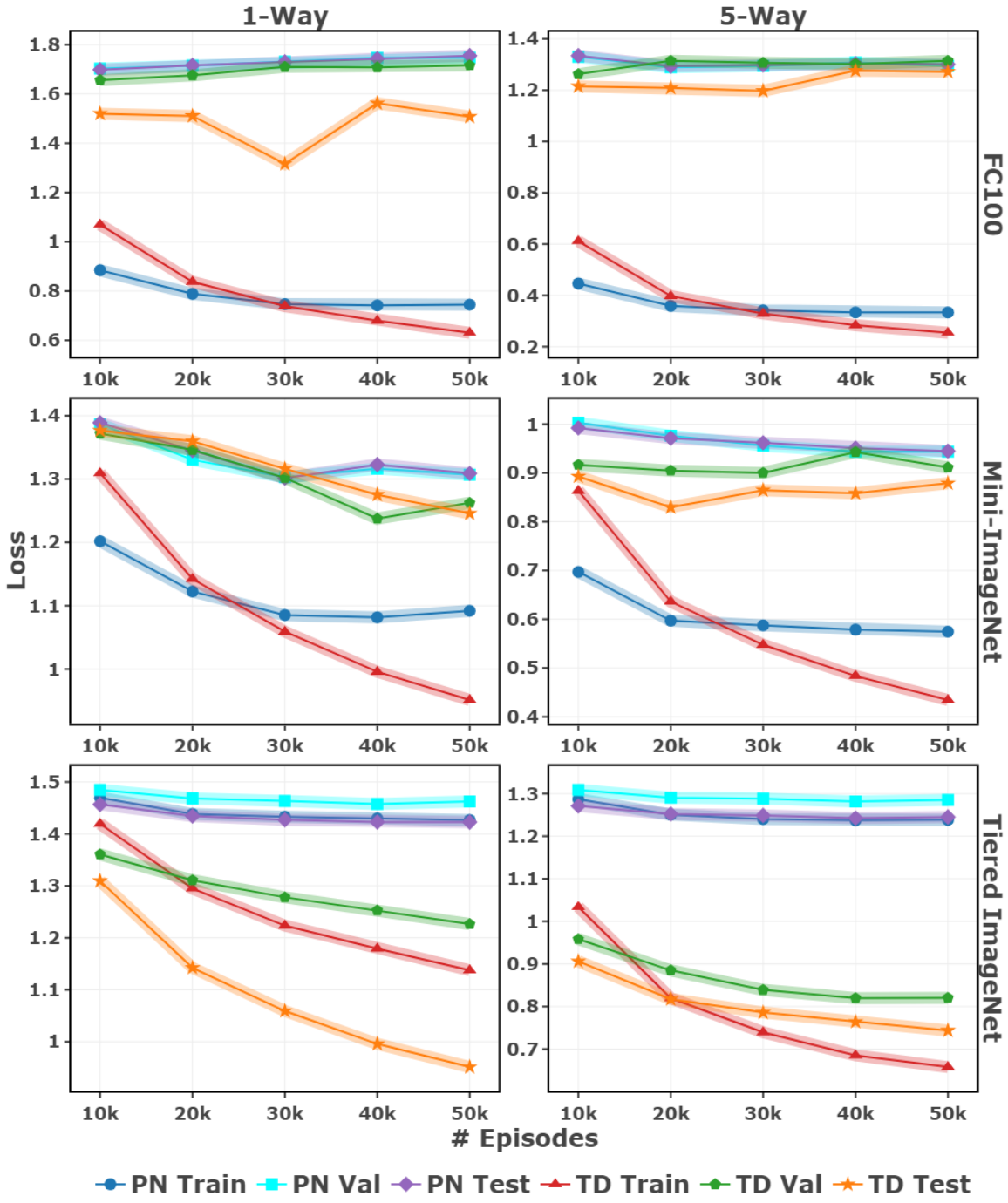


Figure 5.3: Loss curves for Prototypical Networks (PN) and Task-Decoder (TD) across different datasets, plotted against the number of episodes.

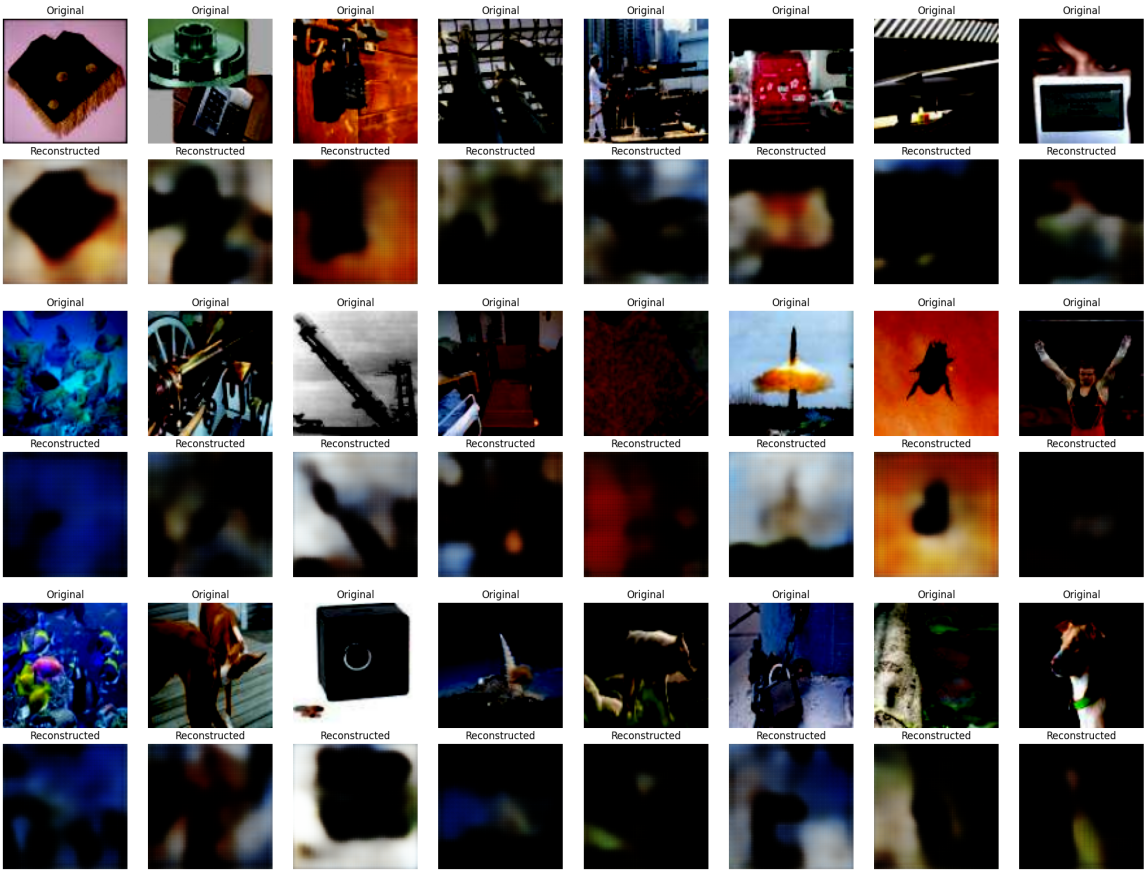


Figure 5.4: The reconstruction results of the Prototypical Network integrated with the Task-Decoder, trained on TieredImageNet, are evaluated using MiniImageNet images.

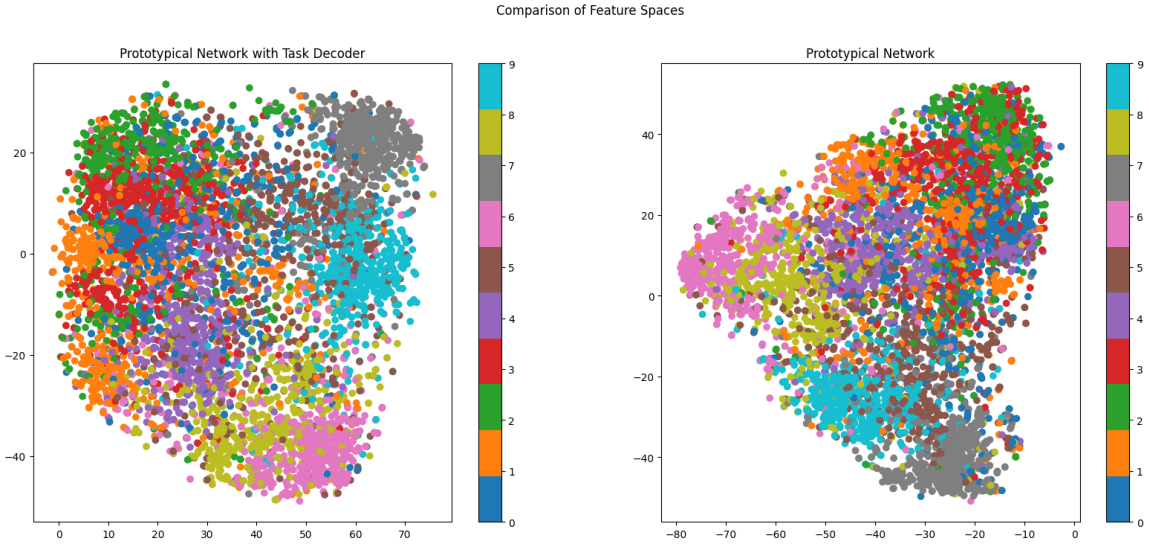


Figure 5.5: illustrates the t-SNE projections of the first 10 classes of MiniImageNet for Prototypical Networks with (left) and without (right) the Task-Decoder. The integration of the Task-Decoder demonstrates enhanced class separability, indicating its effectiveness in improving feature representation.

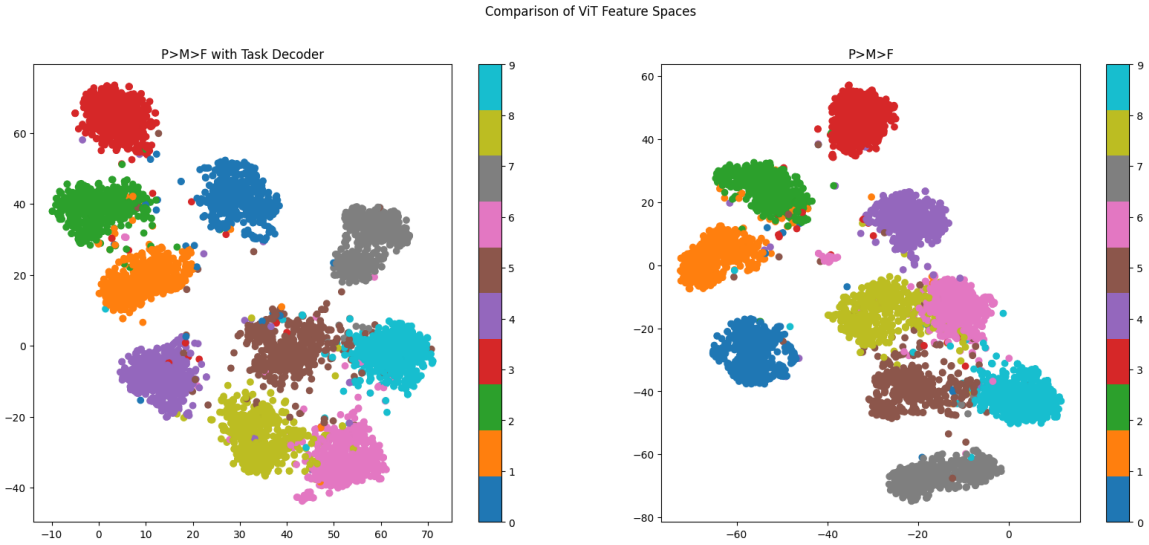


Figure 5.6: illustrates the t-SNE projections of the first 10 classes of MiniImageNet for P>M>F with (left) and without (right) the Task-Decoder. The integration of the Task-Decoder demonstrates enhanced class separability, indicating its effectiveness in improving feature representation.

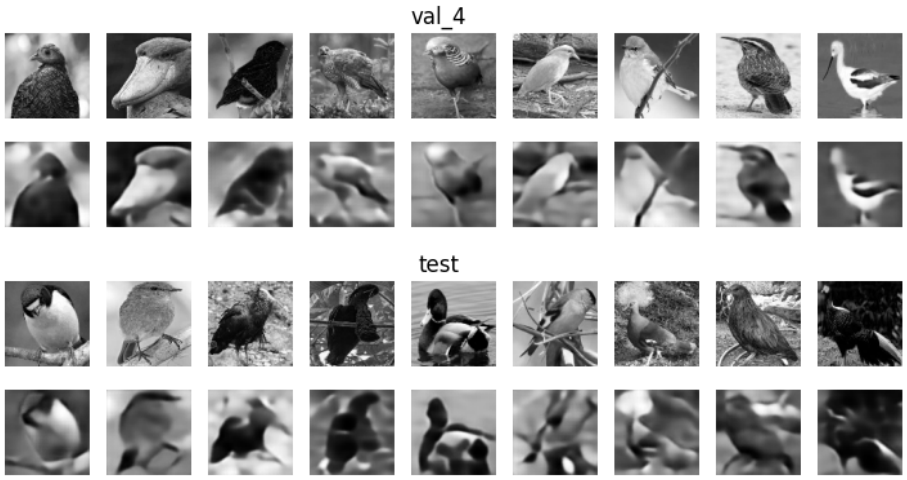


Figure 5.7: Reconstruction results ResNet Task Decoder before few-shot training.

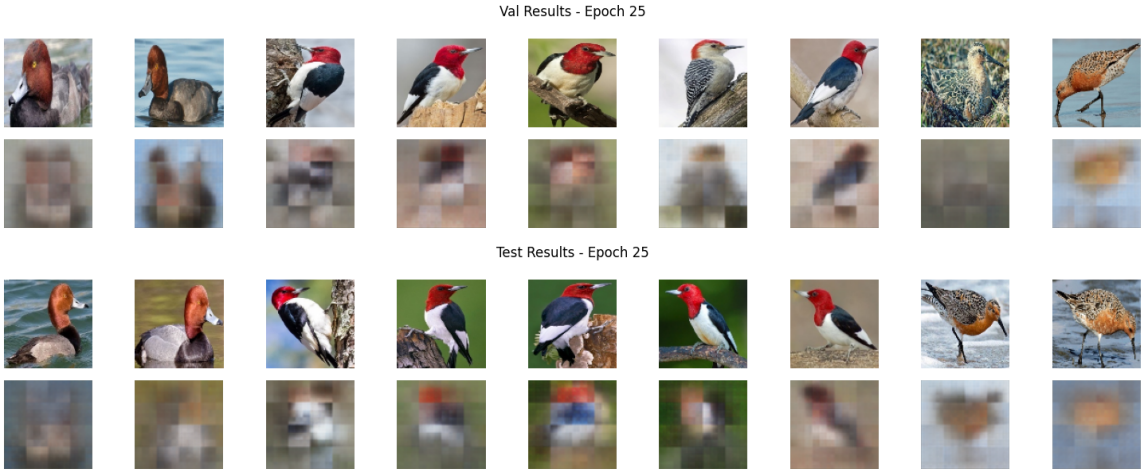


Figure 5.8: Reconstruction results Task Decoder before few-shot training.

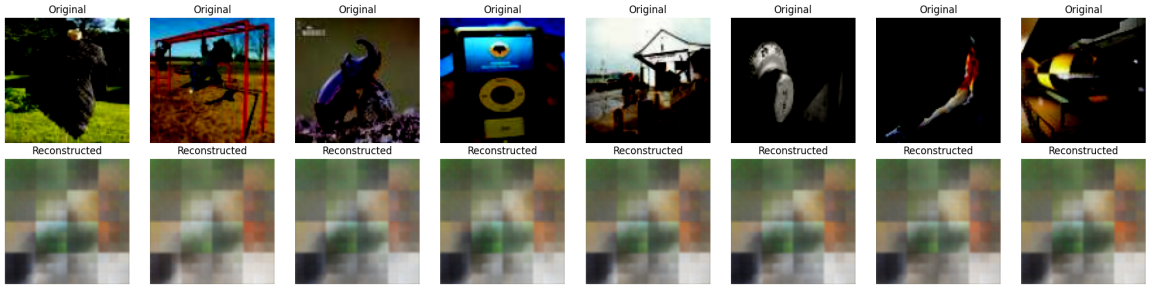


Figure 5.9: Reconstruction results of P>M>F with ViT Task Decoder trained on MiniImageNet images.

Chapter 6

Conclusions and Future Work

6.1 Summary of Contributions

This dissertation has presented a three-stage framework for developing wireless foundation models, focusing on self-supervised representation learning, large language model (LLM)-based reasoning, and meta-learning for rapid adaptation. The integration this dissertation establishes is both architectural and, in one concrete case, fully operational: all three modules, namely an RQ1 representation (DenoMAE2.0), the RQ2 DiSC-AMC reasoning pipeline, and RQ3's Meta-Task adaptation, were composed into a single system and evaluated end to end, together, on one task (RadioML AMC classification; Section 5.5). That joint run is itself a reported result, not merely a proposal: adding the RQ3 adaptation module on top of the RQ1+RQ2 composition gave no net accuracy gain over that composition alone (22.2% vs. 23.0%). This is a genuine three-module integration with an honest, bounded finding. What remains unevaluated jointly is breadth, not existence: CDRF's drone-detection evidence for RQ1 and Meta-Task's vision few-shot evidence for RQ3 are each validated on their own separate tasks and have not been folded into this same wireless

pipeline, so the dissertation's claims should be read component-by-component and, for the one case where composition was tested, exactly as that test reports, rather than generalized to every task the framework could plausibly address. Our work makes the following primary contributions:

1. **DenoMAE and CDRF: Self-Supervised, Vision-Based Representation Learning (Chapters 2–3)** DenoMAE is a self-supervised learning framework, co-developed with Atik Faysal (lead), that learns robust and transferable representations from unlabeled wireless signals by treating noise as an explicit modality and jointly processing time-domain IQ samples and constellation diagrams. The CageDroneRF (CDRF) benchmark complements it with large-scale, real-world over-the-air evidence (39 classes, 500+ GB of raw I/Q) that vision architectures (YOLO for detection, ResNet for hierarchical and open-set classification) transfer effectively to RF perception. Together these works address RQ1, establishing that image-based self-supervision gives label-efficient RF features (82.4% on a shared 10-class benchmark; 0.986 mAP@0.5 on real drone spectrograms), with clear bounds: performance collapses under extreme noise (-30 dB) and transfers weakly, without adaptation, to an unseen dataset (RadioML).

2. **Plug-and-Play and DiSC-AMC: LLM-Based Reasoning for Open-Set AMC (Chapter 4)** Plug-and-Play first demonstrated that a pre-trained LLM, given statistical signal summaries as text, can perform open-set modulation classification without wireless-specific training. **DiSC-AMC** [12], this dissertation's central contribution to RQ2, resolves Plug-and-Play's token-efficiency bottleneck through feature discretization, self-supervised context retrieval, and candidate-space pruning: a QLoRA fine-tuned 7B model reaches 83.0% in-distribution and 82.5% out-of-distribution accuracy on a synthetic 10-class benchmark, surpassing all supervised baselines, while

a training-free 7B configuration reaches 71.9% accuracy at 0.5K tokens, exceeding the proprietary o3-mini baseline given 2.9K tokens of raw statistics. On cross-dataset transfer to RadioML, DiSC-AMC amplifies a frozen or fine-tuned encoder's own classification head by up to 8×. Together, Plug-and-Play and DiSC-AMC provide the dissertation's affirmative answer to RQ2, bounded by the quality of the encoder whose embeddings the pipeline retrieves over.

3. **Meta-Task: Rapid Adaptation through Meta-Learning (Chapter 5)** We introduced Meta-Task, a method-agnostic approach to learnable regularization in few-shot learning, addressing RQ3. On vision few-shot benchmarks, Meta-Task delivers consistent, if uneven, gains (e.g., 65.4%→71.3% for Prototypical Networks on Tiered-ImageNet 5-way 5-shot, converging within 20k episodes versus more than 50k without it). Tested directly as a wireless deployment through the DiSC-AMC pipeline on RadioML, however, Meta-Task gives *no net gain* over plain fine-tuning (22.2% vs. 23.0%), a robust null result confirmed under two independent protocol fixes. RQ3's answer is therefore *partly* affirmative: the mechanism helps where the base learner's inductive bias is weak and the domain is in-distribution, but is not a substitute for a better representation under cross-dataset wireless transfer.

6.2 Key Findings

Each research question is answered following the same structure: hypothesis, controlled evidence, a difficult finding, and a bounded answer.

- **Answer to RQ1 (Represent) is yes, within bounds:** Image-based self-supervision (DenoMAE, Chapter 2) gives label-efficient RF features: 82.4% on a shared 10-class synthetic benchmark (+22.1 points over no pretraining at -10 dB) and 0.986

mAP@0.5 for real drone detection (CDRF, Chapter 3). The bound is explicit: representations collapse to near-chance below roughly -30 dB, and features trained on synthetic constellations transfer weakly, without adaptation, to an unseen dataset (RadioML).

- **Answer to RQ2 (Reason) is yes, with compact symbols and retrieved examples:** DiSC-AMC (Chapter 4) shows that a 7B LLM performs open-set AMC competitively with or better than dedicated supervised encoders (83.0% in-distribution, 82.5% out-of-distribution) when fine-tuned, and remains competitive training-free (71.9% at 0.5K tokens, versus 69.92% for the proprietary o3-mini model at 2.9K tokens). The bound: accuracy is only as good as the retrieval embedding it reasons over, and also collapses near -30 dB.

- **Answer to RQ3 (Adapt) is partly:** Meta-Task's learnable regularization (Chapter 5) helps few-shot vision benchmarks (e.g., +5.9 points for Prototypical Networks on Tiered-ImageNet) but gives no reliable gain on cross-dataset wireless AMC through the DiSC-AMC pipeline (22.2% vs. 23.0%, a robust null result). The lesson: on wireless transfer, the representation is the bottleneck, not the regularizer, directly connecting RQ3 back to RQ1. An independent few-shot channel-estimation study (Section 5.6) points the same way: MAML-based meta-learners did not beat a non-meta deep estimator or a classical LMMSE baseline under noise and domain shift.

6.3 Synthesis: What the Three Answers Add Up To

The three research questions are not independent lines of inquiry that happen to share a testbed; they form a single modular pathway, and one composition between them is already demonstrated end to end rather than merely proposed.

- **RQ1 · Represent:** Image-based self-supervision gives label-efficient RF features: 82.4% on shared synthetic data; 0.986 mAP@0.5 on real RF (Chapters 2–3).
- **RQ2 · Reason:** An LLM with compact symbols and retrieved examples performs open-set AMC: 83.0% in-distribution, 82.5% out-of-distribution, fine-tuned 7B (Chapter 4).
- **RQ3 · Adapt:** A learnable auxiliary task improves few-shot vision tasks (65.4%→71.3%, Prototypical Networks) but gives no net gain on cross-dataset wireless AMC (22.2% vs. 23.0%, Chapter 5).

Composed. DiSC-AMC (RQ2) uses an RQ1 encoder's embeddings for both context retrieval and candidate shortlisting inside its LLM reasoner, lifting RadioML accuracy by up to $8\times$ over the encoder's own classification head (Section 4.6.9), a concrete, already-run demonstration that the framework's components compose rather than merely coexist as independently validated pieces. This composition also inherits failure, not just success: when the RQ1 encoder is DenoMAE2.0 rather than DINOv2, its SynthMod-10-specialized embeddings retrieve uninformative neighbors on RadioML, and RQ2's reasoning gains vanish along with them (Section 4.6.9). The Jev comparison (Section 4.6.10) adds a second, narrower composition result: the DiSC-AMC prompt format transfers to a non-generative decision mechanism for coarse classes, but RadioML's fine-grained, order-based class distinctions still require a model capable of reasoning over the symbolic evidence rather than scoring fixed per-class descriptions, which is evidence that RQ2's benefit is specifically *reasoning*, not just the prompt's compact representation. The wireless validation experiment (Section 5.5) extends the composition to all three research questions at once: an RQ1 encoder, the RQ2 DiSC-AMC reasoner, and RQ3's Meta-Task adaptation were run together end to end on RadioML AMC, making this the dissertation's one fully three-way-composed system; its result is the honest finding that this particular third addition did not

improve on the two-module composition.

Connected. The RQ3 wireless test points back to RQ1 rather than standing alone: on a dataset the representation was not trained on, representation quality limits accuracy far more than adaptation strategy does, so a learnable regularizer cannot substitute for a better encoder. Read together, RQ1 supplies the substrate, RQ2 supplies the reasoning that converts that substrate into decisions (and inherits its limits), and RQ3 clarifies where adaptation can and cannot compensate for gaps in the substrate itself. The qualitative reasoning traces in Section 4.6.11 add a final, cross-cutting caveat to all three: even where the pipeline's numbers are strong, its natural-language rationales are fluent whether or not they are correct, so interpretability and reliability are not the same property and should not be conflated when this framework is deployed.

6.3.1 A Familiar Arc: From Hand-Crafted Features Back to Representation

Placed side by side, this dissertation's chapters retrace, in miniature and on one narrow task, a progression the broader machine learning field has walked through at much larger scale. Plug-and-Play (Chapter 4) begins with hand-crafted statistical features (moments, cumulants, skewness, kurtosis) linearized directly into an LLM prompt, in the spirit of classical AMC's feature-engineering tradition. DenoMAE and DenoMAE2.0 (Chapter 2) respond to that approach's ceiling by replacing hand-crafted features with a *learned*, domain-specific representation, trained self-supervised specifically to avoid the labeled-data cost of end-to-end supervised deep learning while still specializing to wireless signal structure (treating noise as an explicit modality; adding local patch classification in the 2.0 revision). DiSC-AMC and Meta-Task (Chapters 4–5) then add machinery *on top of* that representation to extract more value from limited labels and to reason flexibly about open-set and few-shot

scenarios: retrieval-augmented symbolic reasoning in one case, a learnable auxiliary regularizer in the other.

The dissertation's cross-dataset evidence closes this arc in an unexpected direction. DenoMAE2.0, the domain-specialized representation that wins in-distribution on SynthMod-10 (82.40% vs. DINOv2's 65.33%, Table 4.3), is the *weaker* representation once the task shifts to RadioML, a dataset it was never trained on: DINOv2 (a generic, broadly pretrained visual encoder with no wireless-specific tuning at all) reaches 76.25% through the DiSC-AMC pipeline at $+20$ dB, while DenoMAE2.0 reaches only 21.25% in the same pipeline, and its own fine-tuned head alone reaches just 11.25% (Table 4.5). The representation that was built to specialize away from generic, broadly pretrained features generalizes *worse* than those generic features once evaluated out-of-distribution. Neither of the two machinery layers built on top of representation can repair this: DiSC-AMC's reasoning amplifies DINOv2 by up to $8\times$ but cannot manufacture class signal that DenoMAE2.0's narrower embeddings do not carry (Section 4.6.9), and Meta-Task's learnable regularizer, tested directly on this exact failure case, gives no net accuracy gain over plain fine-tuning (Section 5.5). A separate channel-estimation study (Section 5.6) shows the same ordering on a different wireless task: a classical estimator built on explicit channel statistics stays robust under data paucity and shift, while learned and meta-learned estimators degrade.

The lesson is not that domain-specific self-supervision or downstream adaptation are mistaken directions, since both deliver real, demonstrated gains within their tested scope (DenoMAE2.0 in-distribution; Meta-Task on vision few-shot benchmarks). The lesson is that each successive layer this dissertation adds (features to representations, representations to reasoning, reasoning to adaptation) was built to compensate for a limitation of the layer beneath it, and none of those later layers substitutes for what the earliest layer is actually responsible for: whether the representation itself captures invariances that hold beyond the distribution it was trained on. Having moved away from hand-crafted features specifically

to gain that generality, and then specialized the resulting representation back toward one dataset to gain in-distribution accuracy, this dissertation's evidence arrives back at representation generality as the open problem, the same problem the move away from feature engineering was originally meant to solve. This is the dissertation's clearest instance of a pattern that recurs through machine learning's own history at every scale: architecture, reasoning, and adaptation can be stacked arbitrarily high, but they inherit, rather than correct, the generalization limits of what they are built on top of.

6.4 Limitations

Several limitations bound every result in this dissertation, each surfaced directly by the experiments rather than assumed in advance:

- **Extreme noise:** Image-based representations (DenoMAE, CDRF) and the retrieval they enable (DiSC-AMC) collapse to near-random accuracy at approximately -30 dB SNR, where constellation diagrams become effectively featureless. This is a shared failure mode across every vision-rooted component of the framework, not an isolated weakness of one method.

- **Dataset transfer:** Encoders trained on synthetic AMC data (DenoMAE2.0, DINOv2) transfer weakly to the real-world RadioML benchmark without target-domain adaptation; DiSC-AMC's reasoning amplifies a weak or unadapted encoder but cannot substitute for a representation whose nearest neighbors are uninformative on the new dataset.

- **Wireless adaptation:** Meta-Task gives no net gain on RadioML AMC (22.2% vs. 23.0%); its demonstrated gains are confined to in-distribution, vision few-shot benchmarks. Rapid, few-shot wireless adaptation therefore remains an open problem for

future work, not a solved component of the framework.

- **Scale and scope of validation:** Experiments are conducted primarily on 10- and 24-class synthetic/RadioML benchmarks plus one real-world drone RF benchmark (CDRF); broader modulation taxonomies, multi-antenna/MIMO scenarios, and hardware-in-the-loop deployment remain untested.

- **Statistical uncertainty on close comparisons:** Several headline comparisons (e.g., DiSC-AMC's 83.0% versus DenoMAE2.0's 82.4%, Section 4.6.7) rest on single evaluation runs of 200–240 queries without reported confidence intervals or repeated-seed variance. Margins of a few points in these comparisons should be read descriptively, not as established statistical superiority; Section 4.6.7 flags this explicitly where it applies.

- **Dataset split provenance:** CDRF's released splits are not accompanied by a recording/session/device-level lineage manifest proving that augmented or windowed descendants of the same raw recording never cross a train/test boundary (Chapter 3). This is an absence of documented evidence, not a demonstrated leakage finding, but it means the hierarchical and open-set results in that chapter cannot currently be independently audited for recording-level independence.

- **Artifact and provenance release:** No chapter in this dissertation currently ships a table-to-run manifest (exact command, config, seed, checkpoint hash, and raw predictions for each reported number). This limits independent reproduction of specific rows even where the underlying method and code are available, and is a release-process gap to close rather than a claim about the correctness of the underlying results.

6.5 Scope of the Cost-Effectiveness Claim

The dissertation's title describes the proposed framework as cost-effective, and that word carries a specific, bounded meaning that this section states explicitly rather than leaving implicit.

The design principle is to reuse an available pre-trained base and concentrate new investment only where it addresses a specific task limitation, rather than training a wireless-specific model from scratch. The component studies give numerical evidence for particular instances of this principle: DiSC-AMC's prompt redesign reduces rounded context from 2.9K to 1.3K tokens (Section 4.6.8) while changing reported accuracy; its fine-tuned configuration separately reaches 83.0%/82.5% at 0.5K tokens (Section 4.6.7). These are each configuration-level, context- or token-efficiency results, evaluated under the specific conditions reported for that scenario; they should not be merged into a single dominant operating point, since different scenarios change the model, the prompt, and the training regime simultaneously.

What is *not* measured is a complete deployment-lifecycle accounting. Write a simplified lifecycle cost as

$$C(n) = C_{\text{prep}} + n\, C_{\text{query}} + C_{\text{maintenance}}(n), \tag{6.1}$$

for n queries, preparation cost C_{prep} (data collection, fine-tuning, index construction), per-query cost C_{query} (inference latency, token cost, energy), and maintenance cost over time. If one configuration requires additional preparation $\Delta C_{\text{prep}} > 0$ but saves $\Delta C_{\text{query}} > 0$ per query, the simple break-even query count is $n^* = \Delta C_{\text{prep}} / \Delta C_{\text{query}}$, ignoring maintenance differences. No numerical break-even estimate is offered anywhere in this dissertation, because matched latency, memory, energy, and monetary measurements across configurations

were not collected. Table 6.1 summarizes, per contribution, which resources are reused from pre-trained or inherited components and which represent genuinely new investment; it is an accounting map, not a numerical cost comparison.

Table 6.1: Resource responsibilities by contribution. Reused resources lower preparation cost; the right column lists what each contribution newly invests.

Contribution	Reused / inherited resource	New or retained investment
Plug-and-Play	Pre-trained LLM, in-context learning capability	Statistical feature extraction, prompt construction, exemplar curation
DiSC-AMC	Self-supervised visual encoder, retrieval index, optionally a pre-trained LLM	Discretization pipeline, FAISS index, candidate-retrieval classifier, optional QLoRA adapter
Meta-Task	Existing meta-learner (PN, MAML, MetaOptNet, P>M>F), shared representation	Task-Decoder (reconstruction head) and its training overhead
CageDroneRF	Pre-trained vision architectures (YOLO, ResNet), COCO initialization	Data acquisition, annotation, augmentation pipeline, per-task fine-tuning

The supported cost-effectiveness claim is therefore specific: the framework organizes reuse around selective, targeted specialization rather than isolated from-scratch training, and the component studies demonstrate particular context- and parameter-efficiency benefits (Table 4.7, Table 4.8) and adaptation tradeoffs (Chapter 5) in support of that principle. A complete energy, latency, or monetary ranking against lightweight, task-specific RF classifiers is a separate measurement this dissertation does not provide, and the title's economic claim should be read at this, more limited, level of evidence.

6.6 Contributions and Authorship

Several works underlying this dissertation are collaborative, and academic integrity requires stating authorship plainly rather than letting the framework narrative blur it. Table 6.2 lists, for each work used as evidence in this dissertation, the author's specific role.

Table 6.2: Authorship and role of each work used as evidence in this dissertation.

Work	Role	Venue	Used for
Plug-and-Play AMC	First author	IEEE WOCC 2025	RQ2 core
DiSC-AMC	First author	arXiv:2510.00316 (2026)	RQ2 core; RQ1+RQ2 composition
Meta-Task	First author	arXiv:2402.18599	RQ3 core
CageDroneRF (CDRF)	Shared first author (with A. Faysal)	arXiv:2601.03302	RQ1 real-world evidence
DenoMAE	Co-author (A. Faysal first)	IEEE Commun. Letters 2025	RQ1 representation
DenoMAE2.0	Co-author (A. Faysal first)	IEEE Trans. Commun. 74, 2026	RQ1 representation; DiSC-AMC encoder
Channel-estimation study	Second author (R. Gh. Roshan first)	IEEE WOCC 2026	RQ3 corroborating evidence

For DenoMAE and DenoMAE2.0 specifically: the core idea and primary contribution are Atik Faysal's; the author's role was experiment implementation and paper writing. This dissertation uses both works as the pre-trained representation that other chapters build on and compare against, never as a first-author claim. This distinction matters for assessing the dissertation's novel contribution: the three first-author works (Plug-and-Play, DiSC-AMC, Meta-Task) and the shared-first-author CDRF benchmark are what this dissertation claims as its own; DenoMAE and DenoMAE2.0 are evidence the dissertation builds on and is transparent about not having originated. The channel-estimation study is likewise

second-author work used only as corroborating evidence for RQ3 (Section 5.6).

6.7 Future Research Directions

The framework validated in this dissertation leaves several directions open, some already probed with a first (negative) result rather than left purely speculative.

6.7.1 A First Fusion Attempt, and Why It Did Not Help

The most natural integration test, layering Meta-Task's learnable regularization on top of the DiSC-AMC pipeline on RadioML, was already run (Section 5.5) and gave no net gain (22.2% vs. 23.0% for plain fine-tuning). This is a useful negative result, not an unexplored opportunity: it shows that naively stacking the framework's adaptation module onto its reasoning module does not help when the underlying representation is the bottleneck. A more promising fusion direction, left untested here, is feeding Meta-Task-adapted *encoder* features (rather than a frozen or vanilla fine-tuned DenoMAE2.0/DINOv2 backbone) into DiSC-AMC's retrieval stage, so that adaptation improves the exemplars being retrieved rather than only regularizing a classification head that DiSC-AMC's reasoning already bypasses.

6.7.2 Vision-Language Models for Richer Reasoning

DiSC-AMC currently reasons over discretized *statistics* of the constellation diagram, discarding the image itself after the self-supervised encoder reduces it to an embedding. A vision-language model (VLM) that reasons jointly over the constellation image and the discretized context could recover information the current pipeline discards: for instance, visual symmetry or point-cloud density patterns that are only coarsely captured by moment- and

cumulant-based statistics. This is a natural next step for RQ2 that keeps the framework's modularity: the VLM would replace only the LLM reasoning stage, not the retrieval or discretization stages.

6.7.3 Open Gaps Identified During Evaluation

Several specific, scoped gaps were identified while running the DiSC-AMC and Meta-Task evaluations and are left as concrete follow-up work rather than folded into the main results:

- **Shots-per-class sweep:** The sensitivity of DiSC-AMC and Plug-and-Play to the *number* of in-context examples per class (1 / 2 / 4 shots) was not swept; only the choice of *which* examples to retrieve (FAISS vs. centroid) was ablated (Section 4.6.8).
- **A fair non-generative-model test:** The Jev comparison (Section 4.6.10) shows the DiSC-AMC prompt transfers to a non-generative scoring model, but the test itself has confounds: a prompt not tailored for non-generative decoding, criteria that cannot carry query-specific evidence, and a bug that sent null option descriptions in default runs. A fair comparison requires query-specific criteria and repeated runs at larger sample sizes than the 200–240 queries tested here.
- **Explicit SNR-aware prompting:** Performance degrades gracefully but monotonically with SNR across every LLM-based configuration; none of the current prompts explicitly condition on an SNR estimate, which could let the LLM calibrate its own confidence rather than reasoning as if every query were equally reliable.

6.7.4 Additional Promising Directions

Beyond these, several secondary avenues warrant exploration:

- **Scaling to Broader Modulation Taxonomies:** Extending pretraining and evaluation to 50+ modulation schemes, including emerging waveforms (OTFS, NOMA) and legacy standards (analog FM/AM), to build truly general-purpose encoders and reasoning pipelines.
- **Multi-Task Foundation Models:** Training or prompting unified pipelines that perform modulation classification, SNR estimation, channel equalization, and interference detection simultaneously, leveraging shared representations across tasks.
- **Real-Time Edge Deployment:** Quantifying the latency and memory footprint of the training-free DiSC-AMC configuration (7B open-weight model, 0.5–1.3K-token prompts) on embedded GPU or FPGA targets, where the token-efficiency gains demonstrated here would translate directly into deployment feasibility.
- **Cross-Domain Transfer:** Evaluating whether the representations and reasoning pipelines developed for AMC and drone detection in this dissertation transfer to radar, satellite, or biomedical RF applications, testing the generality claimed by the foundation-model framing.

6.8 Closing Remarks

This dissertation has laid the groundwork for a new generation of wireless AI systems based on the principles of foundation models. By integrating self-supervised learning, LLM-based reasoning, and meta-learning, we have demonstrated a path toward building general-purpose, adaptable, and interpretable models for the wireless domain. The vision is to move beyond specialized, black-box classifiers and toward a future of truly cognitive radios that can understand, reason about, and adapt to the complexities of the wireless world.

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